Survey Research 2019-08-02

*2019 K-Innovation ODA Program with Tunisia*

# Policy Consultation on the Support for the Establishment of an Effective National System of Technology Transfer (NSTT) in Tunisia

STEPI SCIENCE AND TECHNOLOGY POLICY INSTITUTE

RÉPUBLIQUE TUNISIENNE
MINISTÈRE DE L'ENSEIGNEMENT SUPÉRIEUR ET DE LA RECHERCHE SCIENTIFIQUE

## 2019 K-Innovation ODA Program with Tunisia

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|---|---|
| Title | Policy Consultation on the Support for the Establishment of an Effective National System of Technology Transfer(NSTT) in Tunisia |
| Institute | Science and Technology Policy Institute (STEPI) |
| Partner Country | Republic of Tunisia |
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# Executive Summary

Tunisia (officially the Republic of Tunisia) is a country in the Maghreb region of Northern Africa covering 163,610 square kilometers.[1] It is bordered by Algeria and Libya and by the Mediterranean Sea to the north and east. The population of Tunisia is 11.69 million as of 2019.

Tunisia became the cradle of the "Arab Spring" (2011-2012). The years following this revolution were characterized by strong political and security instabilities. Despite the progress made after 2015, Tunisia still faces substantial challenges caused by weak job creation, high unemployment, and unsustainable public finances. Tunisia continues to create economic opportunities on a sufficient scale to consume the growing pool of youth as social tensions and development inequality at the regions remain a major threat.

Tunisia's economic model is oriented toward exports, and the country has gradually become a solid trade partner of Europe. Tunisia has relatively low Research and Development (R&D) performance; specifically, its gross domestic expenditure on R&D accounted for only 0.63% of GDP in 2015. Nonetheless, Tunisia is still characterized by a higher level of R&D performance compared to the average for North African and Arab Countries. In 2015, R&D funding sources accounted for 65% of GERD from the Government, 20% from the business enterprise sector, and 15% from international cooperation.

The insufficient investment in R&D by the private sector remains a major challenge of the

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[1] MOFA 2018.

Tunisian Research and Innovation (R&I) system. Moreover, the country suffers from very limited collaboration between industry and research laboratories and centers. In recent years, the political debate has focused on the need for more concerted activities as well as for a more inclusive decision-making process to shape the Tunisian R&I system and foster R&I performance.

At present, Tunisia has great potential to migrate to a new development model because of its highly qualified human resources and its young population of trainable talent, enabling it to advance its economy to climb up the value chain. This transition is key to enhancing development at the local, regional, and national levels and providing more employment opportunities for university graduates. A key player in this transformation is the national system of Science, Technology, and Innovation (STI), a platform for the generation of innovative products and services.

Following the 10th development plan of Tunisia (2000-2006), the Tunisian government promoted scientific research and innovation development. Under this plan, institutional arrangements have been established to expedite the link between research activities and business demands and market requirements as well as the promotion of scientific knowledge.

To bolster the mobility of scientific knowledge and the vitalization of the research outcomes, the government of Tunisia has focused on the creation of techno parks. The objective is to settle and facilitate specialized research and training institutions in a promising sector as well as economic enterprises.

Thus, in 2012, the National Agency for Scientific Research Promotion (ANPR) had newly launched a network of Technology Transfer Offices (TTOs) in research centers and universities with a central unit within the Agency to coordinate and support this network through detailed conventions.[2] Today, there are 25 TTOs that have joined this network.

[2] K-Innovation 2018.

The missions of TTOs require extensive competence that is not generally available to one person, and TTOs need diverse supports from all partners to carry out their duties. This network of TTOs helps in raising awareness of the importance of technology transfer in universities. Still, ANPR needs more resources to reinforce the TTO network.

Among the Maghreb regions (Algeria, Morocco, and Tunisia), the government of Tunisia has more effectively activated legal, institutional, and financial channels to establish a national innovation system (STEPI, 2018 K-Innovation, 2018). Tunisia has more robust R&D capabilities compared to its neighboring countries, but it still requires more competency and systems set side by side with the world’s most innovative countries. Therefore, Tunisia has constructed international cooperative relationships with a number of countries -- for example, France, Germany, Korea, and other African countries -- to search for opportunities to join the international networks for innovation (STEPI, 2018 K-Innovation, 2018).

This year’s study visit program for the Tunisian delegation aimed to reinforce the capacity of technology transfer practitioners of Technology Transfer Organizations (TTOs) by partaking of Korean experiences in regulatory and institutional frameworks as well as promoting capacity building in the fields of technology transfer and commercialization. This program has been a distinguished opportunity for Tunisian stakeholders in technology transfer and commercialization to experience Korean models as a basis for advancing national policies to strengthen R&D capabilities and to enhance technology transfer and utilization of research for commercialization. In linkage with the Tunisian-Korean scientific cooperation potential, the two countries will be capable of creating good opportunities to further research collaboration and projects for mutual benefits.

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# Chapter 1. Project Overview

## 1. Introduction

Tunisia is a Northern African country bordered by Algeria, Libya, and Mediterranean Sea to the north and east. The area is 163,610 square kilometers, with an estimated population of 11.65 million in 2019. Tunisia became the cradle of the "Arab Spring" (2011-2012).

The country has a manifold economy, varying from agriculture and mining to manufacturing and tourism. In 2018, the GDP totaled $39.86b ($137,36b purchasing power parity)[3]. The per capita GDP (PPP) reached $11,096 in 2018, one of the highest in Africa. The industry sector including the construction industry accounts for 24% of the GDP, and the agricultural sector makes up 9%.[4] The manufacturing sector is mainly composed of clothing and footwear, parts for the automobile industry, and electrical machinery. Despite posting annual growth of 4.5% on average for the years 2000-2010, Tunisia has achieved only 1.5% average growth after that. The main engines of growth, specifically tourism and phosphate industry, still face difficulties attributable to the major cause of the current low economic growth.

The Tunisian National Innovation System (NIS) has been regarded as one of the best constructed and most compelling in Africa and MENA region. It has made a number of achievements in R&D investment and fostered a number of researchers aside from knowledge creation.

Nonetheless, the latest diagnosis of the Tunisian NIS (2015) found that the main cause is

[3] WDI 2018

[4] WDI 2018

the lack of contribution of the NIS in the invention of value and qualified jobs. The following are the identified causes: inadequacy of governance, inadequacy of the innovation system, and shortfalls regarding technology transfer and research vitalization[5]. The Ministry of Higher Education and Scientific Research (MHESR) of Tunisia is the ministry responsible for making policies and enhancing initiatives from all the various innovation stakeholders for scientific and technological development.

Based on a stable partnership with the Science and Technology Policy Institute (STEPI) of Korea in the field of STI policy, the Ministry asked STEPI to suggest recommendations for the improvement of the National Technology Transfer System and enhancement of research vitalization in Tunisia by handing in a project proposal to STEPI in 2017. Within the project implementation, MHESR and STEPI signed an MOU on STI in June 2018 for further collaboration.

The 2018/19 K-Innovation Project with Tunisia placed emphasis on supporting the construction of an effective National System of Technology Transfer (NSTT) in Tunisia. It was suggested and discussed as a follow-up action of the two series of KOICA training programs held between the two countries, first in 2009-2010 and then in 2015-2017.

The 2018 K-Innovation Project with Tunisia focused on the diagnosis of the current situation and key challenges of the technology transfer system in Tunisia. It focused on the micro level issues such as its legal system, the related institutes, and its main partner targeting the ministry level. For the second year, the 2019 K-Innovation focused more on stakeholders who are in the technology transfer organizations (TTOs) in universities and research institutes, etc. It also focused on disseminating the project results with more stakeholders in Tunisia, especially on collaborating with the Korean Embassy in Tunisia to commemorate the 50th anniversary of establishment of diplomatic relations between Korea and Tunisia.

[5] WDI 2018

## 2. Objectives

The title of the project is "Policy Consultation on the Support for the Establishment of an Effective National System of Technology Transfer (NSTT) in Tunisia." The program runs for approximately 24 months from January 1, 2018 to December 31, 2019.

The objectives of the 2018-2019 K-Innovation Project with Tunisia are as follows:

- To investigate problems and diagnose the challenges of the national technology transfer system and policy in Tunisia

- To share Korean experiences in the establishment of the national technology transfer system

- To hold local capacity-building workshops

- To provide technical advice on the establishment of a national technology transfer system that can effectively form research-industry linkage and allow better vitalization of research products

More specifically, this project aims to support effectively the improvement of the Technology Transfer System of Tunisia through the diagnosis of its policy, current system, and performance and capacity-building training sessions in the following aspects:

1. Policy diagnosis of the Technology Transfer System
    - 1.1 Evaluation of institutional framework of TTS
    - 1.2 Evaluation of legal framework
    - 1.3 Diagnosis of the management process
    - 1.4 Evaluation of the performance of TTS
2. Promoting good governance of TTS
    - 2.1 Define a reference framework for TTs
    - 2.2 Suggest the effective management of TTS
3. Providing customized training for the human resources of TTS

3.1 Develop appropriate capacity-building programs for policymakers, researchers, and TTO personnel

3.2 Designing advanced training sessions for TT professionals

4. Sharing Korean experience relevant to Tunisia

In 2018, the first year of the project, the project focused on analysis and diagnosis of the current situation and existing national system of technology transfer in Tunisia. It assisted in the establishment of a system by providing comments and advice for overcoming the challenges suggested by the NIS diagnosis. An invitational study visit program in Korea and a local capacity-building workshop were hosted with the aim of providing the relevant Korean experience, strategies, and methodologies for the enhancement of research vitalization and/or other topics jointly identified by STEPI and MHESR. Afterward, the final report for the 2018 activities was submitted to MHESR at the end of the first year.

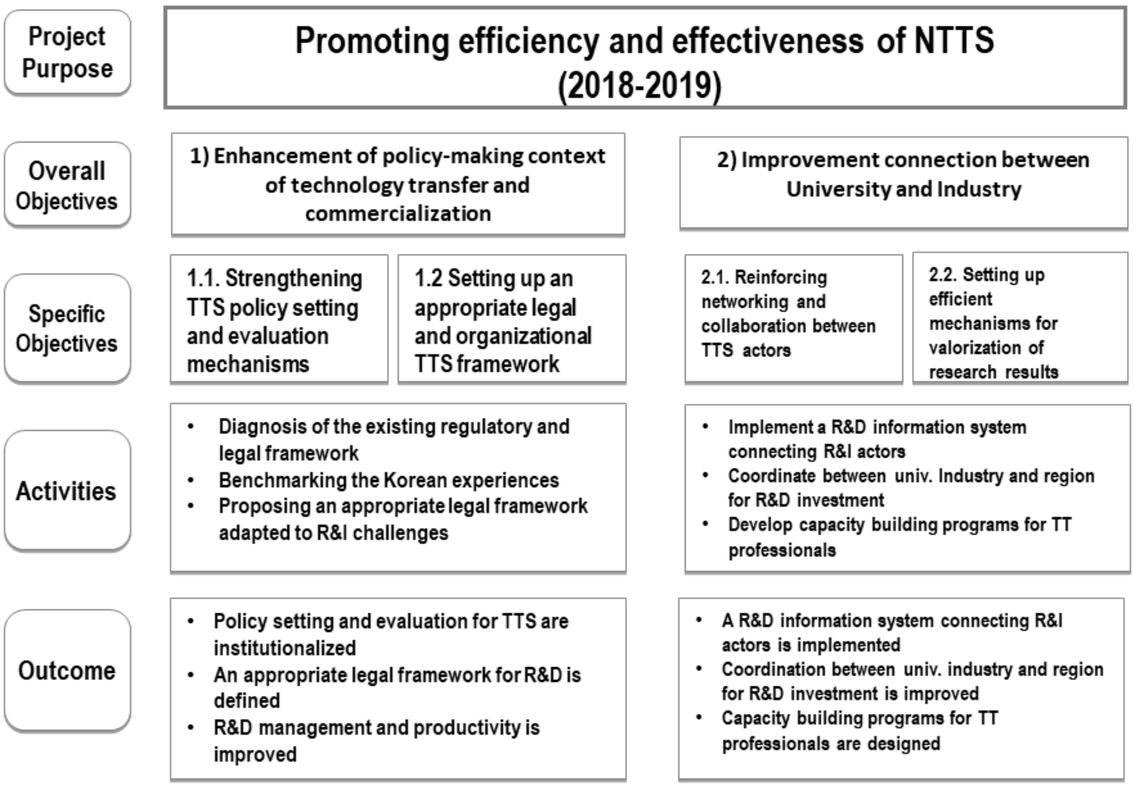
In 2019, the second year, this project focused more on building capacity at the practical level especially for those who work in Technology Transfer Organizations (TTOs). With the support of ANPR and its network, the project focused on the main demands of TTOs to set the objective of the second-year project. From the survey conducted at the kick-off workshop, the TTOs said that they want to learn the initial strategic and management plan to operate. Based on these objectives, this project focused on the capacity building of TTOs and stakeholders as a second-year project goal and implemented a number of activities such as invitational training workshop and dissemination workshop.

## 3. Project Framework

This project sought to strengthen the capacity of policymakers, practitioners of technology transfer offices, and other technology transfer stakeholders to support the Tunisian government's endeavors to advance the national system of technology transfer. The project provides Korean experiences in regulatory or administrative and institutional frameworks and vitalization of research for commercialization.

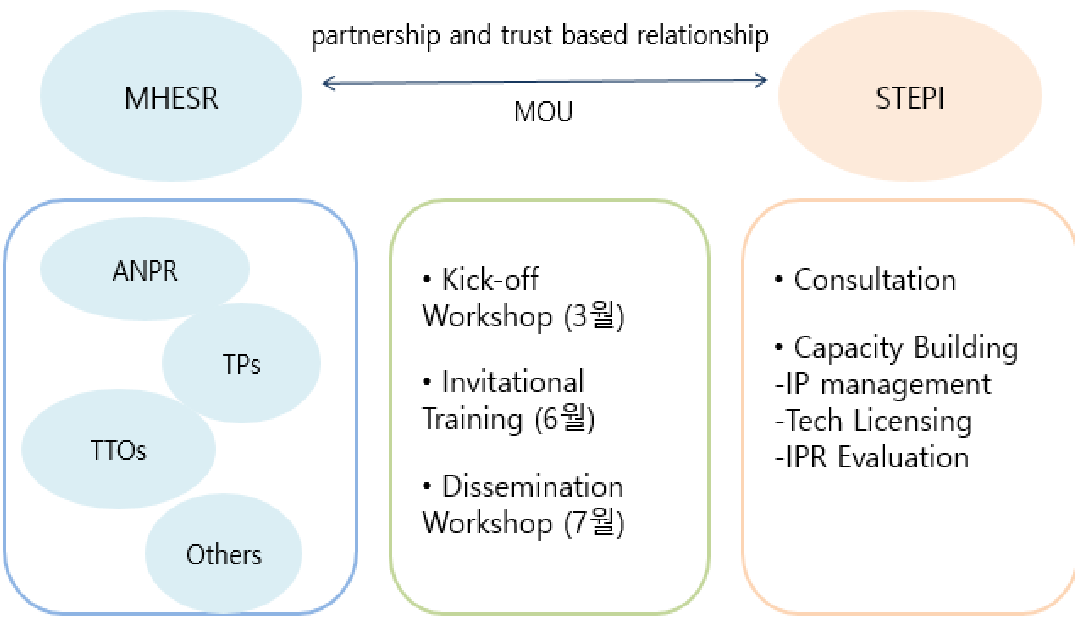
The current status of Tunisia's technology transfer system should be analyzed and diagnosed in order to improve the existing system; after analyzing the Strengths, Weaknesses, Opportunities, and Threats, action plans for the construction of an efficient, effective system are necessary. Finally, mechanisms and programs to enhance Tunisia's national technology transfer should be established in order to advance the technology transfer policy.

[Figure 1-1] Conceptual Framework of the Project




As seen in Figure 1-1, the project purpose of this project is to promote the efficiency and effectiveness of NTTS; hence the two overall objectives of 1) enhancing the policymaking context of technology transfer and commercialization and 2) improving connection between university and industry in Tunisia. Thus, the specific objectives are to strengthen technology transfer system policy setting and evaluation mechanisms, to set up an appropriate legal and organizational TTS framework, to reinforce networking and collaboration between TTS actors, and to set up efficient mechanisms for the vitalization of research results.

The first year of this project focused on diagnosing the existing regulatory and legal framework, benchmarking the Korean experiences, and proposing an appropriate legal framework adapted to R&I challenges. It presents policy recommendations to improve NSTT in Tunisia and provides a study visit program for Tunisian policymakers and experts to learn from the Korean experience. In 2019 as the second year of the project, it places emphasis on implementing an R&D information system, coordinating with TTOs especially with universities and research centers, and building capacity for those related stakeholders in Tunisia. It provides consultations on strategies and recommendations that TTOs can adopt to operate their offices, especially capacity building on IP management, technology licensing, and IPR Evaluation. Since the project needs to be implemented in collaboration between Korea and Tunisia, the partnership and trust-based relationship are exceptionally significant for this project, as indicated in the picture below.




## 3.1 Expected Outcomes

The two-year project will suggest a diagnosis of the national technology transfer system of Tunisia based on the research analysis including the results of interviews and meetings with key stakeholders of technology transfer. This result will provide the basis for policy recommendations and specific guidelines for diverse stakeholders to improve and build relative capacity with regard to the technology transfer system effectively as well as to link research with industrial competitiveness.

As with the first year, a study visit program for the capacity building of TTO stakeholders will be prepared this year, which will be a great opportunity for the participants to get some insights from the lectures by Korean experts and study visit to relative institutions. On top of that, a dissemination workshop and a local training workshop will be held in order to share the findings and enlarge its impacts by providing more opportunities to learn and disseminate related information. The dissemination workshop in Tunisia aims to share the findings, lessons learned, and insights from this project based on Korean experiences with as many as STI stakeholders in technology transfer, which enlarges the impact. It will also help in understanding the key challenges and the way forward for the technology transfer system in Tunisia.

The 2018 and 2019 K-Innovation Program with Tunisia will surely reinforce the stable partnership between Korea and Tunisia especially in the field of STI and ensure strong networks among key stakeholders and experts in technology transfer, particularly those who are involved in relative policy development and implementation.

# 4. Project Team

The research team of 2019 has not been changed from the year 2018. It consists of two full-time experts such as a project manager (Ms. Eunjoo Kim) and a project coordinator (Ms. Jiyong An) from STEPI. The other two experts from external institutions are Prof. Myungdo Oh and Dr. Hyun Woo Choi.

## 4.1 STEPI and Korean Experts

The research project team is accountable for the diagnosis of the current situation and demand of the technology transfer organizations (TTOs) within the National System of Technology Transfer (NSTT) in Tunisia in partnership with ANPR to provide the strategies and operational needs for TTOs to learn and adopt in operating their offices. To share the relevant Korean experience during the Study Visit Program in Korea and Tunisia and provide constructive recommendations, the research team invited additional experts deemed most appropriate in terms of educational background, experience, and expertise.

[Table 1-1] STEPI's Research Team

| Name | Organization | Position & Area of Expertise | Contact Information |
|---|---|---|---|
| Eunjoo Kim | STEPI | Project Coordinator<br>Head of Global Training Team | ejkim@stepi.re.kr |
| Jiyong An | STEPI | Researcher<br>Project Coordinator | freesoul13@stepi.re.kr |
| Myung-do Oh | University of Seoul | Professor<br>Technology Transfer System Expert<br>(Legal & Institutional Framework) | mdoh@uos.ac.kr |
| HyunWoo Park | KISTI | Senior Research Fellow<br>Technology Transfer System Expert<br>(Technology Transfer Policy) | hpark@kisti.re.kr |

## 4.2 Tunisian Team (MHESR and ANPR)

The Tunisian research team is accountable for setting the National System of Technology Transfer Program in Tunisia. As a counterpart, the MHESR and ANPR worked in cooperation with STEPI and the Korean team. The major beneficiaries of the capacity-building program for the second year are MHESR, ANPR, and universities and research centers in ANPR's TTO networks along with other key STI stakeholders.

[Table 1-2] Tunisian Team

| Organization | Name | Email | Contact Information | Position |
|---|---|---|---|---|
| MHESR | Khalil Amiri | khalil.amiri @tunisia.gov.tn | 216-71-84-22-05 | Secretary of State for Scientific Research |
| | Malek Kochlef | malek.kochlef@gmail.com | | Director General of International Cooperation (MHESR) |
| | Slim Choura | slim.choura@mes.rnu.tn, s_choura@yahoo.com | 216-71-847-772 | Director General |
| | Maha Hammami | hammamimaha2@yahoo.fr | 216-97-071-379 | Director of Bilateral Cooperation, MHESR |
| | Anis Rouissi | anis.rouissi@mes.rnu.tn | 216-71-84-1380 | Deputy Director (GDIC, MHESR) |
| | Samia Charfi Kaddour | s.charfikaddour@gmail.com | | Director General of Research Vitalization |
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| | Najla Bouden Romdhane | najla.romdhane@mes.rnu.tn / najlaromdhane@hotmail.fr | 216-97-193-407/ 216-57-478-177 | Director General of the project to modernize higher education in support of the employability of young graduates |
| | Fakher Jamoussi | fjamoussi@yahoo.com | 216-71-833-378 | Professor, Director of Programs and Structures of Research Vitalization |
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| | Souad Boussaid | souad.boussaid@gmail.com | | Engineer |

# 5. Main Activities

## 5.1 Activity 1: Preparation of the project

From January to March 2019, the STEPI team focused on setting up the second-year project objectives, targets, and main activities on the National Technology Transfer System of Tunisia. STEPI and MHESR had conducted Skype meetings online to discuss selecting and planning the major activities and its targets prior to Activity 2. There had been changes of the director in the counterpart from the previous project. Moreover, concept notes of the new project ideas developed in the 2018 project were developed and reviewed by the two teams online.

## 5.2 Activity 2: Kick-off workshop and field study

The STEPI team visited Tunisia on April 7 (Sun.) – 13 (Sat.) 2019 for the kick-off meeting and field study.

The research team provided an introduction of STEPI. The ANPR and MHESR counterpart provided a presentation on an i) overview of intellectual Property and Technology Transfer in Tunisia. Afterward, there were some discussions on the expectations for the project in order to identify areas of interest by the ANPR and TTOs so as to build consensus between STEPI, ANPR, and MHESR on the expected results and outcomes. It was strongly recommended that the participating TTOs submit a checklist of the survey provided through ANPR to STEPI to set up the objectives and project scope for the second-year project.

STEPI visited key STI stakeholders, especially the partner TTOs and related institutions, to discuss and diagnose the needs and current status of the TTOs as part of the Tunisian National Technology Transfer System. The relevant institutions and key innovation actors, especially the participants to the kick-off meetings including ministries, public organizations, academic members, private sector entities, and research institutes, were identified and invited by ANPR and MHESR. Meeting arrangements were also made by

ANPR and MHESR.

A closed meeting was held for sharing the findings of the kick-off meetings and interview. Moreover, based on the kick-off meeting findings, the project scope, direction, activities, and expected outcomes were concretized through discussion with MHESR and ANPR. The Korean Embassy in Tunisia also discussed with STEPI on preparing this year's project and co-hosting seminars to commemorate the 50$^{th}$ Anniversary of Establishment of Diplomatic Relations between Korea and Tunisia.

## 5.3 Activity 3: Study Visit Program for Tunisian Delegations

STEPI hosted the study visit program (Training of Trainers Program) of the Tunisian delegation and operated this program on June 9-15, 2019. The purpose of this program is to support Technology Transfer and Commercialization in Tunisia, especially in Policy Capacity Building for Technology Transfer Organizations such as Universities and Research Institutes in Tunisia by sharing Korean experiences in the development of strategies and operations for TTOs, Technology Transfer and Vitalization, and legal and institutional frameworks and human resource development policy. This program has been an outstanding opportunity for the Tunisian Trainers to benchmark the Korean experience as a basis for the development of strategies and operations for TTOs to enhance Technology Transfer and vitalization of research for commercialization.

This program will provide Tunisian trainers in technology transfer organizations and policymakers with a great opportunity to benchmark Korean experiences as a reference for policymaking and adoption of strategies and operational management skills aimed at enhancing technology transfer and commercialization in TTOs in Tunisia.

For this program, experts and scholars of the field were invited as lecturers of each session. There were lectures on technology transfer commercialization such as technology transfer, IP management, technology transfer strategy and operation methods, technology transfer valuation and evaluation cases, and exercises on technology transfer evaluation.

The program also included visiting a number of STI institutes in Korea such as the Korea

Institute of Science and Technology (KIST) History Hall, Korea University Industry-Academe Cooperation Group, and ancient major startup-related facilities (Maker's Lab, Innovation Café, etc.). Major institutions such as SK Tium were visited, and satisfaction surveys were conducted through lecture evaluations and program evaluation surveys. We got satisfaction scores of 4.5 or higher out of 5 on average.

## 5.4 Activity 4: Dissemination Workshop, High-Level STI Policy Dialogue, and Technology Valuation Workshop

To disseminate the project results of 2019 and to discuss future collaboration, STEPI hosted three events for the 15$^{th}$~17$^{th}$ in the third week of July 2019. This year is the 50$^{th}$ anniversary of establishment of diplomatic relations between Korea and Tunisia, and the Korean Embassy in Tunisia highlighted this week as a special celebration week of Science, Technology, and Innovation week and supported the workshops.

The first event held on July 15 was the high-level STI Policy Dialogue, a special event hosted by STEPI and MHESR, which shared the Korean experience in economy and technology transfer system. It was a meaningful time, sharing knowledge and experience and discussing the implications for Tunisia. On the same day, the workshop on Technology Evaluation – Technology Valuation: Policy and Practice was co-hosted by STEPI and ANPR for trainers of TTOs to learn the technology valuation system and enhancement of the technology transfer system in Tunisia. On July 17, the Dissemination Workshop on the Improvement of National Technology Transfer System (NTTS) was co-hosted by STEPI, ANPR, MHESR, and Korean Embassy in Tunisia. It provided an opportunity to share and disseminate the project achievements for the second-year project and future collaboration as a follow-up.

Moreover, STEPI delegation visited the MHESR, ANPR, KOICA, Korean Embassy, University of Tunis El Manar, National Engineering School of Tunis (ENIT), and Borg Cerdric Bio Center.

## 5.5 Activity 5: Develop PCPs for future collaborations

The K-Innovation Project with Tunisia is a two-year project, and there were a number collaboration demand and project ideas discovered during the last two years thanks to the vigorous support and keen interest of the counterparts of Tunisian partners. The team developed and submitted several project concept papers through KOICA.

## 5.6 Activity 6: Final report

STEPI submitted the final report for its evaluation in Republic of Korea by November. As it was the final year of the two years project, the final report included action plans drafted by the Tunisian side which includes recommendations from Korean experts and lessons learned through the program.

# 6. Project Schedule

The entire schedule of the project is as follows:

[Table 1-3] Project Schedule

| Activities | Jan. 2019 | Feb. 2019 | Mar. 2019 | Apr. 2019 | May 2019 | Jun. 2019 | Jul. 2019 | Aug. 2019 | Sep. 2019 | Oct. 2019 | Nov. 2019 | Dec. 2019 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Review of the proposal and Skype meetings | | | | | | | | | | | | |
| Following up on a new project | | | | | | | | | | | | |
| Kick-off meeting | | | | | | | | | | | | |
| Review of TTS report and follow-up | | | | | | | | | | | | |
| Study visit program | | | | | | | | | | | | |
| Mid-term report | | | | | | | | | | | | |
| Dissemination workshop/high-level STI policy dialogue | | | | | | | | | | | | |
| Follow-up of new projects | | | | | | | | | | | | |
| Submission of final report | | | | | | | | | | | | |

## 7. Project Outputs

STEPI provided the following deliverables:

- **Deliverable 1:** Revised Project Proposal (January 2019)
- **Deliverable 2:** Presentation materials and report of the Kick-off Meeting (April 2019)
- **Deliverable 3:** Presentations and training materials for the Study Visit for the Tunisian Delegation (June 2019)
- **Deliverable 4:** Presentations and training materials for the Dissemination Workshop, High-Level STI Policy Dialogue, and Technology Valuation Workshop (July 2019)
- **Deliverable 5:** Comments on the Technology Transfer Organization Report (September 2019) – Draft to be submitted by ANPR, reviewed by Korean experts
- **Deliverable 6:** Final report submission (November 2019) –report on the diagnosis of the National Technology Transfer System and recommendations to be submitted. Diagnosis and lessons learned to be aligned with the program.
- **Deliverable 7:** Submission of PCPs for the new project - New Demands on the STI Policy had been identified. STEPI reviewed and supported the inputs for PCPs from the Korean side for MHESR. Two~three PCPs were submitted to the KOICA Office in Tunisia by MHESR.

# CHAPTER 2

## Status of Technology Transfer Offices in Tunisia

Policy Consultation on the Support for the Establishment of an Effective National System of Technology Transfer (NSTT) in Tunisia

# Chapter 2. Status of Technology Transfer Offices in Tunisia

## 1. Status of Technology Transfer Offices in Tunisia

### 1.1 Tunisian Economic Context

Since the 1970s, Tunisia had opted for an economic model geared toward export and industrialization, backed by a proactive policy of public investment in physical and human capital and attraction of foreign direct investment through laws (72/73) favoring wholly exporting companies.[6] Mainly dominated in the early 1960s by the agricultural sector, the structure of Tunisia's economy has changed significantly in favor of industry and services.

The industrial fabric is composed of 5,401 companies with workforce of 10 or more employees. The distribution of enterprises by sector is as follows: agro-food 20.4%, building materials 7.7%, chemical industry 10.6%, electrical, mechanical, and electronic industries (MEI) 18.4%, textile and clothing 29.4%, and other industries 13.5%.[7]

Taking advantage of its geographical and cultural proximity, Tunisia has gradually strengthened its relations with the European Union (EU), its main industrial partner and primary client. The Association Agreement signed in 1995 established a free trade zone between the two parties effective January 1, 2008 for industrial products. The launch of the national program for upgrading the industry in the late 1990s has

6 World Bank, World Development Indicators (2016), http://data.worldbank.org/country/tunisia

7 http://www.tunisieindustrie.nat.tn/fr/tissu.asp

improved the competitiveness of Tunisian companies for better integration into global value chains (GVCs). In this context, major international contractors have established subsidiaries in the country and/or developed subcontracting agreements, leading Tunisia to participate more and more in the global economy. In 2017, there were a total of 2,361 wholly exporting industrial companies, becoming the source of 509,818 jobs.[8] In this respect, two sectors are particularly significant: textiles and clothing since the 1970s and, more recently, MEIs. While textiles are suffering a certain decline linked to international particularly Asian/Chinese competition, the MEI sector has undergone a significant evolution over the last fifteen years with the development of automotive and aerospace component activities.

The sector's exports grew up to 18% on average per year between 2000 and 2016. Since the early 2000s, the development of information and communication technologies (ICT) has led to the emergence of new service activities and increased integration of Tunisia into GVCs[9]. Call centers have thus been developed, as well as -- to a lesser extent -- other outsourcing services (outsourcing of accounting services, for example).

Such gradual integration into GVCs has fostered growth in Tunisia, contributing to the creation of many jobs and exports. In 2017, the textile sector accounted for more than 21% of industrial exports, and the EMI sector, for more than 51%.[10] Nonetheless, this development model is running out of steam, and its impact on the Tunisian economy seems limited today. Indeed, the jobs created concern activities with low value added and consequently unskilled personnel. At the same time, the establishment of majority of export-oriented companies near the export logistics zones (ports and airports) has accentuated regional disparities.

The low ratio of management to staff has not favored technology transfer and recovery of value chains, thereby limiting the development of activities. Thus,

[8] http://www.tunisieindustrie.nat.tn/fr/tissu.asp

[9] http://www.tunisieindustrie.nat.tn/fr/conjoncture.asp

[10] http://www.tunisieindustrie.nat.tn/fr/conjoncture.asp

imported inputs constitute a significant portion of Tunisian exports, although this varies according to the products and mainly applies to intermediate products. According to a study by the AfDB (2012), the level of sophistication of Tunisian exports has been declining for several years. Finally, the constraints of the 1972 Export Companies Act severely limited their impact on the rest of the economy as the local market hardly considers a potential customer or supplier.[11]

The socio-economic difficulties of recent years have slowed Tunisia's integration into the world economy, but many opportunities exist in the medium term. In 2008, the Ministry of Industry, Energy, and SMEs published a document titled the "National Industrial Strategy for 2016."[12] Based on an analysis of international trends and good practices of countries having successfully The agency itself does integrated the GVCs, this strategy aimed to transform Tunisia into an innovative Euro-Mediterranean center with a high level of competitiveness. In 2010, Tunisia was also preparing a strategy to promote services and outsourcing. With the revolution of January 2011 and the given socio-economic emergencies, these projects are dormant.

Nonetheless, different sectors offer significant development potential for Tunisia, such as on-site processing of products usually exported as raw materials (hydrocarbons or agricultural products) or creation of high value-added niche products from traditional sectors (technical textile, for example). To encourage this development, the Tunisian authorities have to make considerable efforts to, among other things, improve the business climate, strengthen industrial capacities and logistics infrastructure, and reform the education sector. The fight against corruption at all levels will also be crucial. Tunisia has a multitude of support structures for businesses and public and private institutions involved in the creation and development process (technical centers, business centers, chambers of commerce, nurseries, etc.), but their effectiveness remains limited.[13]

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[11] World Bank, World Development Indicators (2016), http://data.worldbank.org/country/tunisia

[12] http://www.tunisieindustrie.nat.tn/fr/doc.asp?docid=1754&mcat=13&mrub=140

[13] World Bank, World Development Indicators (2016), http://data.worldbank.org/country/tunisia

Ongoing negotiations with the European Union on the free trade in agricultural products and services, among others, offer opportunities provided they are accompanied by the necessary upgrades to align production with international standards and subject to the actual mobility of workers. The diversification of partnerships and the development of new markets, particularly through the strengthening of regional integration, are other assets, even for trade with the EU. Finally, the integration of the country into GVCs will only be fully successful if Tunisian local businesses participate fully in the productive system. A reorganization of the sectors should put an end to the duality of the Tunisian economy, involve the whole economic fabric, and allow more inclusive growth. Tunisia's heavy dependence on Europe, which still absorbs more than 70% of its exports, and the inequalities in territorial development linked to the physical constraints of import-export remain the two vulnerability factors to be improved by appropriate policies.[14]

Currently, we expect crucial changes to back up economic policies, targeting to widen structural transformation and reviewing the economic regulations and governance. Strengthening the regulatory environment and governance requires getting rid of obsolete rules and complex bureaucratic procedures, declining controls on business, and supporting market competition.[15]

## 1.2 Technology Transfer Concept

Technology transfer involves transmitting to an industrial partner a technology (product, technique, process). Such transfer can occur in many ways. It can occur for free or via the transfer of ownership of a patent, licensing for commercial use of the technology, or creation of a startup or another joint venture to develop further and commercialize the technological innovation. In any case, technology transfer is the diffusion of technological

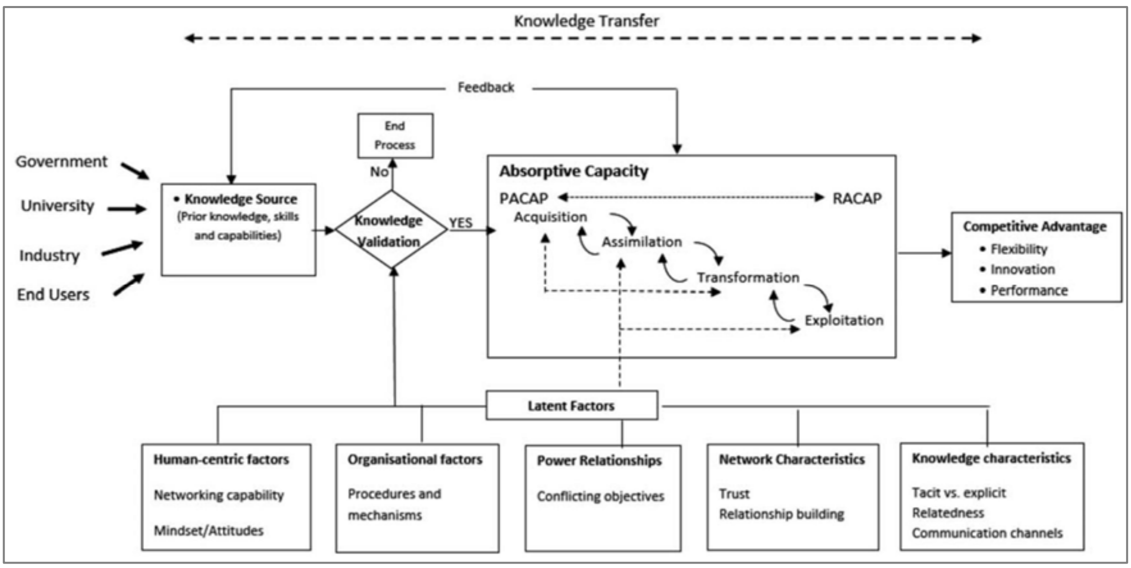
[14] World Bank, World Development Indicators (2016), http://data.worldbank.org/country/tunisia

[15] Kim and Song, K-Innovation 2018

innovation from universities and research centers and laboratories to the industry and the wider socio-economic environment and is the key to improving the competitiveness of enterprises, creating jobs and wealth for society.

Technology transfer needs a public and private environment like Government to give finance, incentives, procedures, instruments for vitalization, protection, and contracts, and for universities to have scientific results for commercialization and for the industry to collaborate with researchers and end users to respond to their needs. To do all this, technology transfer needs power and trust relationships, objectives, mechanisms, and networking capability. This is why the interfacing structure is a good tool for facilitating collaboration, the communication between the two sides. ***(Figure 2-1)***

[Figure 2-1] Conceptual Framework for Knowledge Transfer




Therefore, first, we need to present our knowledge generators in Tunisia as the key actors for technology transfer as well as our institutional framework, who have an important role to support this concept, and our priority axes in research, too.

## 1.3 Knowledge Generators:

Knowledge generators are the scientific results from the research center, universities, technical center, enterprises etc.

[Table 2-1] Knowledge Generators

| | |
|---|---|
| **Universities** | 13 |
| **Laboratories** | 330 |
| **Units** | 300 |
| **Doctoral schools** | 37 |
| **Research center** | 40 |
| **Clusters** | 30 |
| **Technopoles** | 11 |
| **Technical center** | 15 |

## 1.4 Institutional Framework:

The Tunisian national innovation system is composed of stakeholders (ministries, key interfacing structures), key actors, and enablers starting from policy formulation, governance, and then policy implementation to key initiative programs supporting innovation projects.

| Ministries | Key Interfacing Agencies |
|---|---|
| MESRS | ANPR |
| MI and SME | APII |
| MC | CEPEX |
| MTEN | ANCE/CNI |
| MS | |
| MEDD | ANPE/CITET |
| MFPE | ANETI |
| MAP | APIA |
| MT | DGTI |
| MDCI | Offices de développement |

## 1.5 Financial Support:

Each structure indicated in Table N°2 has developed several mechanisms to finance and support vitalization and technology transfer essentially for spin-off, start-up or collaboration between university and industries. The financial support is national or international. Table N°3 present the mechanisms developed by MESRS and ANPR.

| **Ministry** | **MESRS** | **General Directions** | **Enablers** | **Key Actors** |
|---|---|---|---|---|
| Committee | | | National Committee for the Evaluation of Scientific Research (CNEAR) | |
| Support findings | H2020 | H2020 *European Program Management Unity (UGPE)* | European Union | Universities and Laboratories |
| VRR | | DGVR | | Research Center |
| PAQ collabo/PAQ Post PFE | | World Bank | | Technopoles |
| PRF | | | DGRS *General Direction of Scientific Research* | |
| Mobidoc | | ANPR | | European Union |
| PAQ Spin-off | | ANPR | | World Bank |
| Key Interface Structure | | ANPR | | TTO |

## 1.6 Priority Axes for Research in Tunisia:

The criteria adopted in the definition for research priorities in Tunisia were organized into four main axes:

- Actual or potential added value (contribution to sustainable development, positive discrimination, quality of life, employment, exports, scientific excellence...)

- Level of alignment with sectoral strategies, national plans, and international agreements and commitments
- Feasibility (availability of human, natural, financial, and material resources, capacity to implement in the local and / or international context, opportunities for complementarities between various scientific research disciplines)
- Levels of need and urgency (risk for the state or population, terrorism, epidemics, natural disasters, electronic threats...) these criteria enabled distinguishing six major national priorities each subdivided into subpriorities:

**1-Energy, water, and food security**

- Sustainable management of water resources.
- Renewable energy and energy efficiency.
- Preservation of biodiversity and climate change.
- Smart farming and mechanization.
- Fight against epidemics, coastal erosion, and desertification

**2-Societal Project: Education, Culture, and Youth**

- Identity, citizenship, and emerging democratic society
- Education, training, quality assurance, and new pedagogical approaches
- Culture, arts, media, and quality of life
- Youth problems

**3-Citizen Health**

- Drug design
- Development of vaccines and biosimilars
- Governance and health economics
- Epidemics, chronic diseases, and new diseases - E-health and telemedicine

**4-Digital and industrial transition**

- Digital transition

- "Smart cities" and "Internet of Things"
- Security of networks and information systems
- Protection and surveillance of borders and infrastructures
- Nanotechnology and intelligent materials 17

**5-Governance and decentralization**

- Political and economic decentralization
- Local governance and participatory democracy
- Models of development, spatial planning, and quality of life
- Vitalization of the heritage and history of the regions
- Public and private governance

**6-Circular Economy**

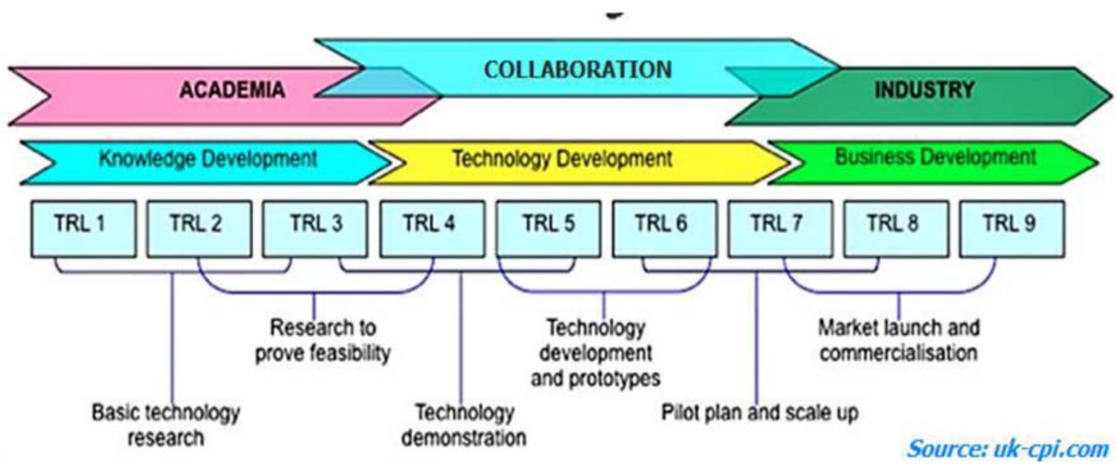
- An agriculture and an industry that respect the environment
- Exploitation of mining resources and useful substances (rare earths)
- Pollution control and its effects
- Treatment and recovery of industrial and household waste

## 1.7 Overview of Tunisia's Innovation and Entrepreneurship Ranking

In this part, we will present the innovation and entrepreneurship ranking of Tunisia for 2018. Actually, the innovation chain includes many steps from generation of the idea, screening and evaluation, experimentation, protection, and commercialization. First, we will present some definitions about innovation and technology readiness level, and then give an idea about the Global Innovation Index (GII) and Global Entrepreneurship Index (GEI) for 2018.

The Definition of innovation is a process of translating an idea or an invention into a good or a service that creates value or for which customers will pay[16]. To be called an innovation, an idea must be replicable at an economical cost, and it must satisfy a specific need. Innovation involves deliberate application of information, imagination, and initiative in deriving greater or different values from resources and includes all processes by which new ideas are generated and converted into useful products (OECD, 2014). In business, innovation often results from the company's application of ideas for further satisfying the needs and expectations of the customers. There are nine technology readiness levels, with TRL 1 as the lowest and TRL 9 as the highest (OECD, 2014)

[Figure 2-2] Innovation Chain: Converting Science into Wealth




Innovation is the translation of the development technology into business technology as mentioned in Figure 2-2.

## 1.8 Global Innovation Index for 2018: GII 2018

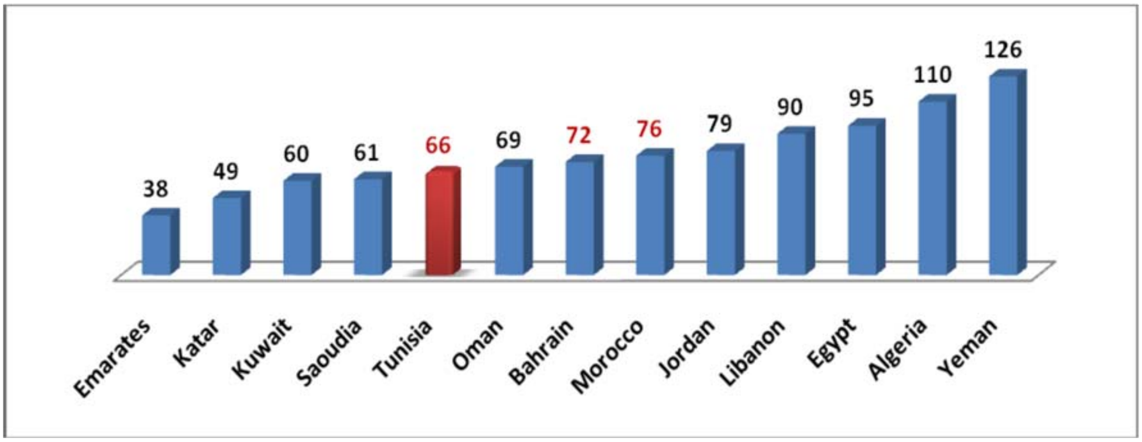
GII 2018 provides metrics about the innovation performance of 126 countries and economies around the world.[17] Its 80 indicators explore a wide vision of innovation, including political environment, education, and infrastructure and business sophistication.

[16] OECD Reviews of Innovation Policy Industry and Technology Policies in Korea, 2014.

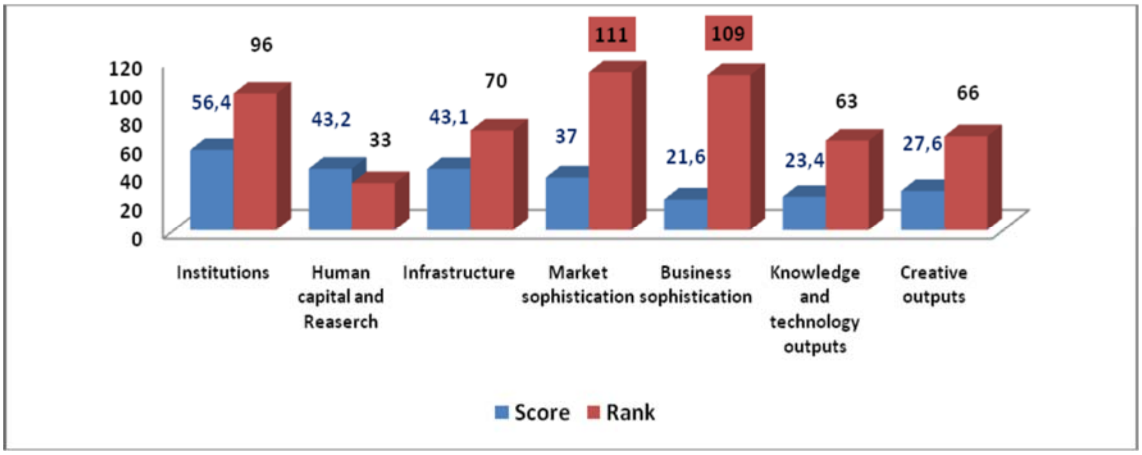
[17] GII, 2018

The 2018 GII, which analyzes detailed indicators for 126 countries, ranks Tunisia 66th in the world, moving up 8 places from its previous ranking of 74th. This is the highest ranking in the African region.

[Figure 2-3] Tunisia's GII 2018 Rank




[Figure 2-4] Tunisia's GII Subpillar Rank 2018

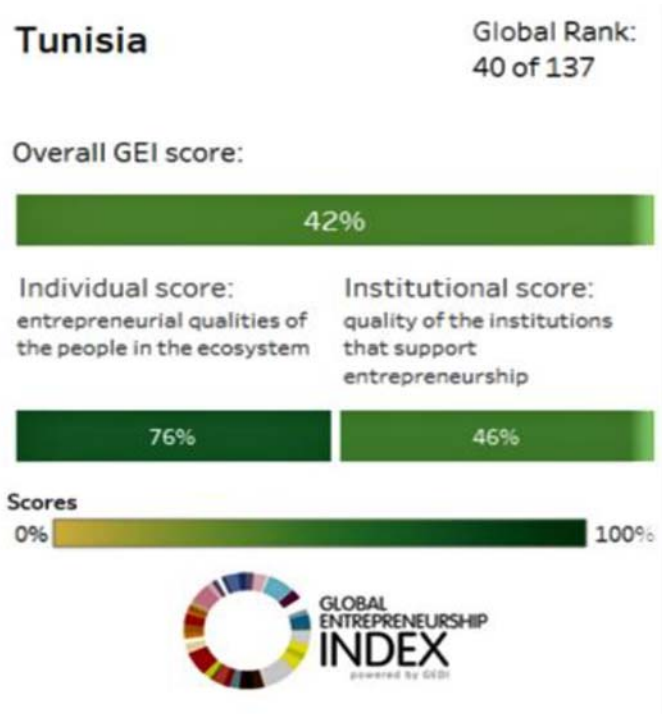


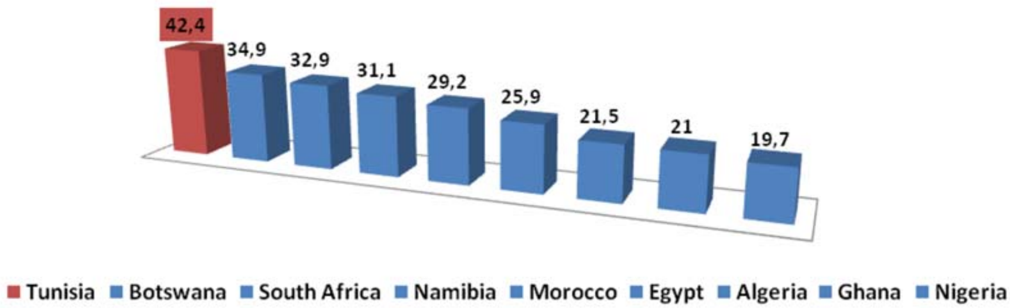

This puts the country in the 33rd spot in the field of human capital and research. In fact, Tunisia belongs to the class of innovative country. Other indicators remain weak and need to be improved, however. As for the sophistication of the market and the sophistication of business, Tunisia ranks 111th and 109th, respectively, out of 126. Its political environment, which is ranked 96th, is regarded as a weak point for Tunisia.

## 1.9 Global Entrepreneurship Index for 2018:

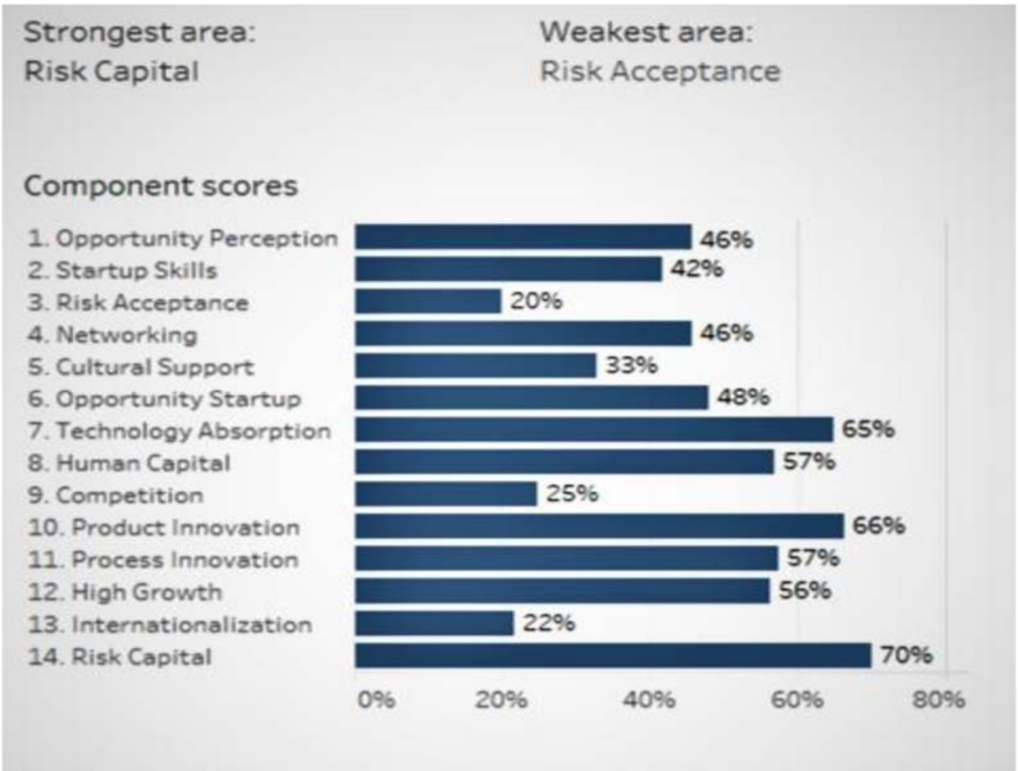
The GEI is a composite indicator of the health of the entrepreneurship ecosystem in a given country (GEI, 2018). The GEI measures both the quality of entrepreneurship and the extent and depth of the supporting entrepreneurial ecosystem (GEI, 2018). We have identified the 14 components that we believe are important for the health of entrepreneurial ecosystems, identified data to capture each, and used this data to calculate three levels of scores for a given country: overall GEI score; scores for Individuals and Institutions; and pillar level (GEI, 2018). The global rank of Tunisia's GEI for 2018 is 40th out of 137 countries as indicated in Figure 2-5.

[Figure 2-5] Global Entrepreneurship Index



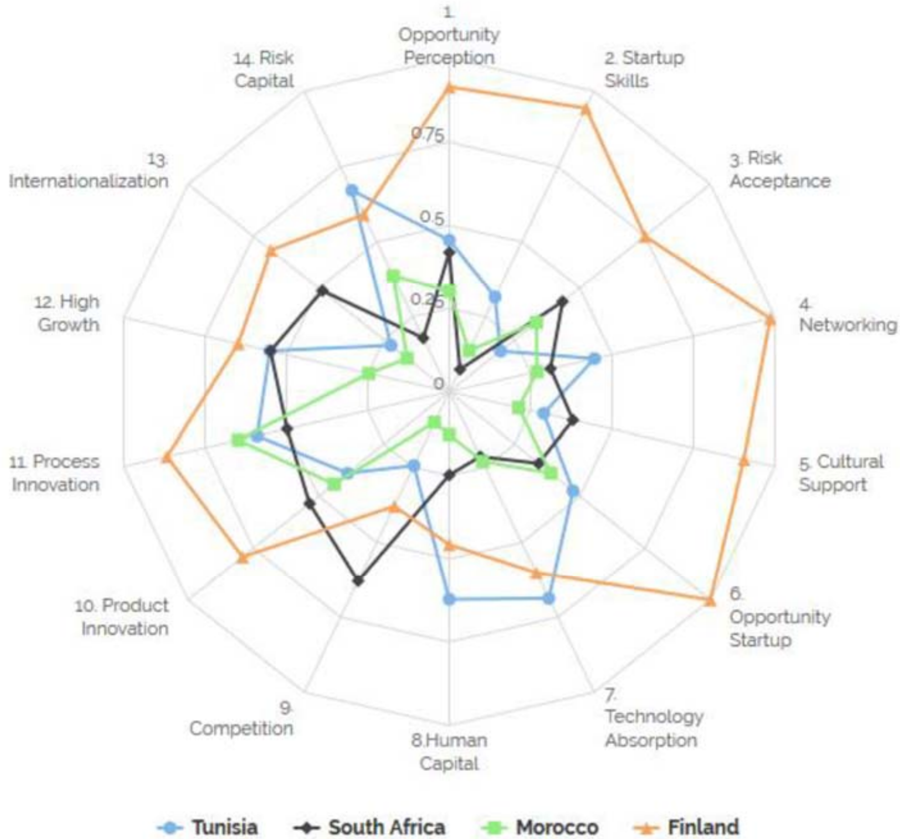



On January 24, a Bloomberg economic magazine study ranked Tunisia among the top 50 most economically innovative countries. Ranked 43rd in the world, Tunisia garnered its best score in the efficiency of the tertiary sector. If we analyze the components of GEI, we find that the weak area as indicated in the figure below is risk acceptance.




Based on this GEI benchmark as in the figure below, we conclude that we have good human capital and process innovation, but the risk acceptance is very weak compared to South Africa and Finland.

[Figure 2-6] GEI Benchmark



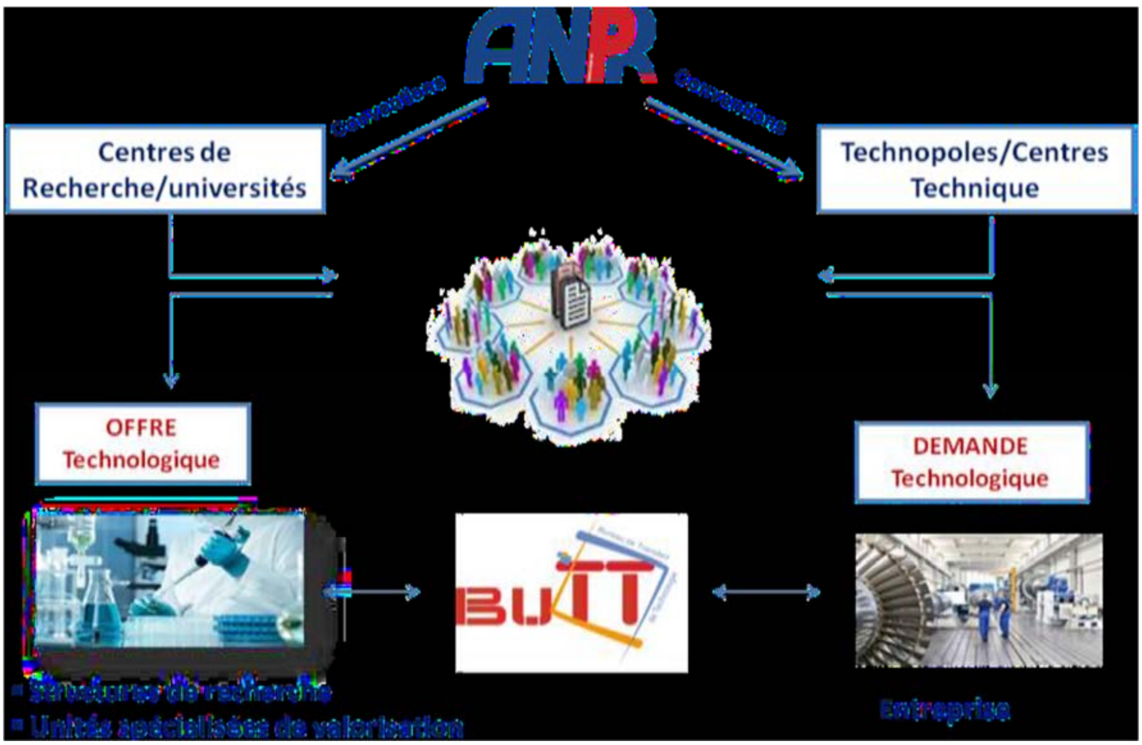

Source: Global Entrepreneurship and Development Institute

Evolution of filing of patents at INNORPI by residents and non-residents

Figure N°9: Evolution of filing of patents

## 1.10 Implementation of TTO: Experience of ANPR

The collaboration between the world of research and the world of industry requires a national and regional coordination structure facilitating systematic exchange between supply and demand. ANPR has established a new technology transfer strategy consisting of making a center of expertise and calling on universities and research centers to become part of this network through his staff. ANPR has established in 2012 this network of TTOs in universities and research centers through specific conventions (Figure N°). Today, there are 25 TTOs in this network which is a bold initiative that the TTO has technical duties in the transfer of technology (Kim et al., 2018). These missions require multiple expertise that are not usually available to one person, since the Agency itself does not have enough resources to ensure assistance to all partners. One of the strong point of this initiative is that it demonstrated the willingness of universities and Public Research Organizations (PROs) to pay attention to technology transfer (Kim et al., 2018). In this respect, TTOs played a role in raising awareness of the significance of technology transfer at universities (Kim et al., 2018).



### 2-Definition:

TTO is a "pole of expertise" serving the vitalization of the results of research, Transfer, and partnership between supply and technological demand. It plays the role of a proximity interface between the research structures and the socio-economic environment.

### 3-Objectives:

The objectives of ANPR are essentially to support the creation of Technology Transfer Offices as well as provide financial support, assist the research structures in the fields of technology transfer through training at the national and international levels, and support them in the resolution of complex recovery cases to contribute to the exploitation of research results.

### 3-Missions:

Each regional TTO should have their own strategy for the region. They should establish their own regional strategy with all stakeholders. They should know the size of the market; they must motivate and attract demand with good visibility of the researchers (experts) and original products. They must know the entrepreneurial cultural environment. They must know the entrepreneurial infrastructure and set up cooperation networks as a network of knowledge. They must know the qualified and experienced researchers, an environment that is friendly to entrepreneurship and researchers, with an institutional culture of openness. They should have the information network for the dissemination of good technology transfer practices and international cooperation projects through which they can promote researcher exchanges.

### 4- Activities:

In fact, the general activities of TTO are presented in the table below.

| | |
|---|---|
| **Inform-Sensibilize** | Raising awareness among researchers about calls for projects, intellectual property, and vitalization of research results. |
| **Meetings** | Research-business events organized at the inter-university level on targeted themes. |
| **Detect-Value** | Project monitoring and detection of valuable research results; identification of business innovation needs. |
| **Collaborate** | Accompany researchers and socio-economic actors in their innovation projects and facilitate inter-university projects |

### 5-Implementation of TTOs:

Based on the figure below, ANPR has adopted many strategies to implement this vision in universities and research center from 2011 until now.


### 2011: Implementation of TTO project with the collaboration of INNORPI through the PASRI project and WIPO project

The project begins with the signing of the contract between ANPR and INNORPI through two projects, which have the same vision: PASRI and WIPO. ANPR has set up a steering committee to define the missions and profile of TTOs. It recruited four engineers and held a convention with universities to put this TTO for the service of university. INNORPI designs its own TTOs in the technopoles and technical center. ANPR has implemented a training cycle for these TTOs to develop their competencies in collaboration with the PASRI project. As the weaknesses associated with this implementation, however, not all TTOs have worked with the same motivation. Resources, budget, procedures, guides, contracts, etc., are also lacking.

- **2014: Implementation of a new strategy of TTOs by ANPR**

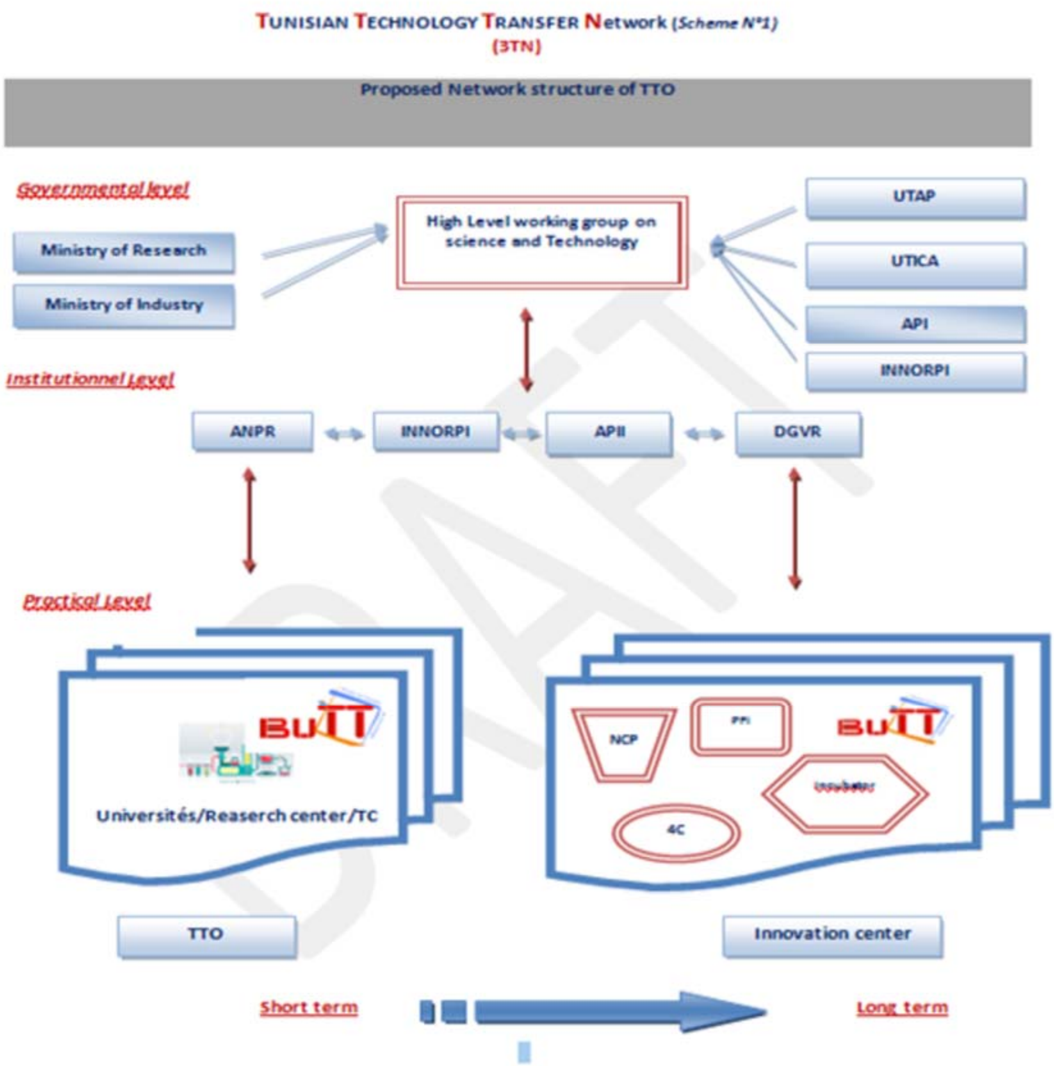
The designation of TTOs involves interested universities from the staff of universities and negotiation of a budget with ministry to support these structures. ANPR launched the new network in 2015 with 17 TTOs and provided them with training in the national and international levels with CLDP as partner and PASRI. The implementation of this new strategy for the creation of technology transfer offices had a budget of 100,000 TD. The organization of a meeting with TTOs requires information, discussion, and debate on the work plan and establishment of the ANPR financing committee. The ANPR supports TTOs in the implementation of a budgeted action plan for TT for the year 2015-2016 based on 5 fundamental axes for the transfer of technology.

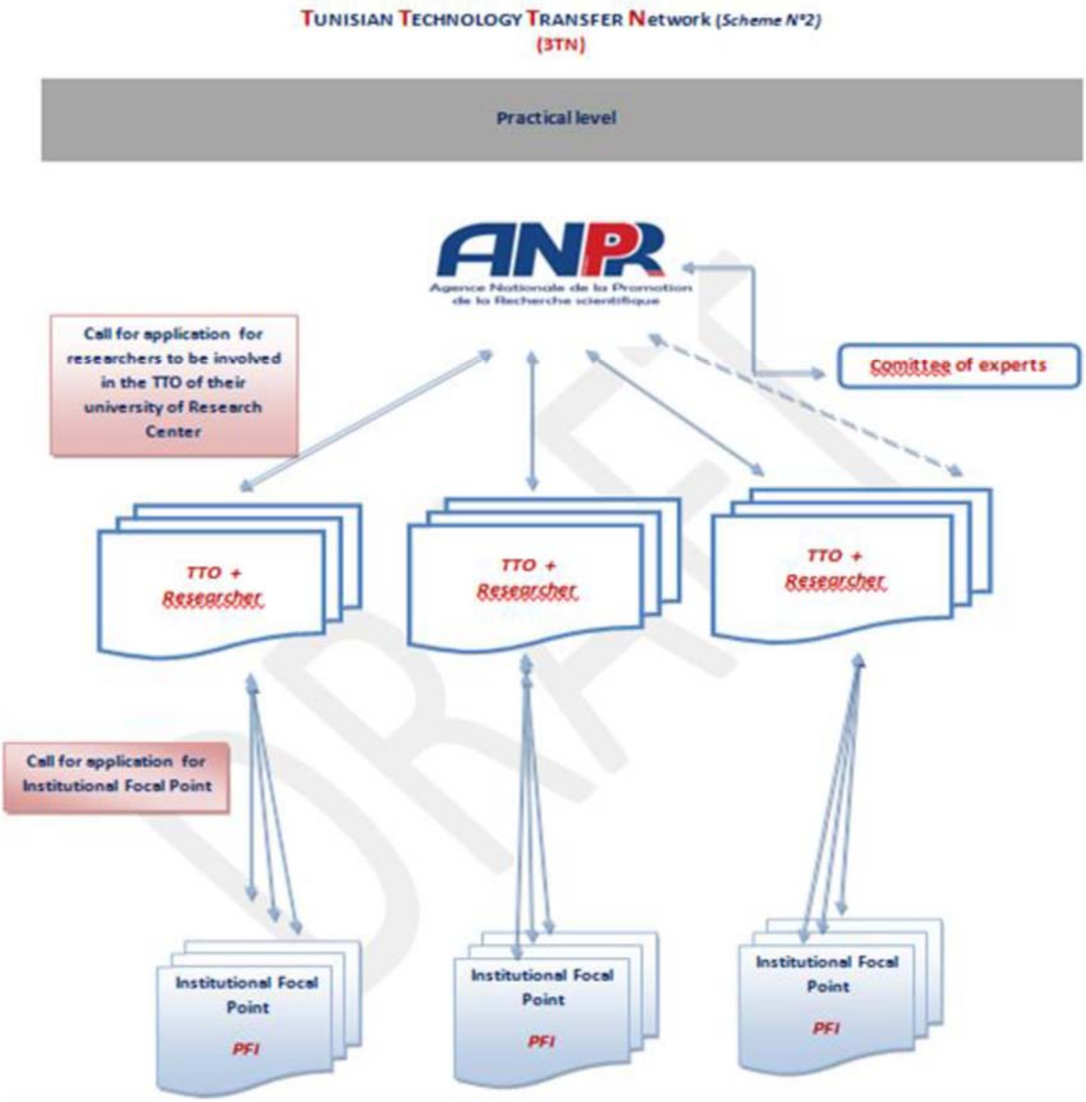
- **2015: ANPR becomes a partner in the ESCWA project to set up a National Transfer Technology Office: NTTO**

The ESCWA Technology Center, in collaboration with ANPR aims to strengthen the capacity of the SNI National System of Innovation through these Technology Transfer Offices linked with universities and research institutes to facilitate the research partnership and with the industry for socio-economic development. These NTTOs were implemented in the MENA region: Tunisia, Egypt, Morocco, Jordan, and Oman. The main activities of this project are: 1) Presentation of the National Innovation System; 2) Establishment of Policy and Strategy for the transfer of technology; 3) Proposal of legislation and policies for the implementation of the roadmap; and 4) Workshops with interested universities and organization of regional forum for networking and exchange of good practices.

- **2017: ANPR holds a large number of conventions with universities and research center**

ANPR establishes a large number of TTOs in universities and research center.




TUNISIAN TECHNOLOGY TRANSFER Network (Scheme N°2)
(3TN)
Practical level
ANPR
Agence Nationale de la Promotion de la Recherche scientifique
Call for application for researchers to be involved in the TTO of their university of Research Center
Comittee of experts
TTO + Researcher
TTO + Researcher
TTO + Researcher
Call for application for Institutional Focal Point
Institutional Focal Point
PFI
Institutional Focal Point
PFI
Institutional Focal Point
PFI

## 1.11 Management of TTO by ANPR

ANPR has set up a Technology Transfer team that includes the following: Chedly Abdelly, the general manager of ANPR whose mission is to approve all TTOs activities; Souad BOUSSAID, the Technology Transfer Offices coordinator responsible for the implementation of the project; a TT and IP Training Manager who does the monitoring and coordination of the project with all ANPR and TTO referents; Karim AZZABI, a jurist of ANPR who drafts contracts; a trainer in contractualization and a member of the Financial Committee; Adel HNID, general secretary and member of the Financial Committee; Moez Safta for Financial; and Mabrouka GUENICHI for the payment of invoices.

[Table 2-2] List of Beneficiaries

| Universities | | | | | | |
|---|---|---|---|---|---|---|
| **Acronyme** | **Président** | | **Signature de la convention** | **Fin du contrat** | **Référent** | **Plan d'action** |
| 1 | UT | Habib Sidhom | Février 2018 | Février 2020 | Saoussen Krichen | 2018 |
| 2 | UTM | Fethi SELLAOUTI | Février 2018 | Février 2020 | Imen Bhibah | 2018 |
| 3 | UMA | Chokri El MABKHOUT | Janvier 2017 | Janvier 2019 | Nadia Amri | 2018 |
| 4 | UCA | Olfa Ben Ouda Sioud | Fevrier 2018 | Fevrier 2020 | Aziza Fertani | 2018 |
| 5 | UKA | Hammadi Messaoudi | Février 2018 | Février 2020 | Marwa Hattab | NA |
| 6 | UJE | Mokhtar Mouhwachi | Février 2018 | Février 2020 | Imen Taghouti | 2018 |
| 7 | UM | | Hedi Bel Hadj Salah | Mai 2016 | Mai 2018 | 2018 |
| 8 | US | Ali Mtiraoui | Mars 2018 | Mars 2020 | Mortadha Graa | 2018 |
| 9 | USF | Abdelwahab Mokni | Avril 2018 | Avril 2020 | Mohamed Bel Hadj | NA |
| 10 | UGA | Kamel Abderrahim | Novembre 2018 | Novembre 2020 | Mejda Bourguiba | 2019 |
| 11 | UGF | | Rached Ben Youness | Mars 2018 | Mars 2019 | 2018 |

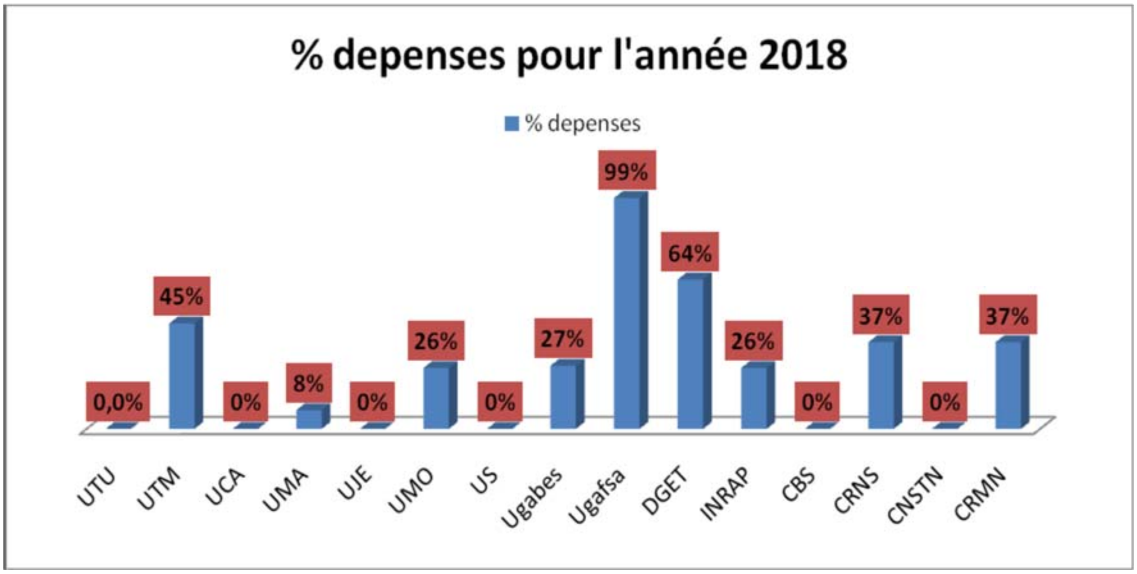
| Research Center and Institutions | | | | | | | |
|---|---|---|---|---|---|---|---|
| Acronyme | Directeur Général | | Date de signature de l’avenant | Fin du contrat | Référent | Plan d’action |
| 12 | INRAP | Mohamed HAMMAMI | Mars 2018 | Mars 2020 | Leila Triki | 2018 |
| 13 | DGET | Ali GHARSALLAH | Juin 2018 | Juin 2020 | Ismail Haddad | 2018 |
| 14 | IPT | Hachmi LOUZIR | Septembre 2016 | Septembre 2018 | Oussema Ben Fadhl | 2017 |
| 15 | CRTEn | Ibrahim BSSAEISS | Juin 2016 | Juin 2018 | Zied Kbaeir | 2017 |
| 16 | IRESA | Elyes HAMZA | Novembre 2018 | Novembre 2020 | Jamal Ben Rabah | 2019 |
| 17 | CITET | Salah Hassini | Novembre 2018 | Novembre 2020 | NA | 2019 |
| 18 | CBS | Sami Sayadi | Fev 2018 | Fev 2020 | Hekma ayadi | 2018 |
| 19 | CRMN | Kamel Besbes | Mars 2018 | Mars 2020 | Samia Dhahri | 2018 |
| 20 | CRNS | Mohamed Jmail | Janvier 2018 | Janvier 2020 | Noomene Hchicha | 2018 |
| 21 | CERT | Naoufel Ben Said | Mars 2018 | Mars 2020 | Hend Ben Hajji | NA |
| 22 | CRNSM | Mokhtar Ferid | Mars 2018 | Mars 2020 | Anis chaouchi | NA |
| 23 | CERTE | Ahmed Gharbi | Fevrier 2018 | Fev 2020 | NA | NA |
| 24 | CNSTN | Mokhtar Hamdi | Fevrier2018 | Fev 2020 | Zohra Azzouz | 2018 |
| 25 | CBBC | | Pas de contrat | NA | NA | NA |

[Table 2-3] List of TTO Managers

| N° | Prénom | Nom | Institution | Fonction |
|---|---|---|---|---|
| 1 | Mortadha | GRAA | Université de Sousse | Dr-Ingénieur |
| 2 | Nedia | Amri | Université de Manouba | Directrice des affaires académiques et de partenariat scientifique |
| 3 | Saoussen | Krichen | Université Tunis | Vice présidente |
| 4 | Imen | Bhibeh | Université Tunis Manar | Admin/Conseiller de l'enseignement supérieur et de la recherche scientifique |
| 5 | Zied | Ben Romdhane | Université Monastir | Ingénieur principal |
| 6 | Marwa | Hattab | Université Kairouan | Chef service rechreche scientifique et évaluation universitaire |
| 7 | | Imen | Taghouti | Chef service rechreche scientifique et évaluation universitaire |
| 8 | Salah | Allagui | Université Gafsa | Vice Président |
| 9 | Mejda | Bourguiba | Université de Gabes | Sous Directeur Recherche Scientifique et coopération internationale |
| 10 | Mohamed | Belhadj | Université de Sfax | Ingénieur général et sous Directeur |
| 11 | Aziza | Fertani | Université de Carthage | Chef service rechreche scientifique et évaluation universitaire |
| 12 | Ismail | Haddad | DGET | Directeur |
| 13 | Leila | Triki | INRAP | Ing.en chef |
| 14 | Oussema | Ben Fadhl | IPT | Chargée de TT |
| 15 | Zied Kbaeir | Kbaeir | CRTEn | Chef Service |
| 16 | Dhouha | Tangour | CITET | Chef Service |
| 17 | Jamel | Ben REBAH | IRESA | Sous Directeur de la liaison de la recherche et de la vulgarisation |

| N° | Prénom | Nom | Institution | Fonction |
|---|---|---|---|---|
| 18 | Hekma | Ayadi | CBS | Ingénieur |
| 19 | Samia | Dhahri | CRMN | Enseignant chercheur |
| 20 | Nomene | HCHICHA | CRNS | Ingénieur Principal |
| 21 | Zohra | AZZOUZ | CNSTN | Sous Directrice |
| 22 | Hend | Ben hajji | CERT | Docteur |
| 23 | Anis | Chaouachi | CNRSM | Ingenieur en chef |
| 24 | NA | NA | CBBC | NA |
| 25 | NA | NA | CERTE | NA |

[Figure 2-7] Distribution of the Budget and Expenses for 2018



| **2018** | |
|---|---|
| **ANPR/CLDP** | Participation of 4 TTOs from (DGET, Sfax, Sousse, and ANPR) in AUTM (Association for University Technology Managers) in Floride |
| Participation of 08 TTOs from (Tunis, Manouba, Monastir, Gabes, IRESA, ANPR) for a regional workshop in Morocco (Role of TT in the improvement of the economy) | |
| Participation of TTOs in the workshop (3 days) in North Africa and MENA region on the commercialization of technology (60 participants) | |
| **ANPR/IRD** | Participation of TTOs in the National Day of Technology Managers: Participation of Bond'Innov/ Protis-Valor and IRD |
| Participation of TTOs in the Nord Africain Workshop and MENA région for the technology commercialization | |
| **ANPR/KOIKA** | Contribution of ANPR to the implementation of action plan 2019-2021 for the benefit of trainer of trainers |
| **ANPR/EMORI** | Training on the H2020 program for TTOs |
| H2020 meeting day between TTOs and NCP for the implementation of an action plan | |

# 2. Status of Technology Transfer System

## 2.1 Interfacing Structure

### 2.1.1 Academe

- **Public Universities:**

| | | |
|---|---|---|
| University of Carthage | University of Manouba` | University of Tunis – El Manar |
| University of Gabes | University of Monastir | Virtual University of Tunis |
| University of Gafsa | University of Sfax | Zitouna University |
| University of Jendouba | University of Sousse | University of Kairouan |
| University of Tunis | | |
| DGET: Technological studies branch; Network of ISETs<br>(Higher institutes of technological studies)[18] | | |

13 universities with 203 higher education institutions

316 research labs and 327 units of research

Number of students: 250900[18]

- **Private universities: 68 institutions with 31,304 students**

[18] Office of Studies, Planning, and Programming (2016/2017)

- **Research Centers: The 38 research centers and the main ones are[19]:**

| | |
|---|---|
| **CERTE** | Centre de Recherche Et des Technologies des Eaux |
| **CRTEn** | Centre de Recherche et des Technologies de l'Energie |
| **CNRSM** | Centre National des Recherches en Sciences et Matériaux |
| **CBBC** | Centre de Biotechnologie de Borj-Cédria |
| **CNSTN** | Centre National des Sciences Et Technologies Nucléaires |
| **INARP** | Institut National de Recherche et d'analyse Physico-chimique |
| **CBS** | Centre de Biotechnologie de Sfax |
| **CRNS** | Centre de Recherche Numérique de Sfax |
| **IPT** | L'Institut Pasteur de Tunis |
| **CRMN** | Centre de Recherche en Microélectronique et Nanotechnologie de Sousse |
| **CITET** | Centre International des Technologies de l'Environnement de Tunis |
| **CERT** | Centre d'Etudes et de Recherche en Télécommunication[20] |

[19] (STEPI, 2018 K-Innovation , 2018)

## Interfacing structures[20]

| Interfacing structures | |
|---|---|
| *APII* | ***Agence de promotion de l'Industrie et de l'Innovation (Agency for the Promotion of Industry and Innovation at the Ministry of Industry, Energy, and Mines)***: Provides support services to entrepreneurs and enterprises. Its mission is to spread the culture of innovation among businesses by promoting programs of capacity-building and incentives mechanisms. |
| *ANPR* | ***Agence Nationale de Promotion de la Recherche Scientifique (National agency for scientific research promotion at the Ministry of Higher Education and Scientific Research)***: Provides services to professionalize the management of research activities in partnership with effective socio-economic operators. It plays a crucial role in interfacing and supporting the implementation of the R&D vitalization process and the transfer of research results. |
| *BuTT* | ***Bureau de Transfert de Technologies (Office of Technology Transfer):*** A local interfacing structure within public research and higher education institutions and a skills center facilitating the exploitation of research results, technology transfer, and partnership between supply and technology demand. Twenty-five universities, research institutions, research centers, and technology parks have been selected for the pilot phase of the implementation of the first generation of BuTTs, and their role is to set up a structured process of IP management. They also support the private sector through various financial tools and the microfinance and entrepreneurship ecosystem. |
| *BMN* | ***Bureau de Mise à Niveau (Industrial Capacity Upgrade Office at the Ministry of Industry, Energy, and Mines)***: In charge of implementing the industrial capacity upgrade program and evaluating and delivering the certificate of innovation to industrial companies that will apply for it. This certificate is particularly needed to access public funds supporting R&D and innovation. |
| *APIA* | ***Agence de promotion des Investissements Agricoles (Agency for Agricultural Investment Promotion at the Ministry of Agriculture, Hydraulic Resources, and Fisheries)*** |
| **International cooperation** | |
| AFD | ***Agence Française de Développement (French Development Agency)*** |
| GIZ | ***Deutsche Gesellschaftfür Internationale Zusammenarbeit*** |
| AfDB | ***African Development Bank*** |
| WB | ***The World Bank Group:*** The World Bank strategy focuses on increasing employment opportunities by transforming the Tunisian economy into a higher value-added, knowledge-intensive one. |
| JICA | ***Japan International Cooperation Agency:*** JICA supports activities around the pillars of field-oriented approach, human security, and enhanced effectiveness, efficiency, and speed. In Tunisia, JICA focuses on industrial development and reduction of regional disparities. It supports infrastructure development and strengthening of economic and industrial capacity. |
| KOIKA | ***International Cooperation Agency of Korea:*** KOICA aims to build a better world by helping achieve the Millennium Development Goals and promoting equitable and sustainable development in partner countries. KOICA supports their strengthening. |
| EU | ***Delegation of the European Union to Tunisia*** |

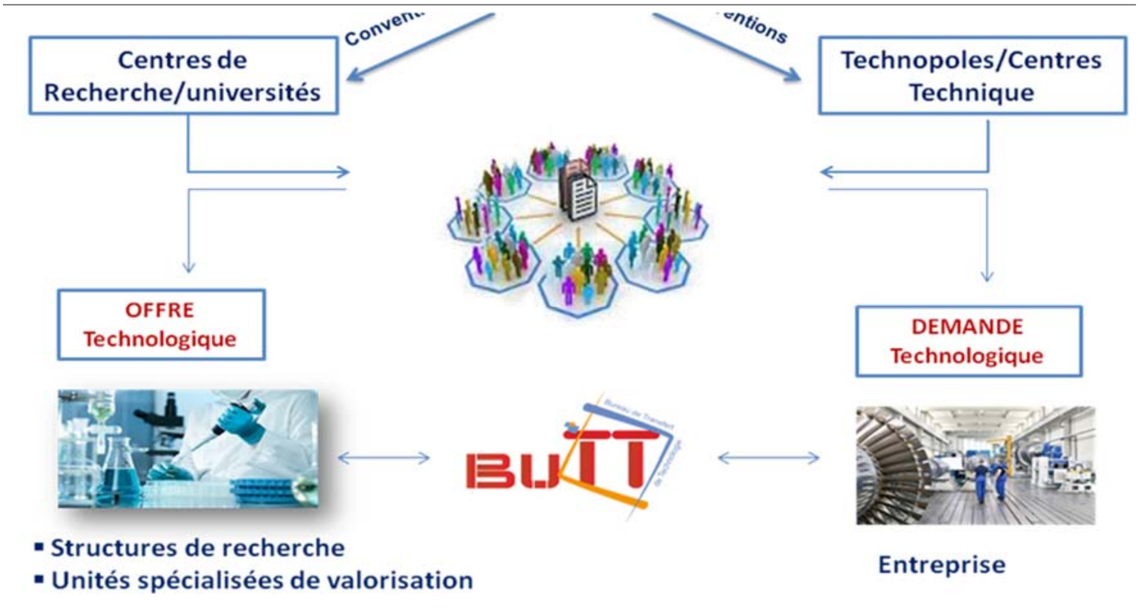
Source: Kim et al., 2018

[20] (STEPI, 2018 K-Innovation , 2018)

## 2.2 Network of TTOs

The National Agency for Scientific Research Promotion (ANPR) established in 2012 a network of TTOs in universities and research centers, with a central unit within the Agency coordinating and supporting this network through specific conventions (Annex 1 and 2). Today, there are 25 TTOs in this network (Figure 2-8) (Kim et al., 2018). This initiative is a bold step as it states that the TTO has technical duties in the transfer of technology. These missions require multiple areas of expertise that are not usually possessed by one person, since the Agency does not have ample resources to be able to ensure assistance to all partners. One of the advantages of this initiative is that it showed the willingness of universities and Public Research Organizations (PROs) to pay attention to technology transfer. In this respect, TTOs played a role in raising awareness of the importance of technology transfer at universities.

[Figure 2-8] Network of TTOs



One of the weaknesses of the initiative is the inability of TTO staff to carry out the tasks entrusted to them (Kim et al., 2018). Note that most of them allocate only about 20% of their time for technology transfer activities and feel that they are unable to play a pivotal role in technology transfer despite their participation in limited training courses (for only 15 days) (Kim et al., 2018). Moreover, these TTO staff can be transferred to other assignments by the university authorities. In this case, the investments in training them and the relationships developed with the Agency would be lost, and they would need to be repeated with the newly appointed staff (Kim et al., 2018). This confirms the lack of legal reference and specialized human resources (Kim et al., 2018).

[Table 2-4] Distribution of TTOs in Tunisia

| Universities | Research Centres | Technology Parks | Others |
|---|---|---|---|
| 1. Tunis University<br>2. Tunis El-Manar University<br>3. Carthage University<br>4. Manouba University<br>5. Jendouba University<br>6. Kairouan University<br>7. Sousse University<br>8. Monastir University<br>9. Sfax University<br>10. Gabes University<br>11. Gafsa University | 1. Pasteur Institute of Tunis **(IPT)**<br>2. National Institute of Research and Chemical and Physical Analyses **(INRAP)**<br>3. Biotechnology Center of Sfax **(CBS)**<br>4. National Center for Nuclear Science and Technology **(CNSTN)**<br>5. Center of Research Studies in Telecommunications<br>6. Research Center of Water Technologies<br>7. National Research Center for Materials Sciences<br>8. Digital Research Center of Sfax<br>9. Center of Research in Microelectronics and Nanotechnologies<br>10. CERTE | Technopole of Borj Cedria | 1. Institution de Recherche et de **l'Enseignement** Supérieur Agricole **(IRESA)**<br>2. General Direction of Technological Studies **(DGET)**<br>3. International Center of Environment Technologies of Tunis **(CITET)** |

Otherwise, the National Institute of Standardization and Industrial Property (INNORPI), in partnership with WIPO, established in 2016 four TTOs in Ghazala Technological Park, Sidi Thabet Technological Park, Technical Center for Packaging, and Technical Center for Chemistry (Kim et al., 2018). These TTOs were incorporated into the structure of the four institutions (Kim et al., 2018). This is a new experience but is still new and cannot be evaluated yet. *All these initiatives emphasized that technology transfer requires a clear legal framework on the one hand and human resources with appropriate competencies in technology transfer on the other* (Kim et al., 2018).

In this context, ANPR started training program for staff assigned to the TTO. The main activities done by ANPR to support TTOs are listed below (Kim et al., 2018).

- Approximately 30 training sessions in IP and TT for 500 participants
- Two regional workshops in Morocco (CLDP) for 12 TTOs (2018) and in Tunisia (2016) for promoting innovation as well as the entrepreneurship ecosystem in Tunisia through TTOs
- Participation of 8 TTOs in AUTM: Association of Technology Managers' University (CLDP) in 2016 and 2018 in California: https://www.autm.net/events-courses/annual-meeting/2019-annual-meeting/
- Visit to interfacing structures in Strasbourg for 12 TTOs (PASRI project)
- Study visit for 8 presidents of universities to Boston (CLDP) to Northeastern University, Mass Challenge, TechStars to assist Algeria and Tunisia in setting up Technology Transfer offices
- Two guides for intellectual property and contractualization were set up during the PASRI Project and diffused to research structures
- Signature of partnership with the UN-ESCWA (Economic and Social Commission for Western Asia) Project from 2015 to 2017 to be a partner of a network with *Egypt*, *Morocco*, *Jordan*, *Oman,* and *Lebanon* to set up a National Office of Technology Transfer (NTTO) in each country. This network has constituted a major asset for Tunisia as well as for the other member countries for the development of partnership in terms of innovation and strategic economic needs. These actions are largely insufficient to set up an operational TTO with appropriate skills. We present in the following sections the main gaps and some recommendations:

## 2.3 Support Programs

**PMN** (Programme de Mise à Niveau) (Upgrading Industrial Capacity Program at the Ministry of Industry, Energy, and Mines): Launched in 1995 to mark the signing of the free trade agreement with the EU, PMN is a national program that aims to upgrade the Tunisian industry. Implementing a PMN program involves the participation of several stakeholders (companies, PMN Office, technical centers, design offices, etc.). PMN has recorded 5256 approved requests from small and medium enterprises (SMEs), for a total investment of TND8.9 billion, and has committed around TND1.2 billion in grants. The most significant stake of this investment was dedicated to increasing production capacities, and very little was allocated for intangible assets.

**PCAM** (Programme d'Appui à la Compétitivité des entreprises et à la facilitation de l'accès au marché) (at the Ministry of Industry, Energy, and Mines). PCAM is the support program for the competitiveness of enterprises and improvement of access to markets. As a program of technical assistance, training, and awareness for industrial businesses and industry-related service providers, it revolves around coaching services on strategic corporate functions: quality, R&D, marketing, production, information systems, and business intelligence. Technical support and provision of equipment to support the quality of technical centers, testing laboratories, testing, and metrology are also covered by PCAM. With a budget of €23 million, PCAM will benefit from the implementation of its operational structures that started in December 2010. The overall objective of this tool is to facilitate access of Tunisian companies to international markets.

**PASRI** is a project funded by the EU in an amount of €12 million for over four years (2011-2014). It has the ambition of providing solutions to the main problems identified at the different levels and the different actors in the innovation chain. It targets the full range of institutional, administrative, financial, technical, and academic stakeholders supposed to support the transformation of technical knowledge into a product or a tangible service.

**Digital Tunisia 2018** (National Strategic Program of Digital Economy at the Ministry of Technology and Digital Economy)

This project aims to move toward a new development model that encourage change to a digital economy. We can see changes through several pillars, notably enhancing the network communication technology infrastructure by enabling all Tunisian families to have access to Internet services and to reach 80% of Tunisian homes by 2018, providing students with computer tablets to replace books and school supplies and developing in parallel digital public services and e-payment facilitation for all citizens specifically through post offices. The success of this project will have a very important positive impact on all economic activities in the country thanks to the acceleration of the technology transfer and innovation in all areas of activity.

## 2.4 STI Financial Support

### 2.4.1 Incentives for cooperation between companies and research structures:

- **VRR (Research Results Vitalization): http://www.mesrst.tn/**

The VRR program was launched in 1992 by the DGVR as the first attempt to promote the commercialization and application of research results to the social or economic environment. This program aims to place on the market the results obtained by the research laboratories. The VRR projects are funded by the Ministry of Higher Education and Scientific Research (MHESR) based on open call for proposals. The beneficiaries are public research organizations (research centers, research laboratories, research units) with the possibility of partnership with economic enterprises. The duration of realization of the VRR project is 3 years.

- **PNRI (National program of research and innovation at the Ministry of Industry and Small and Medium Enterprises) http://www.tunisieindustrie.nat.tn/**

As the National Program of Research and Innovation, PNRI is a program that finances R&D, innovation projects, improvements in industrial capacities, and modernization of production processes through the consolidation of cooperation and partnership between industrial companies, research structures, and technical centers. The program is under the supervision of the ministry of industry. To be eligible for funding, a project must associate an industrial company with a technical center and a public research organization (center, laboratory, or research unit) for up to two years. It must also demonstrate significant innovation as well as a minimum contribution of 20% of the project cost by the industrial partner (Kim et al., 2018).

- **MOBIDOC: http://www.anpr.tn/ (Kim et al., 2018)**

MOBIDOC was created in 2012 and has been managed by the National Agency for the Promotion of Scientific Research (ANPR Agence Nationale de la Promotion de la Recherche Scientifique). This program enables the establishment of partnerships between economic companies and research centers to carry out research programs in the company as the owner of the project idea; thus, the project responds to the needs of the company. The program specifically allocates grants to young researchers. The second session of MOBIDOC was launched in 2017.

- **Competitive program PAQ-Post PFE managed by the direction of vitalization of research results in MHESR http://dgvr.tn/**

• Scale up funding for young engineers or master's students
• Vitalization of innovative project students
• Recipient: the student, his/her supervisor coming from the university, his/her supervisor from the enterprise
• Budget: 35,000 dinars, 20% from the enterprise and 5% from the engineering school
• Duration: one year

- **Competitive program: PAQ-Collabora managed by the direction of vitalization of research results in MHESR http://dgvr.tn/**

• Scale up funding for collaborative projects between research structures, enterprises, and entities belonging to a technopole

• Post-doc or PhD students should be involved in the project

• Recipients: research structure

• Budget: 300,000 dinars, 10% from the enterprise and 5% from the institution of the research center

• Duration: 3 years

- **« PAQ - Pré-Amorçage et essaimage Scientifique » Competitive innovation fund for the exploitation and vitalization of innovative research results and creation of spin-offs managed by ANPR**

• Awarded to engineers, researchers holding master's degrees, PhD students, and Post-doctoral activities

• Budget: 100,000 dinars, 5% from the institution of the research center

• Duration: 18 months

Incentives for the creation of innovative enterprises

- **FOPRODI): Fund for Industrial Promotion and Decentralization (http://www.tunisieindustrie.nat.tn/)**

FOPRODI aims at the creation of a new generation of entrepreneurs, promotion and development of SMEs in the industrial, services, and small business sectors, and implementation of incentive measures for regional development.

- **IKDAM: Public Seed Fund (www.sages.tn)**

IKDAM is a public seed fund that aims to strengthen the innovative activities of startups at the early stage. The fund operates mainly for exploiting patents, drafting technical and economic studies for projects, developing manufacturing processes before the commercialization phase, and completing the financing scheme at the early stage of the company.

### 2.4.2 Incentives for research

- **PIRD: Grant for investment in research and innovation at the Ministry of Industry and Small and Medium Enterprises (http://www.tunisieindustrie.nat.tn/)**

PIRD was created in 1995 by legislation intended to support investment in activities such as R&D conducted by enterprises. As the first research program dedicated to the needs of the enterprise. It provides a 50% grant of up to US$20k toward a feasibility study as well as US$750k toward the testing or adaptation of new technologies or development and evaluation of prototypes. PIRD now includes public and private enterprises as well as scientific associations from all sectors including health and agriculture. PIRD is part of the strategy of the State to raise the technological integration level of the economy.

- **PRF: Federative research programs at DGRS, administrative management at ANPR**

The establishment of PRF allowed for an advanced step toward the organization of activities of the national R&D system through the mobilization of expertise and creation of synergies between structures for research and their public or private partners concerned with the development of the sector of scientific research and technology. These programs are funded under multi-year agreements that define the supporting structure of the project and associated partners, objectives and expected results, human and material resources to be mobilized, and monitoring and evaluation procedures. This program addresses national priority themes *(Annex 3)* defined in consultation with the various operators in the economic sector concerned.

- **ITP: Priority technological investment at the Ministry of Industry, Energy, and Mines (http://www.tunisieindustrie.nat.tn/)**

In addition to the PMN Grant, the ITP grant ensures the financial support of intangible investments, implementation of a quality management system, and certification. Industries and service companies that have operated business for one year at least are eligible for this program. Since its launch in January 2016, ITP has approved 8166 requests for a total investment of TND417.2 million including TND176 million in grants and TND84.7 million in distributed premiums.

In addition, we find, among the financial incentives, that the Ministry of Higher Education and Scientific Research assumes the totality of the costs related to the registration of patents filed by Tunisian researchers belonging to universities or public research centers.

### 2.4.3 STI Support: Organizations, Programs, and Initiatives

#### 2.4.3.1 Technical center

| | |
|---|---|
| **CETIBA** | Wood Industry and Furniture |
| **CETIME** | Mechanical and Electrical Industries |
| **CTC** | Chemistry |
| **CETTEX** | Textile |
| **CNCC** | Leather and Footwear |
| **CTAA** | Agrifood |
| **CTMCCV** | Building Materials, Ceramics, and Glass |
| **PACKTEC** | Packing and Packaging |
| **4 Agriculture Centers** | Cereals, Potatoes, Aquaculture, Organic Farming |
| **IRA Mednine** | Vitalization of Sahara Product |

As far as we know, the organizational charts of each technical (industrial) center provide for the existence of a vitalization unit. Surprisingly, only two centers have upgraded this unit with the help of INNORPI.

#### 2.4.3.2 Technoparks

| | |
|---|---|
| **Pôle de Compétitivité Monastir/El Fejja (Manouba)** | Textile and clothing; hosting the nascent Textile Cluster |
| **BiotechPole Sidi Thabet** | Biotechnology and healthcare industries: working on a Biotech Cluster |
| **Technopole Borj Cédria** | Renewable energy, water, and environment |
| **Pôle de compétitivité de Bizerte** | Food processing; hosting the Lactimed Cluster |
| **Pôle El Ghazela** | ICT |
| **Sousse Technopark** | Mechanical, electrical, and electronic industries; hosting the Mechatronic Cluster |
| **Sfax Technopark** | ICT and multimedia |
| **Industrial and Technological Park of Gabes** | Multi-sectoral |
| **Pôle de Compétitivité de Gafsa** | Multi-sectoral; hosting the Dates Cluster |

#### 2.4.3.3 Business clusters

Tunisia is still in the early stage in the field of clusters' implementation. The emergence of business clusters formally started in the early 2000s with the first groups in the form of export consortia. After the release of Tunisia 2016, the industrial strategy study published by the government in 2008 and which has retained the idea of clustering as a growth driver, several industrial networks tried to establish, in partnership with technoparks, "innovation clusters" particularly in four strategic sectors (textiles and clothing, agribusiness, renewable energy, and ICT). The mobilization spurred by the strategic study was not accompanied by the means that would then have enabled the emergence of innovation-oriented clusters. IPEMED (Institut de Prospective Economique du Monde Méditerranéen) shows that the clustering strategy implementation took too much time in Tunisia. Therefore, some enabling interventions have been initiated by international agencies to aid in the acceleration of the formation and operation of the clusters. GIZ (AgriFood Clusters), AFD (Mechatronics), and UNIDO (Creative Industry Cluster) massively supported some pilot projects. Furthermore, reflections on the methods of supporting clusters have been conducted under the PASRI program, with the emergence of 10-15 clusters in high potential sectors that will include the territoriality dimension forecast in the coming years. Further experiments are currently outstanding, especially in creative industries with the support of UNIDO, renewable energy, textile, biotechnology, etc., but they are at too early a stage to have visible impact.

#### 2.4.3.4 Public startups

Public start-up nurseries are infrastructure spaces under the public agency APII that are equipped to host entrepreneurs when they start launching a promising business. They now constitute a network of 30 nurseries distributed throughout the territory. They are generally located in the Higher Institutes of Technological Studies, engineering schools, research centers, and science parks. Incubation services are supposed to be provided to incumbents in order to help them realize their innovative ideas and transform them into operational projects. Public start-up nurseries host -- for a definite period of one renewable year -- and help incumbents relocate outside the nursery after the incubation period.

#### 2.4.3.5 Incubators and accelerators

Besides the network of start-up nurseries owned by the State, and which are switching gradually to a business incubator model, to serve better the 250 start-ups hosted per year, Tunisia has seen the birth of several initiatives of the private sector and civil society to support innovation. The following are the most significant entities:

- As the first private business incubator, Wiki Start Up was launched by the Carthage Business Angels network. This start-up has its own seed fund called Capital Ease.
- Start Up Factory/IntilaQ for Growth Fund launched by Telco operator Ooredoo, Microsoft Tunisia, and Tunisian-Qatari Friendship Fund.
- ESPRIT Incubator launched by the private university leader in the field of ICT in partnership with the association Tunisie-Croissance, backed by Tuninvest Fund.

- Yunus Social Business Accelerator launched recently by the Yunus Foundation in partnership with AfDB to promote social innovation and contribute to Social Business Development in Tunisia.

#### 2.4.3.6 Agencies in charge of IP protection in Tunisia

The intellectual property rights (IPR) protection procedure involves different agencies depending on the nature of the IP, e.g., Industrial Property falls under the INNORPI scope, copyright is within OTDAV, and Vegetal Variety Obtainment Certificate is under the responsibility of the Ministry of Agriculture. There is no agency covering IP out of these three categories, and there are almost no local IP Adviser specialists in patent writing to assist inventors or entrepreneurs. This gives rise to many barriers to knowledge dissemination and eliminates opportunities in innovation.

- **INNORPI: Institut National de la Normalisation et de la Propriété Industrielle (National Institute of Standardization and Industrial Property)**

INNORPI is a public entity under the Ministry of Industry, Energy, and Mines with legal standing and financial autonomy. INNORPI's mission is "to undertake any action concerning standardization, quality of products and services, and protection of industrial property."

- **OTDAV: Organisme Tunisien des Droits d'Auteurs et des Droits Voisins (Tunisian Agency of Copyrights and Related Rights)**

OTDAV is a public agency that aims to safeguard copyright and related rights and to defend the moral and material interests of copyright owners.

**Assistant structures in charge of IP Protection:**

**MESRS through DGVR** assists researchers in filing their patent for the 4 first years. In fact, some of the researchers file their patent directly through the INNORPI, so we have different statistics.

**ANPR** supports researchers in IP and gives them advice, orientation, and/or training.

# CHAPTER 3

## Sharing Korea's Experiences in Technology Transfer System and Policy Support

Policy Consultation on the Support for the Establishment of an Effective National System of Technology Transfer (NSTT) in Tunisia

# Chapter 3. Sharing Korea's Experiences in Technology Transfer System and Policy Support

## Part I. Technology Commercialization and Valuation: Policy and Practice in Korea

## 1. R&D Commercialization and Valuation

### 1.1 Understanding the Process of Technology Commercialization

Some technologies fail in commercialization because they are incorporated in products for which the anticipated demand never materializes. Others continue to search for suitable products, sometimes over decades, without being incorporated into any product at all. Then there are those that fail because they cannot live up to their promised capability when being demonstrated, or they attract insufficient interest and resources to have such demonstrations made. Finally, entry into the market itself faces an intractable obstacle in some cases. Some technologies have made a fleeting appearance, never to be heard from again. Their problem was one of positioning and delivery. They could neither gain adequate market penetration nor sustain their commercialization for a variety of competitive reasons.

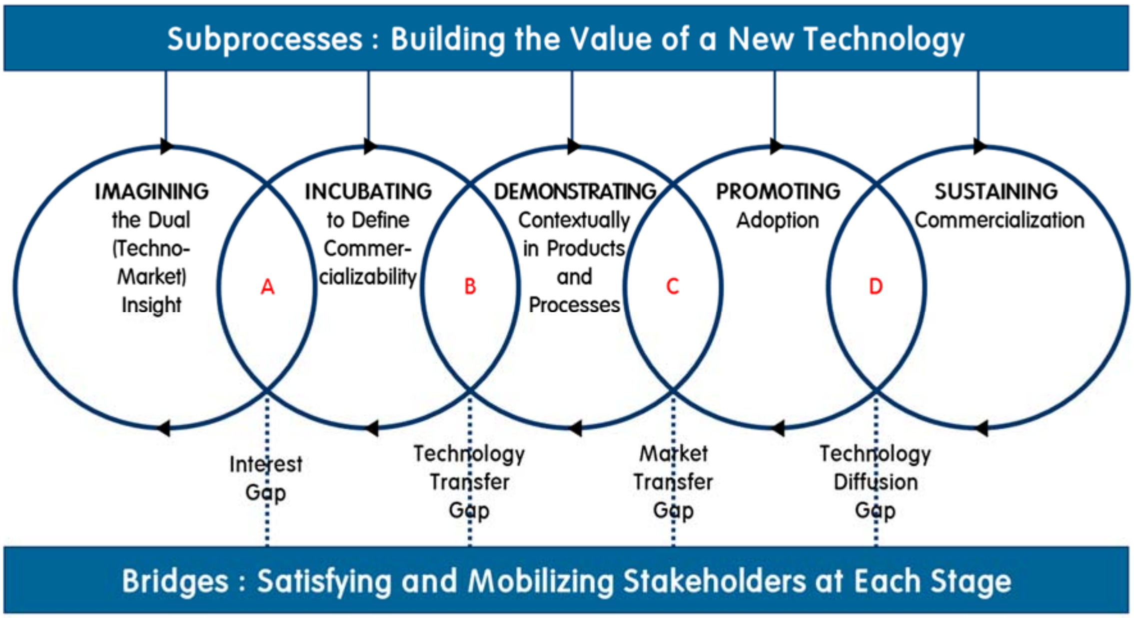
In understanding what went wrong, one needs to know where in the commercialization chain problems occurred and why. In many technologies, the following were the typical activities where things could go wrong:

- Linking of technological discovery to worthwhile, exciting market opportunities

- Having the technology endorsed early by those whose opinion matters
- Incubating the technology sufficiently to understand its true potential, including whether it will ever be cost-effective enough to merit taking further
- Mobilizing adequate resources for its demonstration
- Successfully demonstrating the technology for the context in which it is to be used
- Mobilizing the market constituents needed for gaining market acceptance and delivering the benefits of the technology
- Promoting the final products and processes to an often-skeptical customer group
- Choosing an appropriate business formula for gaining access to the required business system
- Sustaining commercialization so as to realize value from the technology after it has been launched

In other words, technology commercialization is about performing successfully a range of things, each adding value to the technology as it progresses. Being proficient in one or two of them and clumsy in the rest brings down the average result. Worse, it can abort a technology’s progression midstream.

[Figure 3-1] Process of Technology Commercialization




As shown in Figure 1-1, five of these activities constitute the key subprocesses involved in bringing new technologies to market: **imagining** a techno-market insight; **incubating** the technology to define its commercializability; **demonstrating** it contextually in the product and/or processes; **promoting** the latter's adoption; and **sustaining** commercialization. As important as these subprocesses are the four bridges between them (Kim et al., 2018). While the former involve problem-solving of a technical or a marketing nature – doing things to the technology, so to speak – these bridges are associated with mobilizing resources around it. They have to do with satisfying the various stakeholders of the technology at each stage, without whom the technology's value does not get recognized; neither is there an impulse to take it further (Kim et al., 2018). Thus, the bridges are value-creating activities in their own right.

These bridges bring to mind an important reality about the innovation process – that it is fundamentally an exercise in stakeholder management (Kim et al., 2018). Many technologies fail not because of the technical skills of their proponents or because of the market they are targeting. They fail simply because no one got sufficiently interested in them at the right time (Kim et al., 2018).

## 1.2 Commercialization Stages and External Funding

The five value-adding subprocesses correspond roughly to the manner in which funding is structured today as shown in Table 3-1. Competition for funds occurs at each stage, and each stage has its own criteria for attracting resources. Thus, those who imagine and define a technology well in the beginning get to amass a larger war chest of funds. Others have to accumulate resources more progressively in line with the achievements made from one stage to another.

A more recent trend is that different people tend to specialize in different stages of innovation. Separate actors are inventing technologies, defining and incubating them, and demonstrating them in products and processes. There are even some whose sole mission is to help with marketing the latter on a global basis.

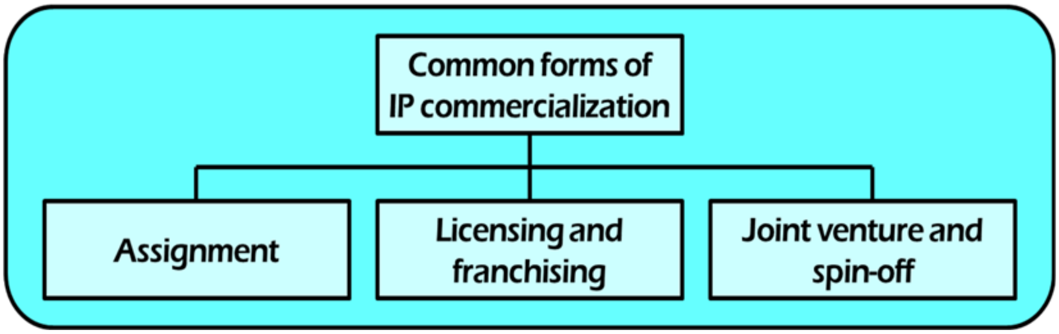
[Table 3-1] Correspondence between the Commercialization Stage and External Funding

| Commercialization Stage | Type of Financing | Source of Funds |
|---|---|---|
| 1. Imagining | Seed or zero-stage financing | Government R&D support, family and friends, special early-stage venture capitalists |
| 2. Incubating | First-stage or start-up financing | Government R&D support and venture capitalists |
| 3. Demonstrating | Second-stage financing | Venture capitalists and government technology development programs |
| 4. Promoting | Third-stage and other expansion-related financing | Banks and other commercial lending Organizations |
| 5. Sustaining | Traditional corporate finance | Banks and retained earnings |

## 1.3 Forms of R&D Commercialization

R&D commercialization is the process of bringing a technology or an intellectual property (IP) to the market in order to be exploited. R&D commercialization can take different forms. The most common forms are summarized in Figure 3-2 below.

[Figure 3-2] Common Forms of R&D Commercialization




Licensing is a route of commercialization wherein an IP rights (industrial property and copyright) holder gives another entity the authority to exploit to make, use, sell, copy, display, distribute, modify, etc. the IP; in return, the licensee will pay royalties, equity, other payment, or other IP (cross-licensing). It is the most popular and sustainable way of commercializing IPRs and is managed through written legally bound agreements. Agreements stipulate the details of extent of rights of exploitation.

Licensing has the following types:

- Exclusive license
- Non-exclusive license
- Sole-license (use clear language)
- Sublicense

A spin-off refers to a separate company established in order to bring a technology developed by a parent company to the market. It is deemed to be a valuable alternative to transform technology into product and service as well as to license out technology. A conventional spin-off company can be set up in two ways:

- A new company is created through the separation from a parent organization (PO) that contributes with its financial, human, and intellectual capital. This R&D spin-off has mainly the scope of further developing and commercializing the IP created at and **assigned** by the PO. Together with the relevant intangible asset, the PO also transfers to the new legal entity the obligations and risks associated with the commercialization of the IP.
- A spin-off can also be a company established, usually by a person external to the PO, with a view of exploiting the intangible **licensed** by the owing organization. Often, in fact, PROs attract venture capitalists interested in investing in the development of the IP created by the organization. This type of company is also referred to as an innovative "startup," more likely to obtain a final marketable product.

## 1.4 What is Technology Valuation?

Technology valuation plays a key role in technology transfer and commercialization. Then what is meant by valuation? Valuation means assigning a quantitative (dollar) value, even to an intangible asset like technology. And why is valuation important? This is because money is the only common language that brings together technologists, inventors, and business people. When is it useful? It is useful when we negotiate transactions involving technology and useful for decision support ("Should we do this R&D project?").

In many cases, valuation is compared with pricing. Basically, technology valuation is conducted by different techniques, which produce different answers. It is an opinion. Meanwhile, pricing is conducted by negotiation, which produces one outcome. It is a commitment. If we carry out technology valuation with a valuation basis, then we negotiate the basis for the pricing of a technology. If we conduct technology valuation without a valuation basis, however, then we negotiate pricing from emotion.

Prospectively, technology valuation is required by deal makers wherein the value of the subject technology is extracted over time, and technology valuation must be win-win results. On the contrary, technology valuation can be needed retrospectively by litigators. In this case, the value is established at a point of time. This shows the need for technology valuation when the parties concerned are adversarial and the result of technology valuation is outcome imposed judicially.

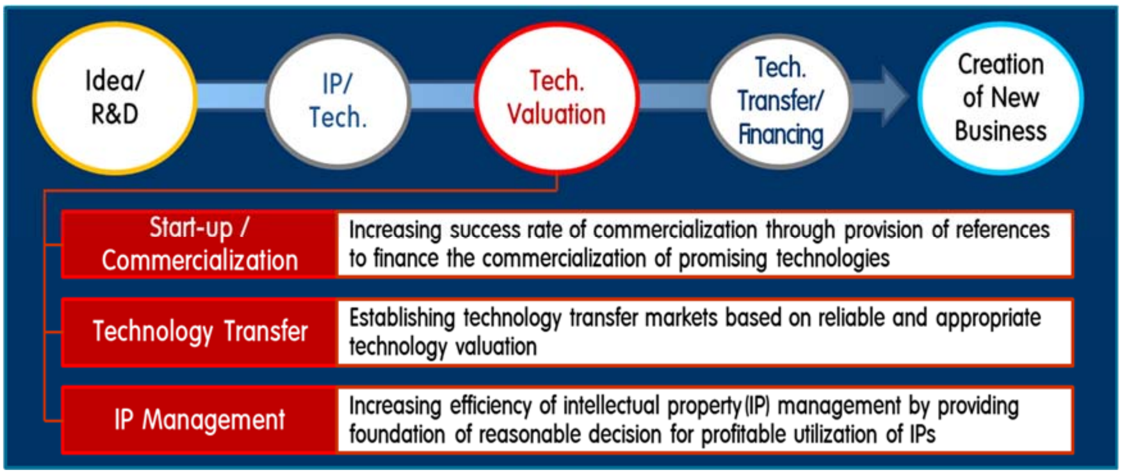
Users of technology valuation include willing sellers of a technology (academe, startups, etc.), willing buyers (startups, pharmaco., etc.), investors (VCs, investment bankers, corporate R&D managers, etc.), lenders (secured creditors), acquirers (investment bankers), monetizers (partnerships), and unwilling buyers or sellers (litigators).

# 2. Policy to Promote Technology Valuation in Korea

## 2.1 Background of the Policy

Technology transfer and licensing, technology-based startup, and investment need to be promoted by well-functioning technology valuation to promote sustainable economic development through technology commercialization. The role and functions of technology valuation are summarized in Figure 3-3 below.

[Figure 3-3] Role and Function of Technology Valuation




As seen in the figure, technology commercialization encompasses and requires both R&D area and market area. It starts from an idea or from R&D, which is transformed into IP and technology, and contributes to the creation of new businesses through technology transfer and financing.

Technology valuation plays the role of bridging the R&D area (idea/R&D and IP/technology) and market area (technology transfer/financing and creation of new business) in the process of technology commercialization.

In particular, technology valuation increases the success rate of commercialization through the provision of references to finance the commercialization of promising technologies, establishes the technology transfer market based on reliable and appropriate information of economic value of technologies, and increases the efficiency of IP management by providing the foundation of reasonable decision for the profitable utilization of IPs.

Korea's technology financing and valuation market has been growing, and public valuation organizations and financial institutions have played a central role in the build-up of technology valuation bases but have yet to move toward promoting the market's trust and boosting demand for valuation.

Valuation for technology transfer and investment purposes is fairly active. Private sector-driven valuation by investment banks and venture capitals need to grow further.

In the long run, private sectors need to lead the technology valuation and financial market system; in the short run, however, the government has to play a more active role in building crucial bases and infrastructures of technology valuation and financial market system.

## 2.2 Policies by Government Departments

The Korean government established "a plan for promoting the credibility of technology valuation" (Oct. 2013, State Council) and initiated policy measures for the establishment of open technology valuation system by market, as shown in Figure 3-4.

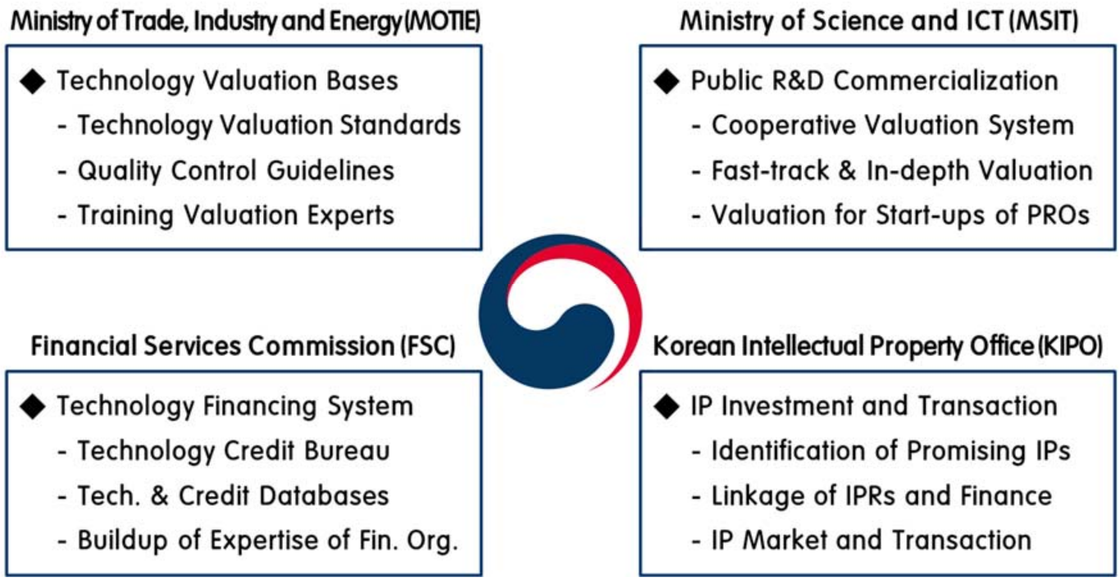
The Ministry of Trade, Industry, and Energy enacted the guidelines for the operation of technology valuation standards and to establish quality control for the valuation report.

The Ministry of Science and ICT is trying to improve the tech. valuation system, enhance the expertise of TLO staff in public R&D organizations, and institutionalize the utilization of technology valuation in the commercialization of public R&D.

The Financial Services Commission designated the Technology Credit Bureau (TCB) and constructed the technology/market DBs to promote technology financing.

The Korean Intellectual Property Office is trying to increase coverage of IPR litigation insurance and arrange exit strategies for financial institutions using various IP funds.

[Figure 3-4] Policies by Government Departments

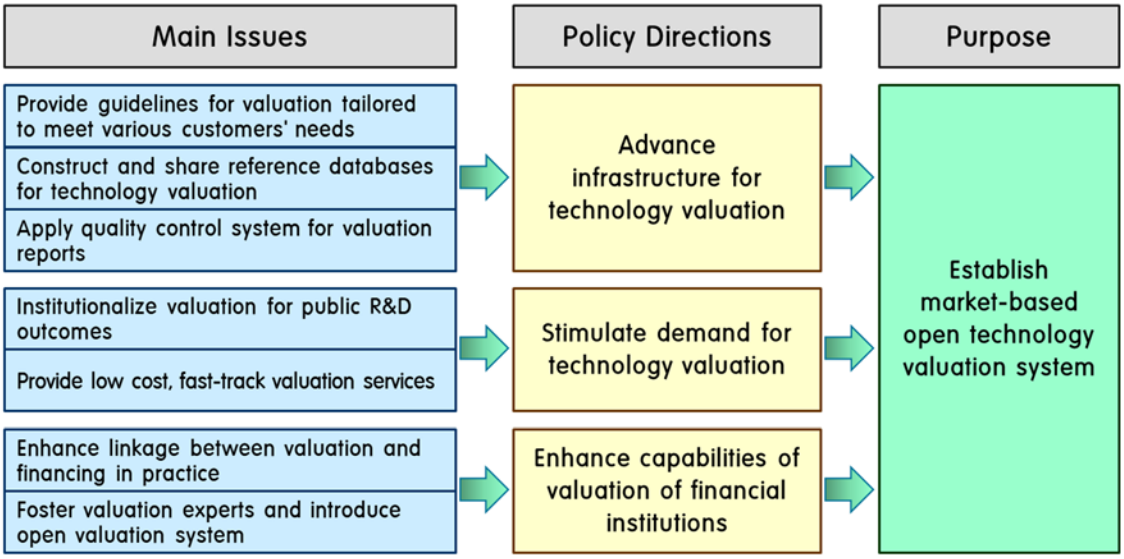



As shown in Figure 3-5, the focus of the policy is to establish a market-based open technology valuation system. To achieve the goal, the Korean government is making efforts to advance the infrastructure for technology valuation, stimulate demand for technology valuation, and enhance the valuation capabilities of financial institutions.

For the policy directions, the government raises and implements the relevant issues. It is providing the guidelines for valuation tailored to meet various customers' needs, constructing and sharing reference databases for technology valuation, and applying the quality control system for valuation reports.

Moreover, the government is making efforts to institutionalize valuation for public R&D outcomes, provide low cost, fast-track valuation services, enhance linkage between valuation and financing in practice, and foster valuation experts and introduce an open valuation system.

[Figure 3-5] Focus of the Policy




## 2.3 Present Condition and Policy Implications

### 2.3.1 Present Condition

■ **Valuation organizations and human resources**

About 30 public organizations are designated and operated as official valuer under related laws. The valuer organizations were designated based on the "Special law to promote venture start-up," "Law on the promotion of technology transfer and commercialization," and "Invention promotion law."

Technology valuation organizations have around 500 PhD holders or highly experienced staff members in all and 10 ~ 30 experts per organization.

■ **Performance of technology valuation**

Valuations performed by public organizations numbered about 4,500 cases in 2018. By purpose, technology transfer accounted for about 50%, strategic decision, about 20%, investment, about 20%, and other purposes, about 10%.

Especially, three organizations including KTFC (Korea Technology Finance Corporation), KIPA (Korea Invention Promotion Association), and KISTI (Korea Institute of Science and Technology Information) have undertaken about half of all the in-depth technology valuation cases (700 cases in 2018).

### 2.3.2 Policy Implications

■ **Linkage between technology and finance**

Linkages between the technology and finance sectors are crucial for improving technology-centered valuation and promoting the technology financing market.

The annual number of technology valuations for the purpose of investment was estimated to be about 900. The existing valuation market has focused more on companies' financial conditions, credit rating, and technological base and capabilities than technology per se.

Financial institutions' technology-based investment and finance need to expand by improving the reliability of technology valuation and applying the valuation results.

■ **Technology Transfer**

The R&D results of universities and government-funded R&D organizations have been growing in a quantitative sense, but the technology transfer rate and royalty revenues relative to R&D inputs need to be increased by using technology valuation more systematically.

For instance, patent registrations (domestic & abroad) by PROs increased from 23,190 cases in 2011 to 35,859 cases in 2017. The technology transfer of public R&D results to private firms increased from 26.0% in 2011 to 37.9% in 2017. Public R&D organizations' technology transfers numbered 7,195, but cases using in-depth technology valuation in determining royalty were about 750 (10.4%).

Low-cost, fast-track valuation for small-sized R&D-funded technologies and in-depth valuation for large-sized R&D-funded technologies are required to select promising technologies from unused ones and to determine the appropriate royalty.

## 2.4 Main Agenda of MSIT

### 2.4.1 Establishment of a Collaborative Technology Valuation System

The first agenda of MSIT (Ministry of Science and ICT) is the establishment of a collaborative technology valuation system. This has the purpose of establishing a collaborative system for in-depth technology valuation by utilizing government-supported research organizations' expertise and experience in various fields of technologies.

This has aimed at promoting reliability and expertise by specializing in cutting-edge and convergence technology including BT, NT, ICT, etc. In order to achieve the aim, the ministry established a collaborative system by designating public R&D organizations with expertise in each technology field as official valuer organizations in charge of analysis of the technological feasibility and strengths of IP rights, led by KISTI as the managing body (2014).

Based on the governing law, the ministry designated five (5) organizations with valuation capabilities such as the Korea Institute of Science and Technology (KIST), Korea Research Institute of Bioscience & Biotechnology (KRIBB), Electronics and Telecommunications Research Institute (ETRI), Korea Institute of Industrial Technology (KITEC), and Korea Institute of Machinery and Materials (KIMM) as official technology valuers specializing in technological analysis.

MSIT also established common valuation standards of the collaborative system between KISTI and other valuation organizations and enhanced the reliability of technology valuation by quality control in the post-examination of valuation results.

Moreover, MIST established a common access system to existing information related to valuation, including technology transactions, technology valuation reports, and technology and market DBs, among technology valuation organizations.

Moreover, MIST is promoting the establishment of a collaborative system with the private sector including the Technology Credit Bureau (TCB) initiated by the Financial Services Commission, including the common use of its experts and technology information.

### 2.4.2 Promotion of Fast-track Technology Valuation

The next agenda of MSIT is the promotion of fast-track technology valuation. This aims to promote the use of Web-based, automated (online) technology valuation system that can deliver low-cost, fast-track valuation for the purpose of selecting promising technology and calculating appropriate royalty.

MSIT is promoting the licensing of the STAR-Value System developed by KISTI, a Web-based (online) self-assessment system for technology valuation, to universities and GRIs that can use the system in their own fast-track valuation of technologies.

For such purpose, MSIT is providing the guidelines and standards for the procedure and methods of using the STAR-Value System with the help of related DBs and supporting training programs financially to use the STAR-Value System effectively for the TLOs of universities and government research institutions that lack valuation expertise to reinforce their own valuation capabilities.

### 2.4.3 Promotion of Public TLO's Expertise in Technology Valuation

The third agenda is the promotion of public TLO’s expertise in technology valuation. For such purpose, MSIT implements programs to train experts, institutionalize technology valuation, and provide financial aid for improving the valuation capabilities of public and private R&D sectors.

For such purpose, MSIT is trying to enhance the public TLO's valuation expertise by implementing professional training programs including valuation theory, practice, and case studies for officially designated valuation organizations in order for them to conduct technology valuation for the technology transfers of public R&D outcomes and allocate valuation expenses to direct cost accounts for large R&D projects.

At the same time, MSIT is promoting the linkage between technology valuation and commercialization. To that end, it is expanding financial support for the foundation of laboratory-based enterprises and promoting financial support for valuation consultation and market survey to TLOs.

## 2.5 Uses of Valuation Results

### 2.5.1 Providing Information for Efficient Commercialization

Technology valuation provides useful information for efficient commercialization. It enables R&D organizations to identify promising technologies by conducting their own valuation and provide the information to potential user firms.

MSIT is encouraging the comprehensive use of technology valuation results and comparable licensing records in determining the royalty rate for technology transfer to promote understanding of technologies' value and determine the fair market price of the technologies.

PROs can calculate the reasonable royalty amount by acquiring reliable and useful reference information on royalty determination for negotiation through technology valuation, especially in licensing heavily funded technologies.

The government is preparing the exemption clauses on the possible differences between the valuation results and actual transaction prices to protect the confidential information of the owner of the subject technology. This measure is able to prevent the decline of negotiation for technology transaction between the transferor (technology holder) and prospective transferee.

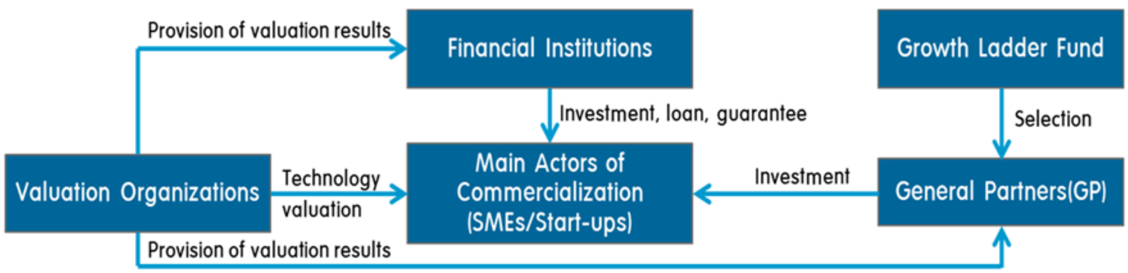
### 2.5.2 Connecting the Valuation Results to Technology Financing

Technology valuation helps connect the valuation results to technology financing. In particular, the Financial Services Commission utilizes the valuation results of technologies in effectively managing public funds to promote financial aid for early-stage investment in technology-based startups and innovative firms.

At the same time, financial institutions are expanding technology value-based finance, Korea Development Bank (KDB) is providing financial support for innovative firms in special R&D zones, and Korea Technology Finance Corporation KTFC) is providing technology guarantee funds.

Figure 3-6 shows the flow of financial support based on technology valuation.

[Figure 3-6] Flow of Financial Support based on Technology Valuation




### 2.5.3 Performance of Technology Valuation and Technology Transfer

Table 3-2 shows the number of technology assessments and valuations conducted by Korean public research organizations' PROs (government research institutions and universities) from 2010 to 2017. In 2017, 273 PROs including 129 GRIs and 144 universities used the services to facilitate their technology transfer and commercialization.

[Table 3-2] Technology Assessment and Valuation

| | | No. of org. | No. of tech. requested | | |
|---|---|---|---|---|---|
| | | | Total | Per org. | Per req. |
| 2010 | | 182 | 3,161 | 17.4 | 52.7 |
| 2011 | | 192 | 5,742 | 29.9 | 82.0 |
| 2012 | | 200 | 6,576 | 32.9 | 87.7 |
| 2013 | | 272 | 3,805 | 14.0 | 63.4 |
| 2014 | | 250 | 4,738 | 19.0 | 70.7 |
| 2015 | | 279 | 3,750 | 13.4 | 46.9 |
| 2016 | | 278 | 2,747 | 9.9 | 28.6 |
| 2017 | | 273 | 1,436 | 5.3 | 18.4 |
| | GRIs | 129 | 195 | 1.5 | 6.8 |
| | Univ. | 144 | 1,241 | 8.6 | 27.0 |

Table 3-3 shows the number of technology transfer contracts by year in Korean PROs. This table shows the consistent increases in the numbers during the period. The most common form of technology transfer contracts is paid licensing. GRIs made 3,246 contracts, whereas universities made 4,280 contracts in 2017.

[Table 3-3] Technology Transfer Contracts (cases)

| | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | GRIs | Univ. |
|---|---|---|---|---|---|---|---|---|---|---|
| Assignment (sales) | 173 | 330 | 383 | 538 | 718 | 1,017 | 1,587 | 1,466 | 363 | 1,224 |
| Licensing (paid) | 2,536 | 2,759 | 3,557 | 3,457 | 4,098 | 5,113 | 5,385 | 4,905 | 2,613 | 2,772 |
| Licensing (unpaid) | 179 | 178 | 282 | 263 | 639 | 322 | 327 | 278 | 168 | 159 |
| Others (OEM, tech. alliances, etc.) | 52 | 153 | 90 | 100 | 107 | 207 | 227 | 546 | 102 | 125 |
| Total | 2,940 | 3,420 | 4,312 | 4,358 | 5,562 | 6,749 | 7,526 | 7,195 | 3,246 | 4,280 |

Table 3-4 shows the amount of royalty revenues by Korean PROs. They recorded the peak in 2015 by acquiring 204,170 million Korean won (about 200 million US dollars). GRIs accounted for more than 50% of the total royalty revenues.

[Table 3-4] Royalty Revenues (million Korean won)

| | | No. of org. | None | < 50 | < 100 | < 500 | ≥500 | Total | Per org. |
|---|---|---|---|---|---|---|---|---|---|
| 2010 | | 222 | 45.9% | 12.6% | 8.6% | 14.0% | 18.9% | 124,514 | 560.9 |
| 2011 | | 191 | 27.7% | 19.4% | 8.4% | 20.9% | 23.6% | 125,812 | 658.7 |
| 2012 | | 198 | 18.7% | 27.3% | 7.6% | 21.7% | 24.7% | 165,180 | 834.2 |
| 2013 | | 272 | 45.6% | 18.4% | 6.6% | 14.0% | 15.5% | 135,353 | 497.6 |
| 2014 | | 250 | 35.7% | 20.5% | 4.8% | 16.9% | 22.1% | 140,332 | 561.3 |
| 2015 | | 279 | 32.6% | 22.2% | 6.1% | 17.9% | 21.1% | 204,170 | 731.8 |
| 2016 | | 278 | 30.9% | 21.2% | 6.1% | 17.6% | 24.1% | 177,113 | 637.1 |
| 2017 | | 271 | 22.5% | 25.1% | 5.2% | 21.0% | 26.2% | 182,718 | 674.2 |
| | GRIs | 127 | 30.7% | 26,0% | 4.7% | 15.8% | 22.8% | 112,683 | 887.3 |
| | Univ. | 144 | 15.3% | 24.3% | 5.6% | 25.7% | 29.2% | 70,035 | 486.4 |

In Table 3-5, we can see the increase in technology-based startups established by PROs in Korea. This table shows the conspicuous increases in the number of startups established every year during the period 2014 ~ 2017. Universities account for 50% of the total number of technology-based startups.

[Table 3-5] Technology-based Startups (cases)

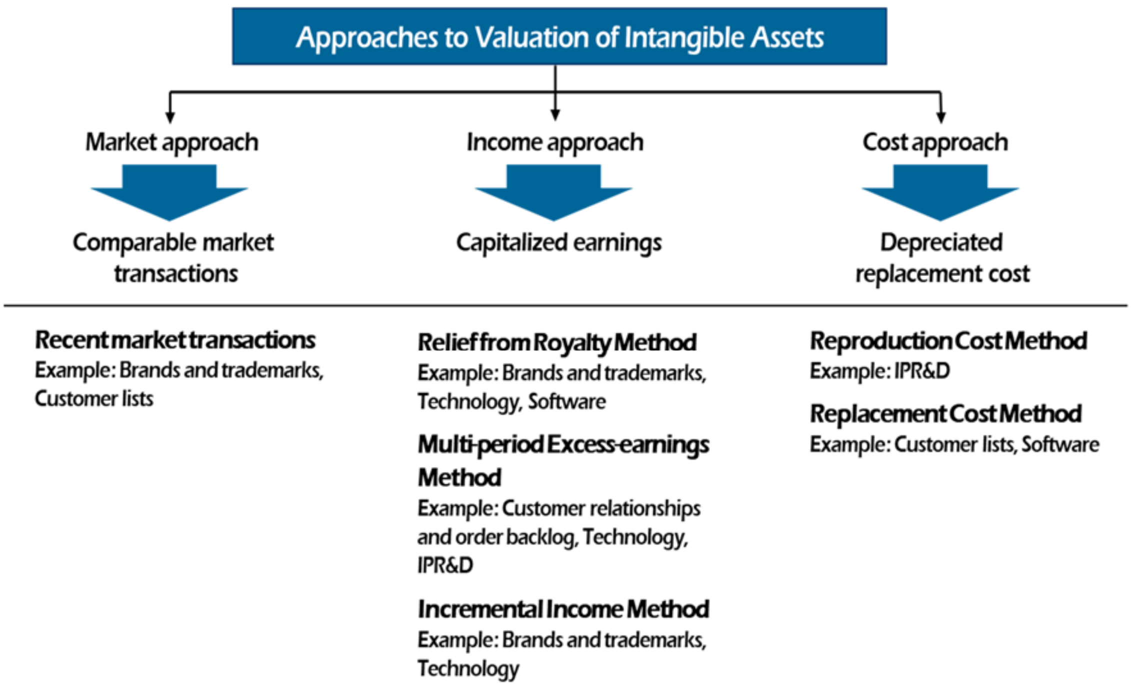
| | | No. of org. | Internal startups | | External startups | | Total startups | |
|---|---|---|---|---|---|---|---|---|
| | | | Total | Per org. | Total | Per org. | Total | Per org. |
| 2014 | GRIs | 115 | 38 | 0.33 | 22 | 0.19 | 60 | 0.52 |
| | Univ. | 135 | 70 | 0.52 | 6 | 0.04 | 76 | 0.56 |
| | Total | 250 | 108 | 0.43 | 28 | 0.11 | 136 | 0.54 |
| 2015 | GRIs | 136 | 75 | 0.55 | 11 | 0.08 | 86 | 0.63 |
| | Univ. | 143 | 99 | 0.69 | 12 | 0.08 | 111 | 0.78 |
| | Total | 279 | 174 | 0.62 | 23 | 0.08 | 197 | 0.71 |
| 2016 | GRIs | 135 | 54 | 0.40 | 8 | 0.06 | 62 | 0.46 |
| | Univ. | 143 | 171 | 1.20 | 33 | 0.23 | 204 | 1.43 |
| | Total | 278 | 225 | 0.81 | 41 | 0.15 | 266 | 0.96 |
| 2017 | GRIs | 129 | 73 | 0.57 | 16 | 0.12 | 89 | 0.69 |
| | Univ. | 144 | 226 | 1.57 | 50 | 0.14 | 276 | 1.92 |
| | Total | 273 | 299 | 1.10 | 66 | 0.24 | 365 | 1.34 |

# 3. Methods of Technology Valuation in Practice

## 3.1 How We Value Technologies – Valuation Approaches

Depending on the perspective employed by each approach as seen in Figure 3-7, approaches used to measure the economic value of a technology are classified into income-based, cost-based, and market-based.

[Figure 3-7] Approaches to the Valuation of Intangible Assets




First, the income approach "looks forward" to value a subject technology by calculating the present value of future cash flows created from the use of the technology asset.

Second, the cost approach "looks backward" to value a technology based on its reproduction cost or replacement cost. A technology's value is calculated based on the cost required in its acquisition due to the difficulty of estimation of future income.

Third, the market approach "looks around" to use prices and other relevant information generated by market transactions involving identical or comparable assets. A technology is valued based on the prices of comparable assets at arm's-length transactions.

Among these three (3) approaches, the market approach is regarded as the first choice for technology valuation. As the market approach has limitations in practical application due to the lack of comparable market data, however, the income approach based on DCF (discounted cash flow) has been used most commonly among practitioners.

The cost approach in not widely used because it does not reflect the future economic value of the valued IP or technology. The market approach also has a limitation since it is difficult to find a comparable technology transaction in the market. Thus, the income approach is generally applied when undertaking technology valuation.

Accordingly, valuation practitioners most frequently use the income approach on grounds that the potential economic value created from the use of a technology asset can be evaluated properly by the approach.

## 3.2 Method and Procedure for Technology Valuation

### 3.2.1 Method of Technology Valuation Used in Korea

There are three approaches to technology valuation – income approach, cost approach, and market approach -- and various methods in the income approach used most commonly in practice.

In Korea, the most widely used method is the Tech. Factor Method based on DCF (Discounted Cash Flow) Model and Relief from Royalty Model (or Royalty Payment Saved Model).

### 3.2.2 Tech. Factor Model

The Tech. Factor Method estimates the present value of free cash flow during a technology's expected economic life multiplied by Technology Factor (TF) as the technology's market value. The basic formula is as follows:

$$V = \sum_{t=1}^{T} \frac{FCF_t}{(1 + r)^t} \times T.F.$$

Where, *V*: Technology value

*T*: Technology's economic life (remaining useful life)

$FCF_t$: Free cash flow during period *t*

*r*: Discount rate

*T.F.*: Technology factor (Technology's contribution ratio)

In applying the Tech. Factor Method to technology valuation, we must estimate the following four determinants:

First, the economic lifespan of technology must be estimated for the calculation of future income flows produced by the commercialization of the technology.

Second, the free cash flow must be estimated for the calculation of business value generated by the commercialization of the technology.

Third, a proper discount rate must be determined in order to convert future cash flows into present values.

Fourth, the level of contribution of technology must be determined in order to identify how much the subject technology is expected to contribute to the total cash inflows.

### 3.2.3 Relief from Royalty Method

Relief from Royalty Model or Royalty Payments Saved Method estimates a subject technology's value by determining an appropriate royalty rate of the technology based on the transaction (licensing) records of similar technologies. The basic formula for the model is as follows:

$$V = \sum_{t=1}^{T} \frac{S_t \text{ X } R - C_t}{(1 + r)^t}$$

Where, $V$: Technology value

$T$: Technology's economic life (remaining useful life)

$S_t$: Sales during period $t$

$r$: Discount rate

$R$: Reasonable royalty rate

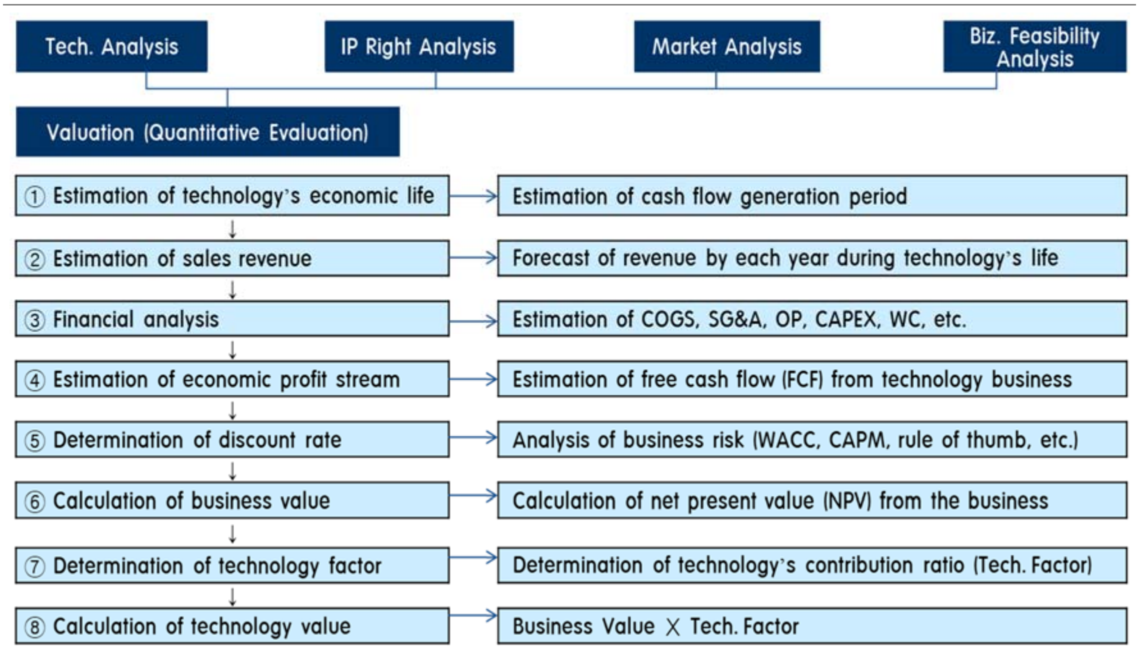
$C_t$: Corporate tax at period $t$

### 3.2.4 Procedure for the Technology Valuation Used in Korea

Technology valuation is conducted by two (2) major steps. The first step is preliminary analyses, and the second step is the quantification of technology value by applying the valuation model as seen in Figure 3-8.

The first step is preliminary analyses, which include the (1) analysis of technological feasibility, (2) analysis of IP rights, (3) analysis of marketability, and (4) analysis of business feasibility. In this step, the first three are about the intrinsic characteristics of the subject technology itself and its product; this means that those three analyses are not relevant to the capabilities and resources of the company that performs commercialization using the technology. The analysis of business feasibility involves considering the commercialization abilities of the company, which bring about different business models or plans when conducting the commercialization of the technology.

[Figure 3-8] Procedure for Technology Valuation




The second step is to quantify the economic value of the technology by applying the valuation model. This step includes the estimation of the technology's economic life (or remaining useful life of technology), estimation of sales revenues, financial analysis (estimation of COGS, SG&A, operating profits, capital expenditure, working capital, etc.), estimation of economic profit stream (or free cash flows), determination of discount rate, calculation of business value, determination of technology factor (or technology's contribution ratio), and calculation of the technology's economic value.

# Part II. Technology Transfer & Diffusion: TTO Point of View

## 1. Technology Transfer & Diffusion: TTO Point of View

### 1.1 Introduction

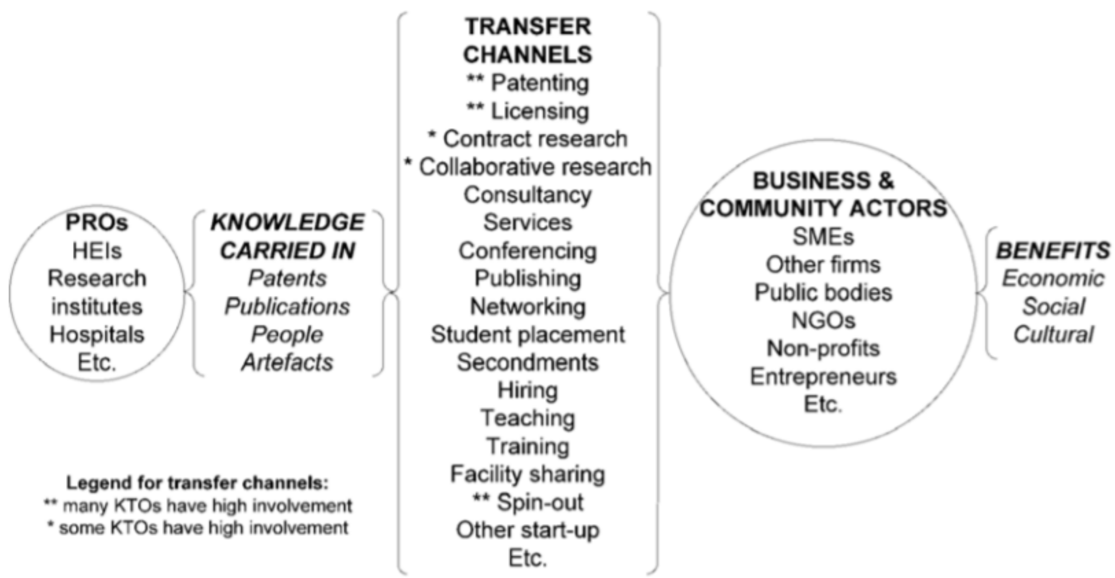
The knowledge-based society, wherein knowledge becomes the source of future competitiveness, has become increasingly interested in creating and utilizing knowledge at the national level. Recently, the paradigm has shifted from the existing closed innovation system to an open innovation one in accordance with the diversification of sources of information and knowledge creation, fusion and combination of technologies, and enlargement of technology development. In order to enable the efficient, systematic acquisition of source technology, advanced companies in Korea and abroad have established cooperative relations with universities, actively collaborating with the industry and academe and transferring technology from universities.

TTO (Technology Transfer Office) has been launched in developed countries as a representative commercialization promotion mechanism of the university creating knowledge and spreading it to the whole world. In the United States, TTOs began to emerge in earnest under the Bayh-Dole Act established in 1980 and provided a full-fledged opportunity for university R & D achievements utilized in the industry. The TTO is an important source of income for major US universities and is responsible for promoting research; it is a very important institution within the university as it is directly under the jurisdiction of the Vice Chancellor.

OECD (2011) defines the TTO and specifies its roles. TTO encompasses different kinds of organizational structures whose common core role is to assist public research organizations (PROs) in managing their intellectual assets in ways that facilitate their transformation into benefits for society. In doing this, the TTO helps bridge the gap between research and innovation. This general role may be broken down into more specific ones including:

- Establishing relationships with firms and community actors
- Generating new funding support from sponsored research or consulting opportunities
- Providing assistance in all areas related to entrepreneurship and intellectual property (IP)
- Facilitating the formation of university-connected companies utilizing PRO's technology (start-up) and/or university people (spin-off) to enhance prospects of further development
- Generating net royalties for the PRO and collaborating partners.

[Figure 3-9] Knowledge transfer from PROs and transfer channels where TTOs may have high involvement




Source: EC Expert Group on Knowledge Transfer Metrics (2009).

According to a report from the Association of University Technology Managers (AUTM), almost all universities in the United States have made technology transfer and commercialization active through TTO organization.

In the United States, AUTM proposes three levels of development stages for university technology transfer organizations: start, stabilization, and maturity.

[Table 3-6] Development stage of university technology transfer organization

| Classification | Beginning Stage (0~5 years) | Stable Stage (5~10 years) | Mature Stage (10~15 years) |
|---|---|---|---|
| Representative | Professor | External expert | Corporate CEO |
| Work process | Contracts, patents, and startups are conducted individually | From the invention to the transfer of technology | Expansion to startup business and industry-academe cooperation in addition to technology transfer |
| Specialist | No expert | PhD, Lawyer | Business professional |
| Standardization of business | No (learn by working) | Progress of work standardization | Completed work standardization |
| Income | Almost none (university support required) | Intermediate stage | Revenues generated |
| Fund | No | Creating a patent investment fund | Establishment of investment fund |
| Organization type | Organization within the university | Organization within the university (independent operation) | Separate organization from university |

Source: ASTM (2006)

In Korea, TTO, which is used internationally, is referred to as Technology Licensing Office (TLO) that fulfills the same purpose and role as TTO.

So far, researchers related to technology transfer have focused on institutional research. Therefore, they emphasize establishment of technology transfer center and training of human resources and insist on expanding the national technology transfer base. Considering the complexity and interconnectivity of the individual themes of technology transfer and commercialization in developed countries, however, it is necessary to analyze the activities and validity of TLOs that are responsible for the micro-level transfer of technologies and commercialization.

In addition to research and development activities, the TLO organization is engaged in a variety of activities such as IPR (Intellectual Property Rights) management, demanding company search and marketing support. As a result, the importance of TLO has been supported by numerous studies in the world. In developing countries, which are in the early stage of technology development, however, it is very difficult to grasp the implementing system reflecting various factors that may occur in the technology transfer and commercialization environment. Especially for the policy executives, great difficulties lie in coping effectively with the technology diffusion problem. Currently, the TLOs of leading universities in Korea seem to be in the stabilization stage, but most of the university TLOs in the early period of industrialization were at the beginning stage. Therefore, when the government supports TLOs, it is necessary to introduce various types of support instead of uniform support methods and to introduce "buffet-type business support methods" wherein universities can select their types considering their capabilities and development stages.

# 2. Korean Experience in Technology Transfer and Diffusion in the Developing Period

## 2.1 Technology Transfer

In particular, in order to understand and analyze the diffusion characteristics of technology transfer in Korea, which has achieved “the miracle of Han River” over a very short period of time, it is necessary to examine the development process of indigenous technology in terms of the technology transfer stage during the developing period.

Having successfully acquired assimilated, and improved mature foreign technologies, Korea aims to repeat the same process with higher-level knowledge in the intermediate technology stage (KimLinsu, 2003). It involves not only activities such as technology transfer and benchmarking yet as well as notable learning through huge investment in research and development (R&D). Many industries in Korea have arrived at this circumstance. If successful, some may eventually accumulate ample indigenous technological capabilities to produce emerging technologies and challenge firms in advanced countries. Innovation is the directing goal in these industries. When a great number of industries attain this stage, the country may be regarded as one of the advanced countries. This oversimplified model provides an accurate explanation of the evolutionary process that took place in the first-tier newly industrializing countries in East Asia, such as Korea and Taiwan. In the 1960s and 1970s, the time when the local technological base was at its initial stage, Korea first acquired and assimilated mature technologies to undertake duplicative imitation of existing foreign products with skilled but cheap labor force. (ChoiKim, 2014) Then the accumulation of technological capability through learning by doing, together with the quality upgrading of the educational system, enabled these countries to undertake creative imitation in the face of rising labor costs and increasing competition from the second-tier newly industrializing countries. [21]

[21] Kim and Choi, 2014

The report by Linsu Kim for UNCTAD-ICTSD (2003) reviews the technology transfer and IPR in the developing period of Korea. Below is the summary.

"Korean firms entered the mature technology stage in the 1960s and 1970s by acquiring, assimilating, and improving generally available mature foreign technology through various mechanisms based on duplicative imitation and evolved into the intermediate technology stage in the 1980s and 1990s through aggressive efforts to strengthen technological capabilities that enabled creative imitation. As the industrialization process unfolded, and Korean firms mastered manufacturing competencies in the duplicative imitation of standardized, low-cost products, they needed to upgrade their indigenous capabilities and manufacture more value-added products in the face of increasing local wages and emerging competitive threats in the labor-intensive production from the second-tier developing countries. This forced Korean firms in the 1980s to shift their emphasis from strategies focusing on labor-intensive mature technologies to those focusing on relatively more knowledge-intensive intermediate technologies across all the sectors.

To tackle challenging new technological tasks, which were beyond their existing capabilities, Korean firms across industrial sectors largely focused their technological efforts on three major areas: foreign technology transfer through formal mechanisms, recruitment of high potential human resources from abroad, and local R&D efforts. In addition, the government invested heavily in upgrading university research and diversifying its research institutions.

Foreign technology transfer played a vital role in building the existing knowledge base of Korean firms. Simple, mature technologies could be easily obtained free of charge through informal mechanisms, because they are readily available in various forms. Even if such technology was patented, foreign patent holders were lenient in controlling such duplicative imitation, as it was no longer useful in sustaining their international competitiveness.

Technologies at the intermediate stage were a lot more complex and difficult to acquire and adopt. To make matters more difficult, foreign patent holders were much more determined to control imitation by developing countries. This is because such technologies continued to play a pivotal role in expanding their international business activities and sustaining their competitiveness. Thus, Korean firms had to resort increasingly to formal technology transfer such as foreign direct investment (FDI) and foreign licensing (FL). This is evident from statistics. FDI increased from $218 million during the period 1967-1971 to $1.76 billion during the period 1982-1986, with royalties associated with FL increasing from $16.3 million to $1.18 billion during the same period. Capital goods imports also increased drastically from $2.5 billion to $50.9 billion during the same period." [22]

Parallel with improved efforts in attaining knowledge-intensive technologies through formal mechanisms and the mobility of high potential human resources at the same time, Korean firms focused on their own R&D activities to build capacity on their bargaining power in technology transfer, fast learning from acquired technology, and mitigate foreign technological dependence. R&D investment has seen a dramatic increase in the past three decades from US$28.6 million in 1971 to US$ 4.7 billion by 1990 and to US$ 12.2 billion by 2000. The Korean economy ranked one of the world's fastest growth rates, but R&D expenditure still rose faster than gross domestic product (GDP). R&D as a percentage of GDP (R&D/GDP) increased from 0.32 to 2.68 percent during the same period, surpassing that of many West European countries.

Consequently, there has been significant structural change in R&D investment. The government played a major role in R&D activities in the early years when the private sector faltered in R&D despite the government's encouragement. More recently, domestic firms have assumed an increasingly large role in the country's R&D efforts in response partly to increasing international competition and partly to a supportive policy environment. While the private sector accounted for only 2 percent of the nation's total R&D expenditure in 1963, this had risen to over 80 percent by 1994, one of the highest among both advanced economies and newly industrializing countries.

The R&D growth rate was the world's highest. The average annual growth rate in R&D expenditure per gross domestic product (GDP) during the period 1981-1991 was 24.2

percent compared to 22.3 percent in Singapore, 15.8 percent in Taiwan, 11.4 percent in Spain, and 7.4 percent in Japan. The average annual growth rate of business R&D per GDP is also the world's highest at 31.6 percent, compared to 23.8 percent in Singapore, 16.5 percent in Taiwan, 14.0 percent in Spain, and 8.8 percent in Japan. Private sector R&D is conducted almost entirely by domestic firms. In addition to intensified in-house R&D, Korean firms began globalizing their R&D activities. LG Electronics, for instance, has developed a network of R&D laboratories in various developed countries. These outposts monitor technological change at the frontier, seek opportunities to develop strategic alliances with local firms, and develop state-of-the-art products through advanced R&D.

The government invested heavily in expanding and deepening university research in the intermediate technology stage. The Korean government and steel corporation POSCO founded three new research-oriented universities specializing in science and technology. The government also enacted the Basic Research Promotion Law in 1989, targeting universities to upgrade their research capabilities. As a result, university research has also expanded substantially. Moreover, the Korean government increased the number of government research institutions (GRIs) from just one, KIST, to over twenty to intensify basic research and serve various industrial needs.

GRIs began to play an important role in strengthening the bargaining power of local enterprises in acquiring increasingly sophisticated foreign technologies. For instance, when Corning Glass refused to transfer optical fiber production technology to Korea in 1977, two large copper cable producers in Korea entered a joint R&D project with a GRI. After 7 years of R&D, the locally developed optical cable was tested successfully on a 35-km route in 1983. Although this local effort eventually came to a halt mainly due to the slow progress in R&D, it nonetheless helped local firms gain bargaining power in acquiring foreign technology from favorable positions.

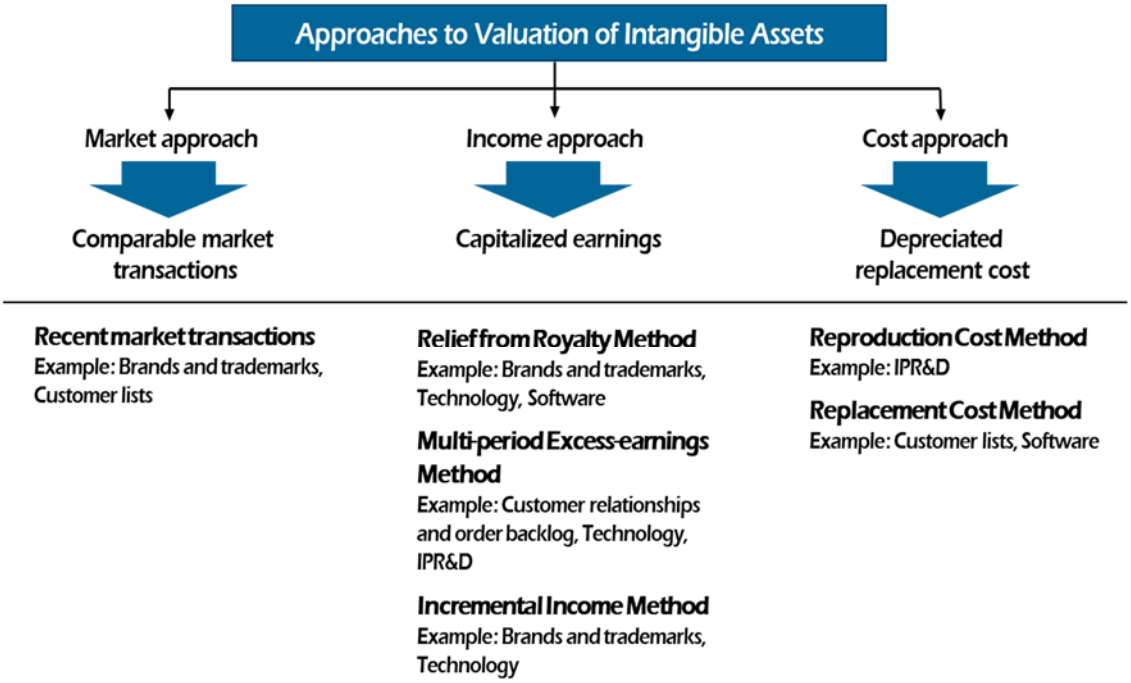
Thus, Korea has rapidly evolved from the mature technology stage -- undertaking duplicative imitation through reverse engineering -- to the intermediate technology stage, undertaking creative imitation through:

- Formal technology transfer
- Recruitment of scientists and engineers with higher potential
- Intensified local R&D activities

In this intermediate technology stage, IPRs became important even for local firms. This can be explained in the patent statistics as shown in Table 1. Patent activity in Korea has increased significantly in the last two decades compared to the first two, increasing a mere 48 percent in the first 14 years (1965-1978) but almost tripling in the next 11 years (1979-1989) and almost tripling again in the next 4 years (1989-1993). Furthermore, the share of Koreans in local patent registration also increased from 11.4 % in 1980 to 69.2 % by 1999, suggesting increased R&D activity.

Korean firms also became active in registering foreign patents. For instance, Korea jumped from 35th place in terms of the number of patents in the US among 36 countries with 5 patents in 1969 to 11th with 538 patents in 1992, representing an average annual growth rate of 43.32 percent. By 1999, Korea had jumped to the 6th spot with 3,679. Samsung Electronics was ranked 4th with 1,545 US patents, after IBM, NEC, and Cannon; thus indicating Korea’s seriousness in securing patent rights at home and abroad.

[Table 3-7] Patents Applied for and Granted in Korea (1981-2000)




Sources: Korea National Statistics Office

Over the decades, a significant number of local firms have managed to grow dynamically from primitive small firms to large modern ones. Most large local pharmaceutical and cosmetic firms and some paper and chemical firms have organically evolved from small firms, imitatively developing their own primitive production processes to become significantly large innovative firms over the decades. For instance, leading local pharmaceutical firms first started as importer/dealers of packaged finished drugs and later entered the drug manufacturing business by packaging imported bulk drugs. They then gradually expanded into more intricate operations, first by formulating imported raw materials and then, through backward integration, producing chemical components. Through this process, they grew in size and in technological capabilities.

## 2.2 Some lessons on the IPR policy in the developing period of technology transfer

The study of Linsu Kim (2003) also offers four important lessons on IPR policy based on the Korean experiences, which should be considered carefully in a developing country.

- Strong IPR protection will hinder rather than facilitate technology transfer and indigenous learning activities in the early stage of industrialization when learning takes place through reverse engineering and duplicative imitation of mature foreign products.
- Countries have accumulated sufficient indigenous capabilities with extensive science and technology infrastructure to undertake creative imitation in the later stage that IPR protection becomes an important element in technology transfer and industrial activities. This suggests that Japan, Korea, and Taiwan could not have achieved their current levels of technological sophistication if strong IPR regimes had been forced on them during the early stage of their industrialization. The same applies to the United States and Western Europe during their industrial revolutions.
- If adequate protection and enforcement of IPRs are genuinely intended to enhance development, policymakers should seriously consider differentiation in terms of the level of economic development and industrial sectors. Otherwise, the “one size fits all” approach is one of the worst approaches for developing countries, particularly for the least developed ones.
- Developing countries should work together to change current trends toward a standardized, all-encompassing multilateral IPR system. They should strive to make IPR policies more favorable to them in the short term. Nonetheless, they should also strengthen their own absorptive capacity for a long-term solution. Local absorptive capacity enables developing countries to identify the relevant technology available elsewhere, strengthen their bargaining power in its transfer to them in more favorable terms, assimilate it quickly once transferred, produce creatively imitative new products around IPRs, and generate their own IPRs.

## 2.3 Technology Transfer Promotion Act in Korea

The national law that has laid the legal foundation for technology transfer and commercialization in Korea is the "Act on Promoting Technology Transfer and Commercialization" in 2000.

The objectives of the law are to promote technology transfer to the private sector and commercialization of technologies developed at public research institutes and to promote smooth transactions, transfer, and commercialization of technologies developed in the private sector (responsible ministry: MOTIE). More intensive actions were intended to fulfill "technology transfer" up to the implementation level.

- Activate the technology transfer path: Fair transfer of development technology from P to P, guarantee of developer's right, obligation of technology market opening by local government, support and incentives
- Secure resources to promote technology transfer: National budget, public funds, financial support targets, and payment regulations for TT
- Establish a technology evaluation system and nurture professional manpower for IT: Technology market expansion, manpower buildup, implementing agency of "Korea Technology Exchange"
- Strengthen government deliberation and coordination for the "Annual Plan for TT Promotion": Private transfer of public technology, promotion of private technology, promotion of investment in technology and commercialization

The Korea government intensively enacted the relevant laws to promote technology transfer as shown in Table 3. Through these promotion laws, Korea could strategically provide comprehensive measures and incentives by focusing on a specific goal. These kinds of government enactment of intensive promotion laws for the specific goals were very unusual and were efficient measures that were hard to find in other DCs. Such experience in Korea demonstrates the importance of proper national intervention in the early stages of industrialization of DC.

The national law that has laid the legal foundation for technology transfer and commercialization in Korea is the "Act on Promoting Technology Transfer and Commercialization." Other relevant laws are the "Special Act on Developing Special R&D Zones," "Special Act on Promoting Venture Companies," and "Act on Promoting Industrial Education and Industry-University Collaboration."

[Table 3-8] Related laws on technology transfer and commercialization in Korea

| Law | Responsible Ministry | Contents |
| --- | --- | --- |
| Special Act on Promoting Special R&D Zones | MSIP | Build the foundation for commercializing the R&D outcome of R&D Special Zones (technology transactions, transfer, and commercialization) (Article 8)<br>Lab-based venture companies (Sections 3~5 of Article 9) |
| Special Act on Promoting Venture Companies | Small and Medium Business Administration | Venture companies specializing in new technologies |
| Act on Promoting Industrial Education and Industry-Academe Collaboration (2003) | Ministry of Education | Industry-University Collaboration Foundation<br>Technology holding companies of universities |

## 3. Birth and Start of University TLO in Korea

As in the case of foreign countries, Korea has various definitions of technology transfer. It has been enacted in the Technology Transfer Promotion Act (2000), wherein technology transfer is defined as the "transfer of technology to another person from technology holders through license or technical guidance of intellectual properties in technology, design and technical information such as patents, utility models, semiconductor layout design, capital goods integrated with technology, which are registered in accordance with the relevant laws such as the Patent Law, etc.

The Technology Transfer Promotion Act enacted in January 2000 requires the establishment of a dedicated technology transfer organization in public research institutes in order to facilitate the transfer of technology developed by public research institutes such as universities to the private sector and commercialization. The research results can be attributed to the relevant institutions.

This is the law that benchmarked the Bayh-Dole Act (1980) of the United States mentioned above and the Act of Japan on the Promotion of Transfer of Research on Technology in Universities, etc. (1998). The Ministry of Commerce, Industry, and Energy established the Korea Technology Exchange (2000) and laid the groundwork for the university technology transfer organization by establishing, deliberating on, and adjusting the technology transfer and commercialization promotion plan annually.

In the science and technology sector, developing and securing source technology and commercializing such technology not only enrich our lives but also become a nation's future source of national competitiveness. Based on this recognition, the Korean government has steadily expanded its R & D investment through university and apprenticeship research and has received a budget of 10 trillion won in 2007 and 18.9 trillion won in 2014. As a result, Korea ranked 4$^{th}$ in the world in terms of the number of patent applications filed by domestic public research institutes in 2014, after China, US, and Japan. Despite the quantitative increase in R & D performance, however, the dormant patent rate is approaching 70%, and the technology transfer rate is only 27.1%. Compared with the major technological competitors, 35.9% in the US and 46.7% in the EU, those are about 50% ~ 90% lower. That is why the transfer of domestic technology and activation of commercialization are urgent. Under such circumstances, the government had been promoting policies and mid- and long-term projects (e.g., Connect Korea) to promote technology transfer commercialization to public research institutes stipulated in the "Technology Transfer and Commercialization Promotion Act" enacted in 2000 (National Technology Bank), and the quality of national R & D.

## 3.1 Operation Supporting Project of SMBA for the Technology Transfer Center of a Private University

In 2001, the Small and Medium Business Administration (SMBA) selected 20 major private universities with massive research funding and SCI papers to support the establishment of the “Technology Transfer Center.” It has started to support the patent applications of the professors by establishing the "Invention Regulations for University Jobs" with a small number of people and has begun to produce the first result of university technology transfer by providing the technology transfer system.

Nonetheless, the technology transfer center as a university-affiliated institution has limited staffing with the contracting staff without full-time staff, not to mention limited linkage with the university's headquarters. It has had many difficulties to overcome. The scale of support provided by the Small and Medium Business Administration, which had promised 5 years of operation support, started at 1 billion won a year, but it was reduced to 4 each year and to 250 million won as of 2005.

Still, the Technology Transfer Center project by SEMA was the first government-supported project that subsidized operational expenses including personnel expenses. It has challenged the concept in the universities by introducing dedicated personnel for university technology transfer and various outputs holding a briefing session on inventive system related to the job, university patent manual publication, university technology transfer briefing session, etc.

This way, private universities support the universities' patents and start the technology transfer business through the support projects of SEMA. On the other hand, the national universities are relatively late in establishing a technology transfer organization. The reason is that universities cannot own professors' research results because there is no legal personality to be a patent applicant. In April 2003, Seoul National University established the "Industry-University Collaboration Foundation" and started to support the patent application and technology transfer of university professors.

In 2004, the Ministry of Commerce, Industry, and Energy and the Seoul National University set up a project to support the construction of a national technology supply platform and started a project of building an integrated technology database by linking national universities such as Seoul National University, Chungnam National University, Chonbuk National University, Kyungpook National University, and Pusan National University. An organization dedicated to technology transfer started to appear at major national universities nationwide.

Although private and national universities set up an organization dedicated to the transfer of technology (2001-2004), the technology transfer organization at this time was a small outsourcing organization with one or two staff members per university. It was a small organization operated by government support projects without independent budget and personnel from University. Universities seemed to follow government policies such as changes in the times and emphasis on new industry-university cooperation systems. Nonetheless, they operated the technology transfer center passively and did not feel confident that university technology would be transferred to the enterprise and commercialized to create new profits.

Unlike the negative attitude of the university, the dedicated staff of the university's TLOs, who have been involved in new field work, have begun to produce results in a pioneering position even in difficult situations. Through its own organization called the "University Technology Transfer Council," the organization modeling and dissemination by learning and networking facilitated the rapid growth of TLOs.

## 3.2 Establishment of Industry-University Collaboration Foundation in Korea

The birth of the "Industrial Education Promotion and Industry-University Collaboration Promotion Act" (which was revised overall by the amendment of the Industrial Education Promotion Act in May 2003 and implemented in September 2003) was somewhat supportive. As a result, the university's industry-academe collaboration has given a boost to change.

This law institutionalized the establishment of an “industry-academe collaboration group” in the form of a special corporation by integrating research contracts, accounting management, intellectual property rights privatization, and technology transfer, which were conducted by the research institutes of existing universities.

By establishing the Industry-University Collaboration Foundation, Korea solved the difficulties of the National University, which had been unable to carry out patent application work because of the lack of legal personality, and made the research administration of the university transparent through enterprise-level accounting management. The government has requested the reorganization of the university so that it can use the indirect income incurred in the management of the research project as promotion fund for revitalizing industry-university cooperation.

At the beginning of the revision of the law, there was resistance from some universities, but the strong will of the Ministry of Education and Human Resources Development had driven the establishment of an industry-academe cooperation group at 330 universities nationwide by 2004. Even the College of Arts and the University of Arts have established an industry-academe collaboration group. Still, the fact that only 96 universities have applied for more than 1 patent application in 2006 indicated somewhat the excessive drive of the government. In addition, the focus on transparent accounting management rather than the intention of strengthening industry-academe collaboration had led to the phenomenon wherein the industry-university collaboration foundation in the early stage of the project started to concentrate excessively on transparent accounting management duties.

There were many cases of the former research institute, technology transfer center, business incubation center, industry-academe support center, etc. Nonetheless, the definition of “industry-academe collaboration” as defined in the promotion law of industry-university collaboration covers all types of industrial research contracts, government research project contracts, patent management, and technology transfer commercialization except the in-school research support project. The impact had also been expanded, looking for stability as a single system of industry-university collaboration group.

“Industry-academe collaboration" refers to the activities of industrial educational institutions and the following activities conducted jointly by the national and local governments, government-funded research institutes, and industries (Article 2, Item 5 of the Act on the Promotion of Industrial Education and Promotion of Industry-University Collaboration):

Development of human resources according to industrial demand and future industrial development

Research and development to create and spread new knowledge and technology

Technology transfer to industries and industrial consultation

The technology transfer organization, which existed as the outsourced center of the university, was united with the administration within the university-industry cooperation foundation at the same time as the start of the foundation and combined with the department performing the research contract of the university. It has secured stable intellectual property rights and shared results in corporate tasks. It has always been a core business in the development strategy and vision of industry-academe collaboration.

Conservative and administration-oriented universities have begun to pay full attention to the university's research results and new sources of funding as a result of the birth of the industry-university collaboration foundation. At the same time, “university-industry collaboration results” were included as a major factor in the evaluation of university professions, and patent registration and technology transfer were major indicators in various national university-sponsored projects such as BK21 project and Nuri project. The professor's perception has changed.

In Korea, most universities are installing and operating TLOs in accordance with the Act on the Promotion of Industrial Education and Promotion of Industry-University Cooperation and the Act on the Transfer of Technology and Promotion of Commercialization.

## 3.3 Development of TLO through “Connect Korea”

In July 2006, the Balanced National Development Committee, Ministry of Education and Human Resources Development, and Ministry of Commerce, Industry, and Energy started the national project dubbed “Connect Korea” and selected 28 leading public research institutes, including leading universities and research institutes, as leading TLOs for technology transfer and commercialization. As a result, TLOs within the selected organizations received rigorous performance evaluations with intensive support from the government for the next five years and played a key role in technological innovation, transfer, and commercialization within the region.

The “TLO Support Program for Universities and Institutes” was a project to support the enhancement of competency such as recruitment of dedicated professional personnel by selecting TLOs that are highly likely to develop among universities and research centers nationwide. This way, universities and research institutes were intended to transfer their own technology to the private sector to raise the royalty income, allocate the royalty income to the researchers and technology transfer contributors from the incentive level, and reinvest in technology transfer and commercialization activities to promote self-reliance. It was also expected that, if this model was successfully settled, it would spread to other universities and research institutes in the area.

The selected public research institutes were 28 universities including 18 universities and 10 research institutes. In terms of balanced regional development, 5 public research institutes (4 universities, 1 research institute) were selected. Since 2006, it had received around 200,000~400,000 US dollars per institution.

The Connect Korea business was only a small budget of 8,000,000 US dollars per year, but the results were quite considerable. High-quality human resources (patent attorneys, PhDs) who are able to use specialized laboratories to discover and manage patents had begun to be put into the university by making it possible to use up to 60%.

Marketing personnel who can fully transfer technology had been reinforced, leading to the expansion of the number of TLO personnel in leading universities to 10 or more. The university that was in the process of patent management tried to produce high-quality technology that can be used in the company along with the recognition change of the professor. The technical PR activities that were only published in the patent booklet and online PR were marketing activities to create an aggressive college-to-enterprise meeting place and to develop a service that customizes the technology that companies need.

The university's technology transfer and commercialization achievement achieved remarkable growth through the "University-leading TLO Support Project (Connect Korea)," which had been implemented since 2006, and the capacity of university TLO had been strengthened.

## 3.4 Organization of TLO

Most TLOs in Korean universities are operated as a subordinate organization of the Industry-University Collaboration Foundation. The Industry-University Collaboration Foundation was established in accordance with the revised "Industrial Education Promotion and Industry-University Cooperation Promotion Act." The Industry-University Collaboration Foundation is a hybrid organization wherein the characteristics as a university subsystem and characteristics as a special corporation with independent legal personality are mixed. In general, the Industry-University Collaboration Foundation is in charge of the overall business related to industry-university collaboration such as research project contracts and research expenses management in addition to technology transfer and commercialization.

The university TLOs were operated under the names of university-industry business team, industry-university cooperation team, industry-university technology transfer team, technology business department, etc.

[Table 3-9] Main tasks of university TLOs

| Classification | Promotion of Research Management | Commercialization of Technology Transfer | Startup Incubation | Others | Sum |
|---|---|---|---|---|---|
| No. of Universities (%) | 53 (46.1) | 55 (47.8) | 0 (0.0) | 7 (6.1) | 115 (100) |

Source: Korean Research Foundation (2010)

In 2009, 115 universities, or 79.3% of the 145 universities that participated in the university-industry collaboration survey, were operating the TLOs they installed. Table 3-9 shows the results of the responses from those 115 TLOs by KRF. Nonetheless, the universities operated as separate departments numbered only 55 (47.8% of the total), answering that they were responsible for technology transfer and commercialization. 46.1% of the universities responded that the university TLOs conducted research management and promotion -- that is, research projects and research expenses management. Most of these universities do not have a separate technology transfer and commercialization department, and it appeared that they were doing the work incidentally.

## 3.5 Manpower of TLO

As of 2009, the average members of university TLOs numbered 4.6, which had remained almost similar since it increased somewhat in 2006. Contract workers are gradually decreasing, and regular workers are increasing.

[Table 3-10] Average number of manpower of university TLO by year (unit: persons)

| Classification | 2005 | 2006 | 2007 | 2008 | 2009 |
|---|---|---|---|---|---|
| Full-time | 2.0 | 2.0 | 2.2 | 2.2 | 3.1 |
| Contract base | 1.9 | 2.4 | 2.2 | 1.9 | 1.1 |
| Others | 0.1 | 0.2 | 0.2 | 0.2 | 0.5 |
| Total employees | 4.0 | 4.6 | 4.6 | 4.4 | 4.6 |

Source: Korean Research Foundation (by year)

Nonetheless, only 44 universities, or 38.3% of the total of 115 universities, had more than 5 dedicated personnel. In particular, 13 universities had no full-time workforce, and 31 universities had one or two full-time workers. Universities with two or fewer people were not able to carry out the technology transfer and commercialization business realistically. Therefore, these universities were likely to perform only the patent application and statistics management tasks of their university professors.

In 2009, TLOs had licensed patent attorneys, technology traders, and corporate technology valuators who numbered 103, or 19.4% of the total. There were 43 universities that employ one person, 37.4%, and 24 universities that use patent attorneys and technology firms. This shows the considerable gap between universities.

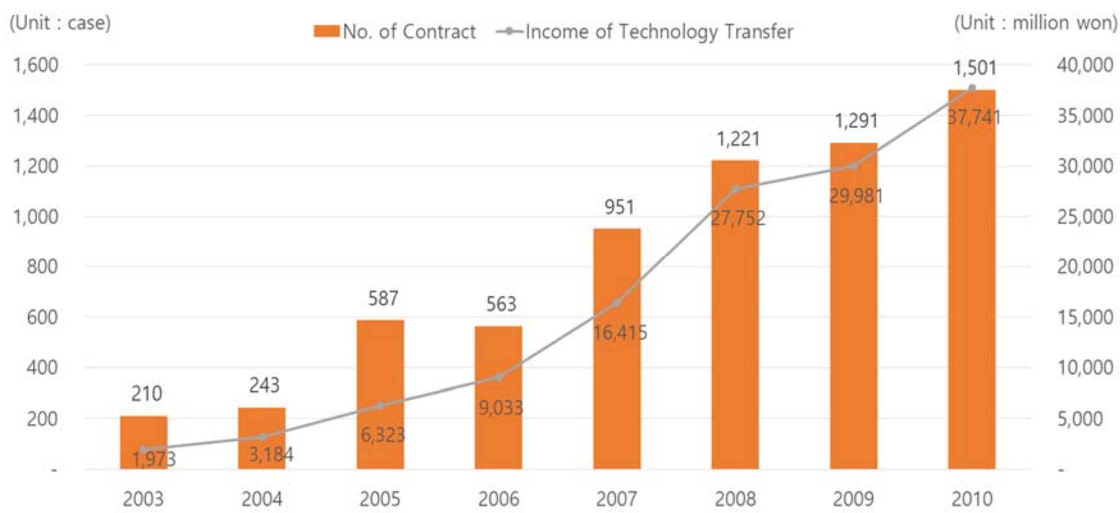
Universities hosted by the "Leading TLO Support Project (Connect Korea)" were relatively well-established universities. These universities utilized 67 patent attorneys, technology traders, and valuation analysts, who accounted for 39% of the total workforce, in 2010. The average workforce was 9.6, which was significantly better than the college average. Nonetheless, there was also a variation in the level of personnel among 18 universities. Hanyang University and Seoul National University employed 3 patent attorneys, but 10 universities were surveyed to have no professional manpower. In the case of Yonsei University, 7 out of 8 staff have professional qualifications, whereas only 1 out of 13 staff in Chungbuk National University have professional qualifications.

# 4. Technology Transfer and Commercialization Status of University TLO in Korea

## 4.1 Technology transfer activities of universities in Korea

The number of contracts for technology transfer and the transfer fee for technology transfer had been steadily increasing since 2003. As shown in [Figure 3-11], the number of technology transfer contracts and technology transfer revenues had grown rapidly for three years from 2006 to 2008 when the "University-leading TLO support project" was launched. It showed a slight downturn in 2009 but had been growing again in 2010.

[Figure 3-10] Status of Technology Transfer Projects in Korea by College Year




The results of the university transfer showed a large deviation between universities. An analysis showed that the number of technology transfer contracts and technology transfer fee from 2008 to 2010 as announced through university alerting (www.academyinfo.go.kr) stood at 8.9%. At least 18 universities account for 48.3% of the technology transfer cases and 70% of the technology transfer fees.

Among the 4-year universities, 96 out of the 202 universities had signed contracts for technology transfer for three years (2008~2010). There were 100 universities with 50% of the total transfer of technology. Among the 202 4-year universities, about 50 were actively carrying out technology transfer activities. There were 46 universities with more than 10 contracts on average for 3 years (2008~2010) and 48 universities with more than 100 million won in technology transfer income.

[Table 3-11] Status of Technology Transfer in Universities for Three Years (2008~2010)

5.1 (unit: case, million won)

| Classification | | 2008 | 2009 | 2010 | 3 year Ave. |
|---|---|---|---|---|---|
| 4-year University Total (202) | Number of Technology Contracts (A) | 1,202 | 1,398 | 1,501 | 1,367 |
| | Income of Technology Transfer (B) | 27,310 | 30,018 | 37,741 | 31,690 |
| TLO Project Host University (18) | Number of Technology Contracts (C) | 608 | 671 | 701 | 660 |
| | Income of Technology Transfer (D) | 20,189 | 20,712 | 24,987 | 21,960 |
| Weight of 18 University | Number of Technology Contracts (C/A) | 50.6% | 48.0% | 46.7% | 48.3% |
| | Income of Technology Transfer (D/B) | 73.9% | 69.0% | 66.2% | 69.3% |

## 4.2 Research Productivity of University and Status of Technical Fees

The average research productivity for 3 years (2008~2010) of the 18 host universities, which were relatively active in technology transfer activities, was found to be 1.3%. That was significantly lower than the 1.9% of Korean public research institutes and 5.3% of US universities (Ministry of Knowledge Economy, etc., 2010). In Hanyang University, which has the highest research productivity, it was less than 3%; only 5 universities recorded more than 2%.

According to the "University-leading TLO support project," the ratio of the current technical fees of the universities is 9.2% of the total technical fee. Most universities had less than 10%. Considering the fact that all universities had 5.3% of the current technical fee rate (as of 2009), the ratio was relatively low compared to the US universities' 76%.

## 4.3 Status of technology transfer income from universities

As a result of analyzing the status of distribution of technology transfer income, 56.0% of the revenues of technology transfer from the university were found to have been used for the researcher 's compensation in 2009, and 18.6% was used as operating cost of the Industry-University Cooperation Foundation. The 2010 data for the university-leading TLO support project, which was based only on 18 leading organizations, showed similar results.

## 4.4 Issues with technology commercialization by universities and GRIs

Korean universities and government-funded research institutes ("GRIs") are still lacking the commercialization capabilities to create new markets and jobs using the knowledge and technology accumulated from various R&D activities. In the case of universities, the sheer volume of papers and patents has increased, but their technology transfer or business incubation performance is quite limited due to the lack of funds. The R&D productivity of GRIs in the science and technology field is less than one third of that of public research institutes in the US.

Despite the steady growth of technology commercialization performance of Korean universities, it still lags behind advanced countries.

When the size of the economy by country is taken into consideration, Korea's technology transfer performance is not on a par with that of advanced countries.

Faculty, students, and technology companies lead business incubation, but their performance is not significant. The R&D productivity of Korean GRIs was estimated to be

2.89% as of 2012, whereas that of the US public research institutes hovered around 10.73% in 2010.

The top six institutions whose technology license fee income is more than 5 billion won account for 76%, whereas the technology license fee income of the bottom 11 institutions is less than 1 billion won, showing a wide gap between leading and lagging groups.

The number of patents owned by GRIs as of 2012 stood at 30,586, with ETRI accounting for about 40%.

Technology incubation by GRIs has been sluggish in general, although their performance over the past three years showed a slight increase.

In the case of lab-based venture companies established by GRIs for the purpose of commercializing technologies, most of them were led by ETRI; the technology incubation performance of other GRIs was minimal.

While GRIs are actively pursuing technology transfer through TLO, they are less active in promoting venture incubation using technology holding companies.

There are obstacles to technology transfer and commercialization by universities and GRIs including lack of expertise and business mindset of organizations in charge of technology commercialization.

There is a wide gap in the size of TLO and technology transfer manpower by institute.

The lack of expertise in TLO often leads to poor performance in technology transfer and commercialization.

Access to funding source such as "Technology Commercialization Special Fund" is limited.

## 5. Task Importance and Priority of TLO of University and GRI in Korea

The study of Kyung-jin Han, et al (2016) examined the importance and priorities of TLO business activities to experts in technology transfer professional organizations and examined the activities that TLO should invest in given the limited time and budget. GRI and University TLO organizations showed similar activity rankings in general but different activity rankings in certain areas. This is because it is necessary to introduce a business process that matches the characteristics of the organization's business and scale, and it is important to establish detailed plans for each business activity. The priorities of organizations dedicated to technology transfer and commercialization were analyzed.

[Table 3-12] Task importance and priority of TLO of University and GRI in Korea (2016)

| Area | Task Activity | Task Importance | Priority |
|---|---|---|---|
| Foundation for Commercialization | Establishment of TLO work process | 0.285 | 1 |
| | Technology transfer system improvement | 0.279 | 2 |
| | IP management system construction | 0.247 | 3 |
| | IP commercialization educational planning | 0.189 | 4 |
| IP Planning & Management | Establishment of IP-R&D patent strategy | 0.325 | 1 |
| | IP application management | 0.260 | 2 |
| | IP portfolio & packaging management | 0.242 | 3 |
| | IP agent management | 0.174 | 4 |
| Exploration of Own Technology | Making SMK | 0.283 | 1 |
| | IP asset inspection | 0.262 | 2 |
| | Grading of technology | 0.245 | 3 |
| | Feasibility analysis & technology valuation | 0.210 | 4 |

| Area | Task Activity | Task Importance | Priority |
|---|---|---|---|
| Technology Marketing | Publicity event | 0.286 | 1 |
| | Operation of technology marketing team | 0.268 | 2 |
| | Customers excavation | 0.262 | 3 |
| | Technology mapping | 0.181 | 4 |
| Negotiation & Contract | Technology transfer negotiation | 0.487 | 1 |
| | Confidential agreements | 0.264 | 2 |
| | Technology introduction | 0.249 | 3 |
| Follow-up | Fee collection and management | 0.601 | 1 |
| | Commercialization | 0.399 | 2 |

## 5.1 Foundation for Commercialization

"Business process" is the most important. University TLO has been the most important activity to establish a systematic business process. This is because it is important for the University to manage the research conducted by each department. The construction and operation of the IP management system are the most important activity of the GRI TLO. In the case of the IP management system, almost all organizations exist as IP systems included in the MIS. Still, there is high demand for a dedicated system that can manage the entire technology transfer process from the creation of IP to technology transfer. Activities to set up the technology transfer business base include the establishment of technology transfer commercialization business process (0.285), improvement of technology transfer commercialization system (regulation) (0.279), and construction and operation of IP management and management system (0.247) 0.189).

## 5.2 IP Planning and Management

"IP-R & D Patent Strategy Development" was the most important activity. At present, the core of technology commercialization is "IP", and it is necessary to hold strong core patents and to hold strong patents to enhance competitiveness of companies and countries. A strong IP creation strategy is needed to identify the need and capability of the company and set the direction of the market environment analysis and commercialization rather than introducing the IP solicitation corporation operation and application / registration screening system. Also, it is necessary to invest in IP R & D activities to create strong IP in order to cope with defensive patents and international patent disputes.

## 5.3 Exploration of Own Technology

Both GRI TLO and University TLO showed that SMK production is the most important. The next highest level of importance in GRI TLO was asset inspection on IP and the business feasibility and technology value evaluation activities were important in University TLO.

## 5.4 Technology Marketing

Both the GRI TLO and the University TLO showed that the most important is the promotion of technical PR events and conferences. The next most important activity was the operation of technology marketing marketer in both institutions.

## 5.5 Negotiation and Contract

Both GRI TLO and University TLO showed that technology transfer negotiations are most important. The next important business activity in GRI TLO was technology introduction before negotiation, while a secret contracting activity was important in university TLO.

## 5.6 Follow Up

Both GRI TLO and University TLO showed that it is important to manage and collect royalties.

Based on the analysis results, it was found that GRI TLO is most important to establish and operate IP management and management system in the field of technology transfer and commercialization. In the case of University TLO, establishment of business process of technology transfer business is the most important.

Both GRI TLO and University TLO showed that IP R & D patent strategy is most important. As the next important task, GRI TLO showed that IP application/registration management was important. In the case of University TLO, IP portfolio and IP packaging activities were important activities.

# CHAPTER 4

## Capacity Development Process

Policy Consultation on the Support for the Establishment of an Effective National System of Technology Transfer (NSTT) in Tunisia

# Chapter 4. Capacity Development Process

## 1. Kick-off Meeting and Field Visit

### 1.1 Introduction

As part of the preparation for the project, from January to February 2019, the STEPI team had a discussion with MHESR to set detailed objectives, target, and project scope for the second-year project. Moreover, there had been follow-up action to develop a concept note on the Korea-Tunisia-Africa Innovation Center (KTAIC) as discussed from the previous year's project in 2018. This agenda was also put on discussion during the visit of the Prime Minister of the Republic of Korea to Tunisia in December 2018. STEPI, MHESR, and ANPR had Skype meetings online to discuss the structure of the main activities prior to this kick-off workshop.

The STEPI team visited Tunisia from April 7 (Sun.) to 13 (Sat.), 2019 for the kick-off meeting and field study. The purpose of the kick-off meeting was to discuss with the Tunisian counterparts to clarify the demand and need for the second-year project through face-to-face meetings and interviews. During the visit, a kick-off workshop was organized with the support of ANPR to invite participants who are directly working in Technology Transfer Organizations (TTOs) at research institutes and universities. Korean experts also shared the relevant Korean experience and its implications for Tunisia. Moreover, after introducing the purpose of the project and sharing the result of the project implemented in 2018, STEPI asked the participants of the workshop to submit a checklist of what is most needed in TTOs to determine the project scope for the 2019 project. The timeline of the project was also discussed to adjust the main schedule of activities of the project. A field study involving various institutions was also included in the program.

[Table 4-1] STEPI Delegation (April 6-12, 2019, including departures and arrivals)

| Name | Organization | Title | Contact Information |
|---|---|---|---|
| Eun-Joo Kim | STEPI | Head of the Capacity Development Training Team | ejkim@stepi.re.kr |
| Jiyong An | STEPI | Researcher | Freesoul13@stepi.re.kr |
| Hyun Woo Park | Korea Institute of Science and Technology Information (KISTI) | Senior Research Fellow | hpark@kisti.re.kr |

■ **Program**

| Date | Departure | Transportation | Arrival | Visiting Institute | Activities |
|---|---|---|---|---|---|
| April 6 | Incheon (23:50) | Airplane | Dubai (05:00) | Departure, Transit in Dubai | Incheon Airport → UAE Dubai Airport |
| April 7 | Dubai (09:00) | Airplane | Tunis (12:25) | Arrival in Tunis | UAE Dubai Airport → Tunis |
| April 7 | Prepare for Kick-off Workshop | | | | |
| April 8 | 10:00~17:00 | | | City of Sciences | Kick off Workshop |
| April 9 | 10:00~17:00 | | | ANPR, MHESR | Prepare for the Study Visit Program to Korea |
| April 10 | 10:00~17:00 | | | ENIT, UTM, KOICA Office | Visit to the Relevant Institute |
| April 11 | 10:00~17:00 | | | Korean Embassy MHESR | Korean Embassy |
| April 12 | 08:00~10:00 | | | MHESR | |
| April 12 | Tunis (14:05) | Airplane | Dubai (23:10) | Arrival In Dubai | Tunis to Dubai |
| April 13 | Dubai (03:40) | Airplane | Incheon (16:55) | Arrival in Incheon | Arrival in Incheon Airport |

## 1.2 Kick-off Workshop

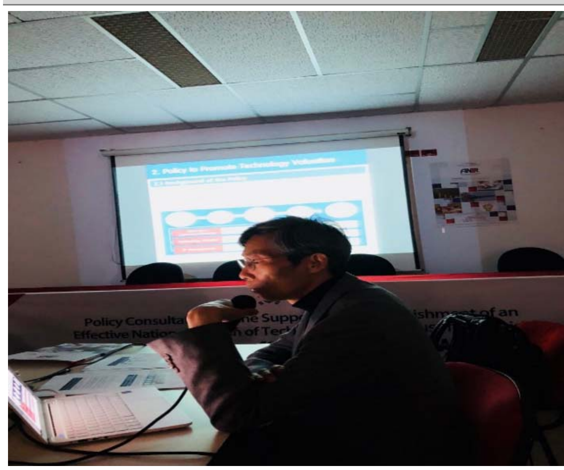
The kick-off workshop was held on April 8 (Mon) at the City of Sciences in Tunis. ANPR, our Tunisian partner institute, invited about 30 participants who work in TTOs. The workshop started with the welcome remarks by Dr. Malek Kochlef, the new Director General of the International Cooperation Division in MHESR, and Dr. Chedley Abdelly, the Director of ANPR in Tunis. Ms. Kim, the project manager of STEPI, gave a debriefing for the project and its results from last year (2018) and shared the objectives of the project for 2019 with the opening remarks. Dr. Park, one of the Korean experts involved in this project since last year, gave a presentation on technology valuation to share the Korean case and its implication for Tunisia. The ANPR and MHESR counterpart provided a presentation on the i) overview of Intellectual Property and Technology Transfer in Tunisia.

About 30 participants who participated in this workshop held intense discussions and Q&As on the technology transfer system and incentives for technology transfer implementation, which is based on the Korean case shared from the presentation. It was strongly recommended that the invited participants who are working in TTOs submit the checklist of the project 2019 to diagnose what is most needed for TTOs to determine the project scope for the project in 2019, which would be reviewed by STEPI and Korean experts. Moreover, applications for the Study Visit Program to Korea were encouraged among the relevant institutes and participants so that they could apply through MHESR and ANPR.

STEPI, ANPR, and MHESR co-examined the project components, schedule, activities, and scope with the Tunisian partners and partner institutions for the second year. Then, expectations for the project were discussed in order to identify areas of interest by STEPI in order to build consensus between STEPI, MHESR, and ANPR on the expected results and outcome.

## Photos

| Opening of the Kick-off Workshop | Group Photo |
|---|---|
|   |   |

| Presentation by Dr. Park | Kick-off Workshop |
|---|---|
|  |  |

## ■ List of Participants

| No. | Name | Organization | Position | Email |
|---|---|---|---|---|
| 1 | Nednia Amri | TTO University of Manouba | Head of TTO and director of scientific international cooperation and with firms | nediaamricyahoo.fr |
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| 3 | Selim M'rad | Monastir University | Researcher- Teacher- TTO-BUTT | selim.mrad@gnet.tn |
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| 6 | Ben Slima Neila | MHESR DGVR | Deputy Director | neila.benslima@gmail.com |
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| 10 | Borhen Louhichi | University of Sousse | Associate Professor | louhichi.ca@gmail.com |
| 11 | Leila TRIKI | INRAP | Head of TTO | inrap.contact@gmail.com |
| 12 | Sinda Zarrouk | Institut Pasteur de Tunis | Head of technological platform | sinda.zarrouk@pasteur.tn |

| No. | Name | Organization | Position | Email |
|---|---|---|---|---|
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| 15 | Khaoula Nasr | National Center for Nuclear Sciences and Technologies of Tunisia | Chemical engineer, PhD student | nasr.khawla@gmail.com |
| 16 | Gaies Emna | Centre National de Pharmacovigilance | Associate Professor | gaiesemna@yahoo.fr |
| 17 | Akila KHLIFI | Center of Research on Microelectronics and Nanotechnology in Sousse (CRMN) | Assistant Professor | khlifi_akila@yahoo.fr |
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| 19 | Somai Myriam | University of Tunis | Staff Administratif | khamrouna@gmail.com |
| 20 | Ismail Haddad | Ministry of Higher Education and Scientific Research | Director of Partnership | ism.haddad@gmail.com |
| 21 | Mortadha GRAA` | University of Sousse | Head of TTO | graa_mortadha@yahoo.fr |
| 22 | Zied Kbaier | Research and Technology Center of Energy, CRTEn | Deputy Director, Head of Tech Transfer and EU Projects Office | kbaier.zied@gmail.com |

| No. | Name | Organization | Position | Email |
|---|---|---|---|---|
| 23 | Lobna Mansouri | Research Center for Water Researches and Technologies, CERTE | Dr., Engineer in Applied Chemistry | lobna.mansouri@certe.rnrt.tn |
| 24 | Karim Azzabi | ANPR | Judicial Affairs and IP | karimazbi@yahoo.fr |
| 25 | Ines ZIDI | University Tunis El Manar-Higher Institute of Medical Technologies of Tunis | Associate Professor in Immunology, Member of TTO Tunis El Manar University. Innovation Creation Center, Winner of Besrour Canadian Prize 2017, Winner of AUF Prize 2019 | ines.zidi@istmt.utm.tn |
| 26 | Souad Boussaid | ANPR | Chief Engineer | |
| 27 | Abdelly Chedly | ANPR | Director | |
| 28 | Malek | MHESR | Director General of International Cooperation | |
| 29 | Anis Roussi | MHESR | International Cooperation | |
| 30 | Hyun Woo Park | KISTI | Research Fellow | |
| 31 | Eunjoo Kim | STEPI | Head of the Global Training Team | |
| 32 | Jiyong An | STEPI | Researcher | |

## Database of focal points for TTOs

| N° | Prénom | Nom | Institution | Fonction | Email |
|---|---|---|---|---|---|
| 1 | Mortadha | GRAA | Université de Sousse | Dr-Ingénieur | graa_mortadha@yahoo.fr |
| 2 | Nedia | Amri | université de Manouba | Directrice des affaires académiques et de partenariat scientifique | nediaamri@yahoo.fr |
| 3 | Saoussen | Krichen | Université Tunis | vice présidente | krichen_s@yahoo.fr |
| 4 | Imen | Bhibeh | Université Tunis Manar | Admin/Conseiller de l'enseignement supérieur et de la recherche scientifique | imen.bhibah@gmail.com |
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| 10 | Mohamed | Belhadj | Université de Sfax | Ingénieur général et sous Directeur | mohamed.belhaj1@gmail.com |
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| 12 | Ismail | Haddad | DGET | Directeur | ism.haddad@gmail.com |
| 13 | Leila | Triki | INRAP | Ing.en chef | Leila.triki@aol.com |
| 14 | Oussema | Ben Fadhl | IPT | chargée de TT | oussama.benfadhel@pasteur.tn |
| 15 | Zied Kbaeir | Kbaeir | CRTEn | Chef Service | kbaier.zied@gmail.com |
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| 17 | Jamel | Ben REBAH | IRESA | Sous Directeur de la liaison de la recherche et de la vulgarisation | jamel.benrebah@iresa.agrinet.tn |
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| 21 | Zohra | AZZOUZ | CNSTN | Sous Directrice | z.berriche@cnstn.mrt.tn |
| 22 | Hend | Ben hajji | CERT | Docteur | hend.benhiji@cert.mincom.tn |
| 23 | Anis | Chaouachi | CNRSM | Ingenieur en chef | anis.chaouachi@gmail.com |
| 24 | NA | NA | CBBC | NA | NA |
| 25 | NA | NA | CERTE | NA | NA |

## 1.3 Field Study

The STEPI team visited the relevant institutions and key STI stakeholders for interviews and meetings to gather information, data, and statistics to understand the context of the Tunisian economy, industry, TTS policy development, law and policy framework, etc. On top of this, visiting the TTOs and its focal points helped increase the understanding of the Korean team and allow them to experience what is most needed for this year’s project. The team visited En National Engineering School of Tunis (ENIT) and University of Tunis El Manor (UTM) and met its relevant stakeholders as well as its students. They also had a meeting with the Korean Embassy based in Tunis and KOICA office to share the discussed activities and expected dates of activities for 2019.

The research team discussed and diagnosed the current state of TTOs in the framework of the Tunisian National Technology Transfer System and analyzed related issues. The relevant institutions and key innovation actors including the ministries, public organizations, academe, private sector entities, and research institutes were identified by MHESR.

A closed meeting was held on April 11 (Thurs.) to share the results of the field study and its planned activities, date, and venue for 2019. In addition, based on the field study results, the project scope, direction, activities, and expected outcome were clarified through discussion with ANPR and MHESR.

The form Checklist of Status of TTOs in Tunisia was circulated to the participants as well as related institutes through ANPR. Those checklists were collected by STEPI after the business trip to Tunis, and this helped STEPI outline the curricula for the study visit program for the Tunisian Delegation.

## ■ Photos

| Meeting in ENIT | ENIT |
|---|---|
|  |  |
| **UTM Group Photo** | **Meeting in UTM** |
|  |  |
| **UTM Group Photo** | **Meeting in UTM** |
|  |  |

## ■ Sample Checklist of the Status of TTOs in Tunisia

### [Checklist of the Status of TTOs in Tunisia]

STEPI and MHESR are carrying out a consultancy project in order to support improvement of national technology transfer system in Tunisia for 2018-2019. This year, the project intends to investigate the status of TTOs in Tunisia and provide some trainings and workshops for TT capacity building of TTOs. Your kind cooperation in completing this checklist would be so much helpful for the Korean experts to have a better understanding of TTOs in Tunisia. You are kindly requested to complete and send the checklist to Ms. Eunjoo Kim (ejkim@stepi.re.kr) and Ms. Jiyong An (freesoul13@stepi.re.kr) by no later than 13 April, 2019. Any of your opinion and suggestions would be very much appreciated.

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*Note : TTO is an an organization that performs tasks such as technology transfer, business incubation, patent application and management, and it means the university's technology commercialization center, industry-university cooperation team, technology commercialization center of public research institute, research result diffusion team and IPR management team.*

---

**1. Organization and personnel**

**o Department in charge**

- Internal execution: Department in charge (Team) or independent corporate

Ex.) In Korea, the RDA (Rural Development Agency) established a 'foundation for the commercialization of agricultural technology' in order to support technology transfer and commercialization.

- Outsourcing: Outsourcing of external organization which has specialty

| |
|---|
| Internal execution : Team in the CRMN (Centre of Research on Microelectronics and Nanotechnology) |

**o Personnel (TT managers and staff)**

- FTE (Full time equivalent) based

| |
|---|
| - A team composed of 5 persons.<br>(managing the TTO : additional task to their basic work) |

**o Expertise of TT managers and staff**

- Academic backgrounds, degree and careers, etc.
- Qualified professionals (lawyer, attorney, accountant, etc.)

| |
|---|
| 1 responsible: Assistant Professor<br>2 researchers (1 Assistant professor and 1 Phd student)<br>1 Engineer<br>1 administrative |

**o Operational budget**

- Budget (annual)
- Status of operational budget input (expenses) (personnel expenses, IP management, technology transfer and tech. licensing and commercialization, etc)

| |
|---|
| - 7000 TND<br>- Trainings, Researchers-Industrials meetings, communication |

**2. Main businesses**

**o Technology and IP management**

- establishing technology DB (status of yearly registration)
- # of holding technologies/invention reporting (yearly)
- Status of analysis of possibilities of technology commercialization

| |
|---|
| The CRMN (Centre of Research on Microelectronics and Nanotechnology) is a new research center in phase of installation (equipment purchase, implementation of platerforms, recruitment, etc.). Thus, the technology transfer office is recent, less than a year, so the main activity, is listing expertises, preparing communication (roll-up, brochures presenting the center, the researchers, the office), trainings to introduce Technology Transfer, intellectual property, marketing of research results |

o **Technology valuation**

- How it is done: internal execution / external commissioning,
- Evaluation method: qualitative (technical grade, etc.) and quantitative (technology valuation, etc)
- Number of technology evaluations performed (by year)

| |
|---|
| |

o **Technology Marketing Activities**

- Technology promotional materials (Marketing Collaterals such as brochures, on-line newsletters, etc..) published
- Participation in technomarts, exhibitions and seminars, identification of technology users

| |
|---|
| _ Roll-up prepared, brochure, booklet (list experts, platforms, projects,..)<br>- Participation to trainings, info days, workshops |

o **Technology Transfer (Licensing) Performance**

- Technology transfer (licensing) Contract performance: Transfer of technology, licensing (grant, fee, etc.), others

| |
|---|
| - |

o **Technology start-up performance**

- Number of technology start-ups (internal, external, etc.)

| |
|---|
| - |

**3. System and Policy**

**q Support for technology transfer and commercialization (TLO) ,legal and institutional policy status**

- Relevant laws and regulations and policy

| |
|---|
| - |

**q Technology Transfer Performance Compensation System**

- Personal evaluation, financial compensation etc.
- Technology transfer profit sharing between researchers, dedicated organization and contributors (intermediaries)

| |
|---|
| - |

**q Education and trainings for technology transfer and commercialization**

- Type of education and trainings (contents, targeted group, etc)
- Number of education and trainings

| |
|---|
| Trainings organized:<br>- Introduction to technology transfer (targeted group : reserachs, PhD students) |

**q Barriers**

- Organizational, personnel, legal and regulatory frameworks and policy aspects
- Others

| |
|---|
| - Duplication of structures and excessive bureaucracy<br>- Need to recruit a full time TTO manager, with a specific background, in addition to the team of researchers and administratives staff who assist him<br>- Need of external leading experts with expertise in law, intellectual property, marketing, to assist researchers |

*Thanks for your kind cooperation.*

## ■ Meeting Minutes

| Minutes of the Korea/Tunisia Meeting<br>Debriefing and Next step of the project |
|---|
| ***Meeting date:*** 11/04/2019 in MESRS |
| ***Committee:***<br>Mr. Malek Kochlef, Mm. Samia Charfi Kaddour, Mm. Rim Said, Mm. Awatef Soltani, Mm. Saida, Mm. Maha Hammami, Mr. Anis Rouissi Mm. Souad Boussaid,<br>Mm. Eunjoo Kim, Mm. Jiyong An |
| ***Agenda:***<br>- Debriefing and synthesis of the activities done in Tunisia in collaboration with STEPI on April 8-10<br>- Next steps for 2019<br>- Another possible cooperation |
| ***Opening Meeting***<br>*Mr. Malek Kochlef* thanked all the participants for their presence especially the Korean delegation for the long-standing assistance between Tunisia and Korea. About 100 participants have benefited from the STI training. Mr. Kochlef kindly asked Mm. Eunjoo Kim to do a debriefing for all the activities done from April 08 to 10 and what are her points of view about all this. He also asked her to give us her plan for the next step for 2019 and the possible practical cooperation in the future based on the previous action plan and decisions.<br>*Mm. Kim* was very satisfied with the workshop done on April 08 in the City of Sciences with about 30 participants between researchers and TTOs. Pr. Park has given an interesting presentation on Technology Valuation, and Souad also presented an overview of TTOs as well. Their visit to ENIT and UTM was very impressive as well. The Korean delegation decided to support the ENIT innovation master's degree when they go to Korea for their study visit. They ask the director of the Master to do a concept note and to send it to them to arrange this activity. Mm. Kim presented the next step, which will be the study visit in Korea from June 10 to 14, 2019. It is a training of trainers for Technology Transfer managers (5 participants and 2 participants from MESRS and from ANPR). It will be a practical training for 5 days. Thus, to prepare the content of this training, they should know through checklist documents the status of the TTOs diffused by ANPR and the training needs from universities and research center. In addition, Mm. Kim demanded an update TTO report from ANPR. Based on this study visit and report, a dissemination workshop will be held in Tunisia on the 10th of July for about 100 participants at Movempick Hotel. This workshop will be co-hosted with ANPR and the Korean Embassy. By the same time in July, a high-level Korean/Tunisian policy dialogue will be held in MESRS with the president of STEPI as well as the Ministry to discuss innovative policy aspects. In fact, the objective is to make practical decisions on Tech. Transfer and innovation in Tunisia.<br>***Discussion:***<br>*Mr. Malek Kochlef* wishes that the dissemination workshop have strong content and good experts. Thus, for excellent cooperation, we need good visibility of the dissemination of Korean experience. This seminar should be a good opportunity for networking and practical achievements (Website, Korean/Tunisian Technology Transfer platform, etc.).<br>Concerning the policy dialogue, he aims to know in advance the topic of the discussion and the output of this high-level meeting.<br>*Mm. Maha Hammami* added as well that this dissemination workshop will be an opportunity to announce the Korean Tunisian Innovation (KTI) center, including the general and specific objectives as well as the activities.<br>*Mm. Kim* supported this suggestion because the embassy encourages similar initiatives in ICT and other sector, so we can add the Technology Transfer, which is a horizontal domain, and we need more efforts to push partners to do this center.<br>*Mr. Malek Kochlef* appreciated such suggestion very much but added that we should prepare by the end of April a concept note to clarify all aspects. |
| **Decisions**:<br>- Trainer of Trainers from April 10 to 14, 2019 in Korea for 7 participants, and MESRS will do the selection<br>- Update TTO report (ANPR)<br>- Dissemination workshop on July 10, 2019<br>- High-level policy dialogue (STEPI/MESRS) |

■ **STEPI-IICC Facebook**

Stepi Iicc

4월 9일 오후 7:38 ·

[2019 K-Innovation _ Korea-Tunisia Improvement of National System of Technology Transfer (NSTT) in Tunisia Kick off Workshop for 2019 and Study Visits]

On April 8, STEPI, MHESR and ANPR hosted a Kick-off workshop of the Policy Consultation on the Support to the Establishment of an Effective National System of Technology Transfer (NSTT) for 2019 in the City of Sciences in Tunis, Tunisia. For this kick-off workshop and study visits, STEPI team visited Tunis, Tunisia for April 6... 더 보기

## 2. Study Visit Program for the Tunisian Delegation

The study visit program (Training of Trainers Program) of the Tunisian delegation was hosted by STEPI and was operated from June 9 to 15, 2019. The purpose of this program is to support Technology Transfer and Commercialization in Tunisia. Especially in the Policy Capacity Building for Technology Transfer Organizations such as Universities and Research Institutes in Tunisia, Korean experiences in building strategies and operations for TTOs, Technology Transfer and Vitalization, and legal and institutional frameworks and human resource development policy were shared. This program has been an excellent opportunity for the Tunisian Trainers to experience Korean success stories as basis for developing strategies and operations for TTOs to promote Technology Transfer and exploitation of research for commercialization.

This program will provide Tunisian trainers in technology transfer organizations and policymakers with a good opportunity to use Korean experiences as reference for establishing policies aimed at promoting technology transfer and commercialization in TTOs in Tunisia.

For this program, experts and scholars of the field were invited as lecturers of each session. There were lectures on technology transfer commercialization such as technology transfer, IP management, technology transfer strategy and operation methods, technology transfer valuation and evaluation cases, and exercises on technology transfer evaluation.

The program also included visiting to Korean institutes such as the Korea Institute of Science and Technology (KIST) History Hall, Korea University Industry-Academe Cooperation Group, and ancient major startup-related facilities (Maker's Lab, Innovation Café, etc.). Major institutions such as SK Tium were also visited.

Satisfaction surveys were conducted through lecture evaluations and program evaluation surveys. On average, we get satisfaction scores of 4.5 or higher out of 5.

## List of Participants:

| | Name | Organization | Position | E-mail |
|---|---|---|---|---|
| 1 | Besma Ben Mesbah | MoHESR/ General Directorate of Research Vitalization | Deputy Director of Technoparks | benmesbeh.besma@gmail.com |
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| 3 | Ines Zidi | University of Tunis El Manar (UTM) | Associate Professor | ines.zidi@istmt.utm.tn |
| 4 | Ismail Haddad | MHESR General Directorate of Technological Studies | Director of Partnership | ism.haddad@gmail.com |
| 5 | Selim Mrad | University of Monastir | Teacher and Researcher | selim.mrad@gnet.tn |
| 6 | Borhene Louhichi | University of Sousse | Associate Professor and Member of TTO | borhen.louhichi@etsmtl.ca |

## ■ Workshop Materials

2019 STI Capacity Development Workshop (Policy Consultation to Support the Establishment of National Technology Transfer System in Tunisia)

<워크숍 자료집>


## ■ Visiting Institutions

① SK Tium

SK Telecom takes you on the journey to that future. We opened the first-generation analogue mobile phone era and led the world in commercializing the CDMA and HSDPA technologies. As the first Korean mobile carrier that commercialized LTE and the first in the world to commercialize LTE-A, we have led the world's telecommunication industry.

Today, SK Telecom is connecting the world with the future. It is with great pleasure that we at Tium invite you to a firsthand experience of SK Telecom's vision to connect new value as well as the future world through passion and innovation.

② Korea Institute of Science and Technology (KIST)

The Korea Institute of Science and Technology (KIST) is a science and technology institute working to realize a better quality of life to all people by preparing for the future. KIST was founded as the first science and technology research institute of Korea in 1966; since then, it has continually played a leading role in national development. Many government-funded research institutes have been established after KIST; as such, KIST has set the standards to become the national think tank of science and technology.

In addition, KIST is concentrating its research capability on making sure that its research outcomes will translate into greater competitiveness for Korean businesses. We believe that science and technology will become the foundation for firmly establishing a creative economy. Since KIST was born out of international development aids, we believe it is timely to return the favor by disseminating the KIST model, a science-based official development assistance (ODA); thus contributing to the betterment of the global community while enhancing Korea's status in the world.

http://eng.kist.re.kr/kist_eng/main/

③ Korea University Research and Business Foundation

KU Research and Business Foundation will continue to strengthen its research productivity to lay the foundation for KU to become a world-class research-oriented university. KU has renewed its commitment to fulfilling its responsibilities as a global leader in the 21st century. In line with KU's commitment, we at KU Research and Business Foundation will dramatically enhance our research competitiveness and create new research areas by strengthening KU's planning capabilities and industry-academe cooperation and by expanding the scope of technology commercialization.

According to the 2018 QS World University Rankings, KU ranked 86th in the world and ranked 1st among the private universities in Korea, marking its highest ranking ever in history. Since undergoing evaluation by the World University Rankings in 2004, KU became the first private university in Korea to break into the Top 80. We will continue to create a researcher-centered support system and improve other relevant systems so that KU, which has already been proven to be the top research and education university in Korea, can compete with the best research universities in the world. We will concentrate our efforts on facilitating technology commercialization, technology venture programs, and international technology cooperation.

We are committed to KU's future-oriented academic development. Your support from within as well as outside the university will be deeply appreciated.

https://rms.korea.ac.kr/nrpt/openHome/index.do

### ■ Program

| | | |
|---|---|---|
| **Arrival: Sunday, 09 June 2019** | | |
| 11:20 | Arrival in Seoul, Incheon airport | Airport pickup |
| 14:00~ | Check-in at the hotel | Shilla Stay Hotel |
| 16:00~18:15 | Cultural Program | |
| 18:15~19:30 | Dinner | |
| 19:30~21:00 | Cultural Program, return to hotel | Shilla Stay Hotel |
| **DAY 1: Monday, 10 June 2019** | | |
| 09:00-10:00 | Program Orientation | |
| 10:00-12:00 | IPR (intellectual property right) Management Process for technical transfer<br>-Kyungmin Chung, Representative Patent Attorney of the Dowool IP Group | |
| 12:00-13:30 | Lunch | |
| 13:30-15:30 | Strategic Management of Science and Technology Park and Promotion of University-Industry-Research Inst. Network<br>-Dr. Deok Soon Yim, STEPI | |
| 15:30-16:30 | Break | |

| | | |
|---|---|---|
| 16:30-17:00 | Move to SK Tium | |
| 17:00-18:30 | SK Tium | |
| 18:30-19:30 | Dinner | |
| **DAY 2: Tuesday, 11 June 2019** | | |
| 10:00-12:00 | Korea Institute of Science and Technology (KIST)<br>-Introduction of the Korea Institute of Science and Technology<br>-Meeting with the Technology Transfer Team | Seongbuk-gu, Seoul |
| 12:00-14:30 | Lunch & Move to Korea University | |
| 14:30-16:00 | Korea University Research and Business Foundation<br>-Introduction of Korea University<br>- Meeting with Prof. Gyeongsoo Shim | Gangnam-gu, Seoul |
| 18:00- | Dinner | |
| **DAY 3: Wednesday, 12 June 2019** | | |
| 11:00-13:30 | Strategies and Operations for early Technology Transfer organization<br>- Prof. Myung do Oh (Seoul University) | |
| 13:30-15:00 | Lunch & Discussion with Prof. Myungdo Oh | |
| 15:00-16:00 | Break | |
| 16:00-18:00 | Technology Commercialization “How it works to catch up”<br>- Dr. Jong Heung Park, ETRI | |
| 18:00-20:00 | Dinner | |
| **DAY 4: Thursday, 13 June 2019** | | |
| 10:00-13:00 | Technology Commercialization and Valuation - Policy and Practice –<br>- Dr. Hyunwoo Park, KISTI | |
| 13:00-15:00 | Lunch | |
| 15:00-18:00 | Discussion and Writing reports by Tunisian Delegations | |
| **DAY 5: Friday, 14 June 2019** | | |

| | | |
|---|---|---|
| 10:00-12:00 | STAR-Value System & how it works<br>- Dr. Jongtaik Lee, KISTI | |
| 12:00-13:30 | Lunch | |
| 13:30-15:30 | STAR-Value System & how it works<br>- Dr. Jongtaik Lee, KISTI | |
| **Departure: Saturday, 15 June 2019** | | |
| 11:00~ | Hotel check-out and move to the airport | |

## ■ Summary of the Lectures

**Title: IPR management for patent transfer**

- Briefly discuss what IPR and IPR valuation are.
- First, compare between IPR and Technology and find out what is so special about IPR. IPR can be traded independently, but its value is highly dependent on technology and business. IPR can be treated as an independent property as a warrant and when licensing and calculating compensation.
- There are various purposes for IPR valuation - trading, finance, contribution, strategy, litigation, and tax issue. The valuation process and method can be adjusted based on the purpose of the valuation. For IPR valuation, there are three known methods – market approach, revenue approach, and cost approach. The method can be selected based on many factors – purpose, technology step, when to use, and kind of IPR.
- 4 factors concerning IPR valuation (Technical, Right, Marketability, Business). Among these 4 factors I as a patent attorney concentrate on Right -- stability, scope of right, and matching.
- Discuss what IPR management is and why do we need it. (1) Offensive interests – protection, royalties (2) Defensive interests – Cross-license (3) Marketing means – advertisement (4) Reduction of R&D cost.
- How to do IPR management – patent portfolio, IP trading, Holding IP analysis, IP marketing strategy, IP risk analysis, Infringe monitoring

**Title: Strategic Management of Science and Technology Park and Promotion of University Industry-Research Inst. Network**

**Title: Commercialization of Innovative Technologies in KIST**

- Brief introduction and KIST's major achievements
- KIST at a glance, research areas, and international cooperation of KIST
- Status of technology commercialization of KIST and TLO leading the Commercialization of Innovative Technologies in KIST
- Introduction of KIST R&BD Program (KIST bridge program)
- Introduction of KIST R&BD Program for Startups (KIST steppingstone program)
- Case studies of technology commercialization in Korea

**Title: University IP Commercialization and Technology Transfer Market**

The mission of a university mirrors neither the mission of a corporation nor its metrics for measuring success or its value system. The mission of a business is to maximize economic return to shareholders. The university's mission is to create knowledge through research and development and to disseminate such IP assets mostly through education for the public good.

Based on those topics, this presentation seeks to provide the key message of how a university's IP assets can be more commercialized and how to evaluate the value of each of the intellectual assets through the comparison between the US and South Korea IP market's policy and university technology transfer process.

**Title: Technology Transfer and Diffusion – TTO Point of View**

- Importance of Technology Transfer and Diffusion
- Understanding Technology Diffusion
- Key Policies of Technology Diffusion Promotion

- Knowledge Transfer from PROs and Transfer Channels
- Stakeholders Involved in the Technology Transfer
- Korean Experience: Promotion Act, Promotion Policy, Industry-University Collaboration Foundation, TLO
- Challenges and Issues for the Development of University TLO
- Suggestions for Tunisia's TTO

**Title: Technology Commercialization "How it works to catch up"**

ETRI, which has contributed to the economic growth of Korea by developing ICT technologies for 43 years, will be introduced briefly. Innovation and catch-up process will follow to explain the technology transfer, R&D, R&BD, startup, and technology commercialization strategy. In addition, cases for global commercialization will be introduced conceptually.

**Title: Technology Valuation in Korea: Policy and Practice**

- The lecture intends to review the roles of technology valuation in technology transfer and commercialization in order to maximize the efficiency in utilizing the R&D results of PROs and private firms in relation to the policy implementation to promote technology transfer and commercialization at the national level and to explain the characteristics and structures of models and methods of technology valuation.
- The roles and uses of technology valuation will be explained, and the recent policies to promote technology valuation on a national level in Korea will be introduced. The characteristics and structure of the practical methods and models will be introduced, and how to use the technology valuation in Tunisia efficiently will be discussed.
- The following are the main topics of the lectures:
  - Roles of Technology Valuation in Technology Commercialization
  - Policy to Promote Technology Valuation in Korea
  - Methods of Technology Valuation in Practice

**Title: Technology Valuation System (STAR-Value System) in Korea**

- The purpose of this lecture is to introduce the STAR-Value system (technology valuation system), which is actively used in areas related to technology transfer, technology commercialization, and technology management/planning in Korea. How the system can be utilized for enhancing technological performance and strategies in Korea, which can help Tunisia's technology valuation, is one of the measures for promoting technology transfer and commercialization.
- The concepts of technology valuation and general methodology will be reviewed by the description of factors reflected to the technology valuation overview above, including the description of evaluation model, information, and functions provided by the system. The high utilization of the STAR-Value system will be explained by introducing an online demonstration of technology valuation in the system with a valuation case.
- The following are the contents of this lecture:
  - Reviewing the concepts and general methodology of technology valuation
  - Overview of the STAR-Value system and its components and functions
  - Demonstration of the STAR-Value system and case studies

## ■ Photos

**First day of Lecture**

**Lecture on IP**

**Lecture**

**Photo at the orientation**

**Korea Univ. Makers' Ville**


**Visit to KIST**

**Group Photo**


**Group Photo after Certification**

## Daily Evaluation Survey Form

**Daily Evaluation Survey**

**on STI Capacity Development Workshop**

**-Policy Consultation to Support the Establishment of National Technology Transfer System in Tunisia-**

| | |
|---|---|
| Name | |
| Organization | |

**Instructions: This questionnaire is designed to evaluate the lecture. The evaluation of lecture is meant to improve the quality of our training program. Please indicate your level of agreement with the items listed below by checking the appropriate box and provide any comments. Thanks for your support!**

| Lecture 1 | IPR Management | | Lecturer | | Kyungmin Chung |
|---|---|---|---|---|---|
| | Strongly Disagree | Disagree | Neutral | Agree | Strongly Agree |
| 1. The topic covered in the lecture was profoundly linked in my work. | ① | ② | ③ | ④ | ⑤ |
| 2. The lecturer's delivery is clear and easy to understand. | ① | ② | ③ | ④ | ⑤ |
| 3. The knowledge learned can be applied directly to my work. | ① | ② | ③ | ④ | ⑤ |
| 4. I would like to recommend this lecture to others and to invite to the upcoming dissemination workshop in Tunisia. | ① | ② | ③ | ④ | ⑤ |
| 5. Please provide any comments or suggestions that might help improve this lecture in the future. | | | | | |

## Evaluation Survey Form

**Evaluation Survey**

**on STI Capacity Development Workshop**

**-Policy Consultation to Support the Establishment of National Technology Transfer System in Tunisia-**

| | |
|---|---|
| Name | |
| Organization | |

*Instructions: This questionnaire is designed to evaluate the process of the training program on how much you agree with the question, on a scale rated one to five. Your valuable opinions would be incorporated in organizing our next programs for the upcoming dissemination workshop in Tunisia. This is intended for capacity development of key stake holders for TT/TC policy development in Tunisia. We appreciate your support and cooperation!*

| Questions | Strongly Disagree | Disagree | Neutral | Agree | Strongly Agree |
|---|---|---|---|---|---|
| 1. The objectives of the training were clearly defined. | ① | ② | ③ | ④ | ⑤ |
| 2. The program was helpful in enhancing my capacity to perform the duties. | ① | ② | ③ | ④ | ⑤ |
| 3. The program met the objective of institutional capacity building. | ① | ② | ③ | ④ | ⑤ |
| 4. The contents of the program were organized well and easy to follow. | ① | ② | ③ | ④ | ⑤ |
| 5. The materials distributed were helpful. | ① | ② | ③ | ④ | ⑤ |
| 6. The topics covered were relevant to my organization's needs for TT/TC capability development. | ① | ② | ③ | ④ | ⑤ |
| 7. Enough participation opportunities and interactions were encouraged. | ① | ② | ③ | ④ | ⑤ |
| 8. The program has influenced in changing my mindset. | ① | ② | ③ | ④ | ⑤ |
| 9. The duration of the program (6 days) was sufficient. | ① | ② | ③ | ④ | ⑤ |
| 10. The meeting room, accommodation and the other facilities were adequate and comfortable. | ① | ② | ③ | ④ | ⑤ |
| 11. What did you like most about this training? What other topics/contents would you | | | | | |

| |
|---|
| suggest to be dealt in the future? |
| |
| 12. How do you hope to change your practice as a result of this training? |
| |
| 13. Are there any suggestions for improvement or requests on this training? If so, please explain in detail. |
| |

## 3. Dissemination Workshop, High-level STI Policy Dialogue, and Technology Valuation Workshop

To disseminate the project results of 2019 and to discuss future collaboration, STEPI hosted three events for the $15^{th}$~$17^{th}$ in the third week of July 2019. This year marks the $50^{th}$ anniversary of establishment of diplomatic relations between Korea and Tunisia, and the Korean Embassy in Tunisia highlighted this week as a Special Celebration Week of Science, Technology, and Innovation and supported the workshops.

The first event held on July 15 was a high-level STI Policy Dialogue, a special event hosted by STEPI and MHESR that shared the Korean experience in its economy and technology transfer system. It was a meaningful time to share knowledge and experience and discuss what can be the implications for Tunisia. On the same day, the workshop on Technology Evaluation – Technology Valuation: Policy and Practice was co-hosted by STEPI and ANPR for trainers of TTOs to learn the technology valuation system and enhancement of the technology transfer system in Tunisia. On July 17, the Dissemination Workshop on the Improvement of the National Technology Transfer System (NTTS) was co-hosted by STEPI, ANPR, MHESR, and Korean Embassy in Tunisia. It provided a great opportunity to share and disseminate the project achievements for the second-year project and future collaboration as a follow-up.

In addition, a STEPI delegation visited the Ministry of Higher Education and Scientific Research of Tunisia (MHESR), University of Tunis El Manar (UTM), National Engineering School of Tunis (Ecole Nationale d'Ingénieurs de Tunis: ENIT), and Borg Cerdric bio center.

## ■ STEPI Delegation

| Name | Organization | Title | Contact Information |
|---|---|---|---|
| Eun-Joo Kim | STEPI | Head of the Global Training Team | ejkim@stepi.re.kr |
| Jiyong An | STEPI | Researcher | Freesoul13@stepi.re.kr |
| Wangdong Kim | STEPI | Head of the ODA Office | |
| Deok Soon Yim | STEPI | Senior Research Fellow | |
| Youngrak Choi | STEPI | Former President | |
| Hyun Woo Park | Korea Institute of Science and Technology Information (KISTI) | Senior Research Fellow | hpark@kisti.re.kr |
| Myung Do Oh | University of Seoul (UoS) | Professor | |

### ■ Detailed Schedule

| Date | Departing from | Means of transportation | Destination | Institutions to visit | Tasks conducted |
|---|---|---|---|---|---|
| July 13 | Incheon (23:55) | Flight | Dubai (04:25) | Departure, via Dubai, | Incheon International Airport → Dubai International Airport |
| July 14 | Dubai (09:00) | Flight | Tunis (12:25) | Arrival in Tunis | Dubai International Airport → Tunis–Carthage International Airport |
| July 14 | **Preparations for the workshops** | | | | |
| July 15 | 10:00-12:00 | | | MHESR | High-level STI Policy Dialogue in MHESR |
| | 14:00-17:00 | | | Hotel | Technology Valuation Workshop |
| July 16 | 09:00-15:00 | | | El Manar Univ | El Manar University |
| July 17 | 09:00-17:00 | | | Hotel | Dissemination Workshop |
| July 18 | 10:00-14:00 | | | Borj Cedria Bio Center | Visit institutes |
| July 19 | 10:00-12:00 | | | MHESR | Meeting with the Secretary of State of MHESR |
| | 15:00-17:00 | | | Embassy, KOICA | Share the project results and the way forward |
| July 19 | Tunis (14:05) | Flight | Dubai (23:10) | Departure via Dubai | Tunis–Carthage International Airport (EK748) → Dubai International Airport |
| July 20 | Dubai (03:40) | Flight | Incheon (16:55) | Incheon | Incheon International Airport |

■ **Korea-Tunisia High-Level STI Policy Dialogue**

In the morning of July 15, 2019 (Mon.), STEPI and MHESR co-hosted the Korea-Tunisia High-Level STI Policy Dialogue at the conference room of the Ministry of Higher Education and Scientific Research of Tunisia, which was a special event to share important knowledge and experience as well as to commemorate the 50th anniversary of the diplomatic relations between Korea and Tunisia.

Over 30 high-level officials attended this meeting including the STEPI team (Eunjoo Kim, Jiyong An, Dr. Wangdong Kim, Dr. Deok Soon Yim, Dr. Young Rak Choi, Dr. Myung Do Oh, Dr. Hyun Woo Park) and MHESR team (Dr. Samia Charfi, Director General of the Technology Transfer Bureau, Mr. Hichem Adouni, Foreign Affairs Director, Dr. Maha Hammami, International Cooperation Manager, Dr. Anis Roussi, International Cooperation Manager, etc.).

Samia Charfi, Director of the Technology Transfer Bureau, gave the welcome remarks and introduced each of the Korea-Tunisia delegations. Choi Young-rak, Ethiopia's Deputy Adviser, gave a presentation on Korea's Economic Development, whereas Oh Myung-do, Professor of the University of Seoul, gave a presentation on Technology Transfer and Diffusion: Enactment Point of View. The presentations are mainly about experiences from the policymakers' perspectives regarding Korea's economic development, technology transfer system, and diffusion policies, and discussions and Q & A were held with various ministries in Tunisia. The Tunisian Ministry of Higher Education and Science jointly held a high-level STI policy dialogue to listen to various experiences in Korea and to explore the areas that could provide implications through policy commentaries or ministries.

## ■ Photos

| First day of Lecture | Lecture on IP |
|---|---|
|  |  |



| Lecture | Photo at the orientation |
|---|---|
|  |  |

## ■ Program

# Tunisia Korea High-Level Policy Dialogue on the contribution of STI in industrial regeneration and economic development

**Tunis, July 15, 2019**

**Introduction**

Based on the cooperation ties between Korea and Tunisia, this high-level policy dialogue on STI and its contribution to industrial regeneration and economic development is a step forward in policy consultations. At the dialogue meeting, participants will share policy measures and trends for future cooperation in the hopes of gaining policy suggestions and insights on further Tunisia Korea collaboration.

**Participants**

From the Tunisian side: H.E. the Secretary of State for Scientific Research, Advisory Board for Scientific Research (including 2 former ministers of Higher Education and /or Scientific Research), former Ambassador of Tunisia to Korea, and selected stakeholders in the STO system in Tunisia

From the Korean side: H.E. Ambassador of Korea to Tunisia, President of STEPI, and two Korean Experts.

The number of participants in the meeting will be 15 to 20.

**Venue**

MHESR headquarters in Tunis.

**Language**

English

**Meeting outline**

| | |
|---|---|
| 10 am | Opening |
| Session1 | *Presentations*<br>This session establishes the intellectual and conceptual framework for the dialogue. Both sides will give presentations.<br>*e.g.* STI and its contribution to industrial regeneration and economic development, *S&T and Economic Development, S&T Policy Tools for Implementation, STI for SDGs roadmap…* |
| Session2 | Guided Discussion |

### ■ Workshop on Technology Valuation “Technology Valuation: Policy and Practice”

In the afternoon of Monday, July 15, 2019, the workshop on Technology Valuation – “Technology Valuation: Policy and Practice” was co-hosted by STEPI and ANPR at the Moven pick Hotel in Tunis for the 50th anniversary of Korea-Tunisia diplomatic relations.

The theme of the workshop was a practical lecture on the STAR-value assessment system developed by KISTI, focusing on technology and valuation systems (STAR-Value system).
The attendees were key staff and trainers from the Tunisian Technology Transfer Organizations. There were approximately 18 local staff who participated to this workshop, including ANPR's Souad Boussaid Engineer, Leila Triki (INRAP Manager), Ines Zidi (UTM Professor), and Besma Sioud (CBBC Technology Transfer Manager).

There were detailed lectures on the STAR-Value system, a technology valuation system developed by KISTI, for the practitioners of the Tunisian technology transfer organization, including a lecture on the policy and implementation of technology commercialization and value. It put emphasis on the role of technology valuation in technology transfer and commercialization.

Moreover, it shared Korea's Technology Valuation Policy and Practices for Technology Valuation and contemplated on Technology Valuation and its application to Tunisian Cases to Explore Technology Valuation at the National Level and Practical Cases and Models. It shared approaches to implementing technology valuations in the marketplace, teaching, and applying policies and practical calculation methods related to technology valuation. Three approaches were compared: income-based approach, price-based approach, and market approach.

## ■ Photos

| Lecture | Photo at the orientation |
|---|---|
|   |   |

## ■ Program

### Workshop on Technology Valuation
### "Technology Valuation: Policy and Practice"

**Date: July 15, 2019, 14:00-17:00**
**Venue: Zurich, Movenpick Gammarth Hotel, Tunis, Tunisia**

**Lecture by Dr. Hyun Woo Park**

Technology valuation plays a key role in technology transfer and commercialization by providing essential information on the economic value of the subject technology for its users. In this presentation, the roles and uses of technology valuation will be examined, and the recent policies to promote technology valuation on a national level in Korea will be introduced. The characteristics and structure of the practical methods and models will be introduced, and how to use the technology valuation in Tunisia efficiently will be discussed. The main topics of the lectures are as follows:

- Concept and Uses of Technology Valuation
  - What is Technology Valuation?
  - Who needs to Value Technology?

- Policy to Promote Technology Valuation in Korea
  - Background of the Policy
  - Policies by Government Departments
  - Present Condition and Policy Implications
  - Main Agenda of MSIT (Ministry of Science and ICT)

- Methods of Technology Valuation in Practice
  - How We Value Technologies - Valuation Approaches
  - Income Approach, Market Approach, Cost Approach

- Web-based Technology Valuation
  - Overview of the STAR-Value System
  - Conceptual Map of the STAR-Value System
  - Web Configuration of the STAR-Value System
  - Underlying Technologies of the STAR-Value System
  - Valuation by the STAR-Value System

- Korea-Tunisia Dissemination Workshop on the Improvement of National Technology Transfer System (NTTS)

On July 17, the Dissemination Workshop on the Improvement of National Technology Transfer System (NTTS) was co-hosted by STEPI, ANPR, MHESR, and Korean Embassy in Tunisia. It provided a great opportunity to share and disseminate the project achievements for the second-year project and future collaboration as a follow-up. This was specially co-hosted by the Korean embassy to commemorate the 50th anniversary of Korea-Tunisia diplomatic relations.

About 80 participants from ministries, research institutes, universities, TTOs, and private sector attended this dissemination workshop including Khalil Amiri, Secretary of State of MHESR, H.E. Gu Rae Cho, Ambassador of the Korean Embassy in Tunisia, Moon-jung Choi, director of KOICA Office, Tunisia, STEPI team, and MHESR team.

The welcome remarks were delivered by Khalil Amiri, the Secretary of State of MHESR. Dr. Young-rak Choi, the former president of STEPI, gave a keynote speech, with the congratulatory speech delivered by H.E. Ambassador Gurae Cho, Korean Ambassador to Tunisia.

Moreover, five participants from Tunisia who attended the study visit program to Korea in June (Prof. Ines Zidi, Dr. Ismail Haddad, Dr. Selim Mrad, Dr. Bourhen Louhichi, Dr. Besma Ben Besmah, and Dr. Besma Sioud) shared their experiences in and implications from attending the program in Korea.

Professor Myung-do Oh gave a presentation on Technology Transfer and Diffusion: Enactment Point of View, whereas Dr. Deok soon Yim gave a presentation on the strategic management of Science and Technology Park and Promotion of University-Industry-Research Inst. Network. Lastly, Dr. Wangdong Kim, the head of office of the STI ODA Project, gave a presentation titled "Toward the Next Phase of Tunisia-Korea STI Cooperation." After the presentations and Q & As, the results of the research and experiences of major invited participants in the Tunisian national system of technology transfer were shared.

MHESR has established an action plan for improving the Tunisian technology transfer and commercialization legal system based on this project and held policy consultation on improving the national technology transfer system. The industry-university technology transfer and commercialization network was established to support the capacity building of 25 technology transfer centers systematically.

## ■ Photos

<table>
<tr><th>Dissemination Workshop</th><th>Group Photo</th></tr>
<tr><td></td><td>
</td></tr>
<tr><th>Congratulatory remarks by the Korean Ambassador</th><th>Opening remarks by the Secretary of State of MHESR</th></tr>
<tr><td></td><td>
</td></tr>
<tr><th>Keynote speech by Dr. Choi</th><th>Moderated by Ms. Kim of STEPI</th></tr>
<tr><td></td><td></td></tr>
</table>

## Program

### Dissemination Workshop

### Policy Consultation on the Support for the Establishment of an Effective National System of Technology Transfer (NSTT) in Tunisia (2018-2019)

***Held on Monday, July 17, 2019 at***

***Hotel Movempick Gammarth Tunis***

| Time | Session Title | Speakers |
|---|---|---|
| 08:30-09:00 | Registration | |
| 09:00 - 09:30 | Opening Session<br>Opening Remarks<br>Welcome Remarks<br>Congratulatory Remarks | <br>Dr. Khalil Amiri, Secretary of State, MHESR<br>Prof. Young Rak Choi, Former President, STEPI<br>His Excellency Koo Rae Cho, Ambassador of Korea |
| 09:30 - 10:30 | Session 1:<br>- What TTOs have learned from the Korean experiences in technology<br>- How should it be applied to their Technology Offices? | Prof. Ines ZIDI: University of TM<br>Ms. Besma SIOUD: CBBC<br>Mr. Slim MRAD: University of Monastir<br>Prof. Borhene LOUHICHI: University of Sousse |
| 10:30 - 11:00 | Coffee Break | |
| 11:00 - 12:00 | Session 2:<br>- Technology Transfer and Diffusion:<br>TTO Point of View<br>- What can be adopted for the Tunisian context?<br>- Promotion of University-Industry-Research | Prof. Myung Do Oh, Professor, University of Seoul<br><br>Dr. Deok Soo Yim, Senior Research Fellow, STEPI |

| Time | Session Title | Speakers |
|---|---|---|
| | Institutes Network | |
| 12.00-12.30 | Closing Session<br>Toward the Next Phase of Korea-Tunisia STI Cooperation | Moderated by Dr. Chedly Abdelly, Director of ANPR<br>Dr. Wang Dong Kim, Director of STI ODA, STEPI |
| 12:30 - 14:00 | Lunch Break | |

## ■ Visiting Institutes

**University of Tunisia El Manar (UTM)**

On July 16, 2019, the STEPI team visited the University of Tunisia El Manar and had a meeting with the president of the University, major faculty members, and researchers. The university gave a presentation to introduce the organization chart, student status, current technology transfer-related research and projects conducted by UTM and its innovation center.

It was great opportunity to explore Korea's cooperation by sharing the demands and limitations related to the current Tunisian technology transfer organization. The STEPI team seeks to link cooperation projects with El Manar University by sharing projects that are expected and scheduled for cooperation with Tunisia. Nonetheless, close inspection and vigorous support for key research through KOICA are needed.

## ■ Photos

| Keynote Speech by Dr. Choi | Moderated by Ms. Kim of STEPI |
|---|---|
|  |  |

**Borj Cedria Bio Center**

On July 18, the STEPI team visited the Borj Cedria Bio Center to meet with the relevant researchers and understand and study information on the center's facilities, major research, and international cooperation. The center mainly focuses on plant research on soil, efficacy of olive oil and grape seed oil, and research on soil culture.

The center was found to be supported vigorously by JICA through loan projects, and this project will be completed this year. The center is also conducting joint research with Sungkyunkwan University in Korea. Many research outcomes, including cosmetics, expect to be able to collaborate with the relevant research institutes through KOICA.

■ **Photos**

| Keynote speech by Dr. Choi | Moderated by Ms. Kim of STEPI |
| --- | --- |
|  |  |

■ **Summary of the Meeting with the Secretary of State**

As a follow-up meeting with internal researchers, the STEPI team had a meeting with the Secretary of State of MHESR and key public officials to discuss local needs and possible cooperation in next year's projects before leaving Tunisia.

On July 19 (Fri.), Khalil Amiri, the Secretary of State, Dr. Anis Roussi of the International Cooperation Bureau, and Dr. Maha Hammami and Souad Boussaid, engineer of ANPR, attended the meeting along with the STEPI team at the meeting room of MHESR.

In particular, the meeting discussed the establishment of the Korea-Tunisia-Africa Innovation Center and the Tunisian-STEPI think tank, which were planned and discussed from last year’s project in 2018. Dr. Khalil Amiri highlighted the fact that Tunisia is North Africa's gateway to Europe, emphasizing the importance of establishing a Korea-African Innovation Center as it has the advantage of being relatively infrastructural among African countries. He also discussed the need for further cooperation with Korea.

The need for the Korea-Africa Innovation Center is to establish a base to harness Africa's potential in the field of science and technology and to support entry into Korea. Africa is an emerging continent, very important in terms of market, science and technology manpower, with an estimated 200 retirement science and technology personnel in Korea who are currently serving in Africa.

Korea's cooperation is concentrated in the traditional ODA areas such as health and agriculture, and there is no platform for systematically supporting scientific and technological cooperation. Therefore, Tunisia, a base for entry into North Africa, will be an optimal base. As a relatively developed country in terms of science and technology system among African empires, it possesses excellent manpower and universities in the field of bio and IT (in South Africa, the technology system is weakening).

As a country with excellent diplomacy and detail, Tunisia easily connects with many African countries and considers supporting the provision of space and linkages with international organizations such as the African Development Bank and the African Union.

**KOICA Office**

On July 19 (Fri.), 2019, the STEPI team visited the KOICA office in Tunisia. There was a discussion on the feasibility of cooperation projects such as the Korea-African Innovation Center and Tunisia-STEPI think tank as discussed with MHESR. In order to develop Tunisia's science and technology for the establishment of Tunisia-STEPI, strengthening policy capacity is a matter of urgency, but the difficult situation cannot be overlooked because there is no policy research organization. If an organization such as Korea's STEPI is established within the University of Tunisia and support is provided for five years, such will lay the foundation for the development of science and technology in Tunisia.

Recommendations include reviewing the necessary processes such as local demand survey and project proposal to select the KOICA business, discussing cooperation with major ministries and embassies, and reviewing various Korean project cooperation plans and follow-up measures for next year's support.

■ **Photos**

| Group photo with the Secretary of State | Meeting with the Secretary of State |
|---|---|
|  |  |
| **Visit to KOICA Office** | **Visit to KOICA Office** |
|  |  |

■ **Summary of Action Plans:**

To establish a high-level executive committee for policy and strategy to decide national research priorities for the advancement of technological innovation in industries;

Enhancement of an institutional framework to promote technology transfer; diagnosis of the existing system; benchmark of international cases of technology transfer in accordance with Korea's strategy and the reform of the relevant legal systems accordingly; increase in research incentives for activities related to technology transfer; and review of support plans for the relevant budget;

Establishment of an institutional framework for technology transfer secretariats to make up for a technology transfer system; clarification of the definition and role of the secretariats; and introduction of the relevant implementation manuals;

Production of training content and provision of training for capacity building; establishment of networks between technology transfer secretariats; and exchange with Korean institutions that visited through the Study Visit program and pursuit of cooperation with Tunisian programs.

## ■ ACTION PLAN (2019-2020)

| Axes/Activities/Description |
|---|
| **1. Policy and Strategy** |
| **1.1 Identify the industry-oriented STI potential in connection with the national research priorities**<br>- Designate a ***High-Level Working Group*** (***HLWG***) in charge of setting up a National Strategy for Technology Transfer aligned with industry STI priorities. The HLWG includes (but is not limited to) representatives from:<br>✓ Ministry of Higher Education and Scientific Research;<br>✓ Ministry of Industry, SME, and Energy (with APII, INNORPI);<br>✓ Ministry of Agriculture (with IRESA);<br>✓ Ministry of Health;<br>✓ Ministry of ICT;<br>✓ Business Unions for Industry and Agriculture: UTICA, UTAP…<br>- Set up a strategy for each priority sector in partnership with the business sector including field-related techno parks and clusters;<br>- Assign a ***Regional Working Group*** for universities, research centers, technoparks, clusters, and local industries in each region;<br>- Identify industry/economy-oriented research strengths and recognize the TT potential for exploitation.<br><br>**1.2 Enhance the regulatory framework for the better promotion of TT**<br>- Diagnose the existing TT legal framework;<br>- Benchmark with international TT experiences with particular focus on the Korean strategy;<br>- Amend TT-related legal texts;<br>- Set up incentives for researchers in relation to TT activities;<br>- Establish financial support mechanisms for TT activities. |

| 2. Organization of the Technology Transfer System |
|---|
| **2.1 Prepare an institutional framework for Technology Transfer Offices**<br>- Appoint TTOs in coordination with universities/research centers/technopoles/technical centers;<br>- Allocate the necessary funds for TTO activities;<br>- Analyze the needs of the business sector;<br>- Set up a database for research projects/patents/sectors;<br>- Establish an ANPR tool for the management of funds associated with TT (patent, licensing …) at the national and international cooperation levels. |
| **2.2 Define the organization and missions of TTOs** |
| **2.3 Manual of procedures of TTOs**<br>- Patent filing procedure: declaration of invention form, prior art search, evaluation, drafting, filing, valuation, contracting, and marketing;<br>- Procedure for creating spin-offs / startups: business plan;<br>- Procedure for identifying industrial needs;<br>- Identification procedure for mature research projects**.** |
| **3. Capacity building** |
| **3.1 Concept of training contents**<br>- 9 training modules (*CF*: MESRS Report: Technology Transfer System) |
| **3.2 Training of trainers' TOT** (*CF*: MESRS Report: Technology Transfer System)<br>- General 6-month training for 40 candidates: 30 already working at TOTs with 5 administrative staff and 5 researchers;<br>- Specialization of each in a theme for 6 months with practical training. |
| **3.3 TTO Networking at the National and Regional Levels (MENA)** |
| **3.4 Blended learning (Online Master of Science Program)**<br>Establishment of an online Master of Science degree in collaboration with the Virtual University of Tunis |
| **3.5 Training of Tunisian startups through STEPI at the Global Seoul Center** |
| **3.6 Internationalization of MOBIDOC (PhD studies on enterprises) in collaboration with Hanyang University (HYU)** |
| **3.7 International event**<br>Organization of international events, workshops, and conferences in the IT sector with companies between Seoul National University and the Tunisian side |
| **3.8 Cooperation: exchange of post-docs, PhD students, and experts**<br>**-** Doctoral internships<br>**-** Scholarships for PhD and master's students<br>**-** Post-doc Mobility |

## TUNISIAN TECHNOLOGY TRANSFER Network (3TN)

Background

Science, technology, and innovation (STI) are the drivers of economic growth, improved productivity and efficiency, job creation, and sustainable development. The main goal of this project is to contribute to the development of a strategy for the Science and Technology Innovation (STI) of the Tunisian Research and Innovation System. More precisely, this project will contribute to developing a policy that enables better impact of the Tunisian Research System at Universities and Research Centers on economic development. This project will give insight into the institutional organization in charge of BuTT (TOT).

At the operational level, the Tunisian Technology Transfer Network of TOTs will play an essential role for linkages among the universities and research institutions with the economic development sector, industry, and relevant government actors. The current project offers a real opportunity for the improvement of the capacities of BuTT staff.

Stakeholders

- Ministry of Higher Education and Scientific Research (DGVR, ANPR, DGRS, and DGCI)
- Ministry of Industry (DGIDT, DGIIT, IINNORPI, APII)
- Business Unions for Industry and Trade and Agriculture (UTICA and UTAP)

Beneficiaries

- BuTT
- Universities, Research Centers, Technoparks
- Industries, Clusters

## Letter of Appreciation from MHESR to STEPI

REPUBLIC OF TUNISIA

***

**Ministry of Higher Education**
**and Scientific research**

***

Tunis, 23 May 2019

**Dr. Hwang Hee Cho, President,**
**Science and Technology Policy Institute**

Mr. President,

I am delighted to express to you and to the entire IICC team my most sincere thanks for the support provided to our Research and Innovation System, especially through the K-Innovation ODA Project that aims to support the establishment of an effective National System of Technology Transfer (NSTT) in Tunisia. I would like to thank you for the interesting recommendations included in the final report of the 2018 Project and your kind words in the preface.

I greatly appreciate what has been achieved in the context of our bilateral cooperation so far and look forward to extending our partnership further, towards the establishment of a center dedicated to capacity building in the area of STI policy with a mission to reach out to other African countries in the region.

I would also like to seize this opportunity to invite you to Tunisia to participate in a High Level Policy Dialogue on July 17th, and in which we plan to discuss the contribution of STI policy in industrial regeneration and economic development.

Looking forward to welcoming you soon to Tunisia, please accept the expression of my highest regards.

**Dr. Khalil Amiri**

**Secretary of State for Scientific Research**

## ■ Press Releases

STEPI-IICC Facebook (date uploaded: April & July 2019)
(https://www.facebook.com/iicc.stepi)

-Press Release in Tunisia (Business News, date released: August 5, 2019)
(https://www.businessnews.com.tn/la-tunisie-et-la-coree-du-sud-organisent-des-activites-sur-le-transfert-tenologique,520,89917,3)

Business News
.com.tn

La Tunisie et la Corée du Sud organisent des activités sur le transfert technologique

*05/08/2019 13:44*


En vue de promouvoir la coopération dans le domaine de la science et de la technologie entre la Tunisie et la Corée du Sud, des activités ont été co-organisées entre le ministère de l'Enseignement supérieur et de la Recherche scientifique et le *« Science and Technology Policy Institute (STEPI) »* à cette occasion.

Plusieurs recommandations ont été élaborées notamment, le gouvernement doit établir un système d'innovation pour aider les entreprises, concernées par le transfert et la commercialisation de technologies, à avoir un accès facile aux informations pertinentes sur les projets de R & D réalisés et les droits de propriété intellectuelle détenus par les centres de recherche et les universités, accroître l'indépendance et la spécialisation des BuTT pour jouer leur rôle pleinement et les universités et les centres de recherche sont appelés à créer une société de gestion des investissements et à la gérer de manière autonome.

*« Aussi, et sur la base de l'expérience de la Corée, certaines suggestions visent la création d'un environnement propice à la diffusion des technologies et à son amélioration au niveau de la Société en général. Elles sont recommandées pour compléter la politique de la science et la technologie en Tunisie. Parmi ces suggestions on trouve : l'élargissement de l'aide fournie aux programmes de diffusion technologique existants et création de nouveaux programmes et la création d'un groupe de consultants pour les PME afin de les orienter sur comment intégrer de nouvelles technologies dans leurs affaires »*, indique un communiqué du département de l'Enseignement supérieur publié ce lundi 5 août 2019.

E.B.A.

-Radio Station Interview on the Dissemination Workshop (date of broadcast: July 2019)

| Photo of Recording Interview at the Radio Station in Tunisia | |
|---|---|
|  |  |

-Press Release in Korea (Et News, date released: July 22, 2019)

전자신문 | etnews

## STEPI, 한국-튀니지 수교 50주년 맞아 과학기술혁신 협력 강화

발행일 : 2019.07.22

<조구래 튀니지 대사가 기술이전 성과확산 워크숍에서 축사하는 모습>

과학기술정책연구원(STEPI·원장 조황희)이 우리나라와 튀니지 수교 50주년을 맞아 과학기술혁신 분야 협력 확대를 위한 다양한 행보에 나섰다.

STEPI는 지난 15~18일 튀니지 수도 튀니스에서 튀니지 고등교육과학연구부(MHESR), 국가과학연구진흥청(ANPR)과 함께 다양한 행사를 진행했다.

15일에는 '한국-튀니지 과학기술정책 고위급 STI 정책대화'를 열고, 우리나라 과학기술혁신 경험과 노하우를 공유했다. 이 자리에는 카릴 아미리 과학연구 특임장관을 비롯해 현지 주요 부처 고위급 공무원 20여명이 참석했다.

이날에는 튀니지 주요 기술이전관계자들을 대상으로 '기술 가치평가 워크숍'도 진행됐다. STEPI는 한국과학기술정보연구원(KISTI)이 개발한 기술 가치평가 시스템(STAR-Value System)을 소개하고, 다양한 기술 가치평가 방법론을 전수했다.

17일에는 '국가기술이전 시스템 개선을 위한 정책 자문 사업에 대한 성과확산 워크숍'을 열었다. 튀니지 정부 대표단 및 기술이전 주요 관계자 70여명이 자리한 가운데 튀니지 국가 기술이전시스템 구축 결과를 공유하는 시간을 가졌다.

조황희 STEPI 원장은 "튀니지는 지정학적으로 아프리카 진출을 위한 거점 국가이자, 과학기술 분야에서 한국의 주요 파트너 국가"라며 "앞으로도 양국 간 과학기술 분야의 협력을 강화하고 튀니지를 중심으로 아프리카의 지속가능한 과학기술에 정책자문 등 의미 있는 지원을 하는데 힘쓰겠다"라고 말했다.

대전=김영준기자 kyj85@etnews.com

-Press Release in Korea (Jungdo News, date released: July 22, 2019)

중도일보

1951년 창간 中都日報 joongdo.co.kr

한국-튀니지 수교 50주년...STEPI, 과학기술혁신 협력 강화

김성현 기자 최종 기사입력 2019-07-22 15:14


과학기술정책연구원(STEPI)은 주튀니지한국대사관과 한국과 튀니지 수교 50주년을 기념하고 과학기술혁신 분야 협력을 확대하기 위한 다양한 행사를 현지에서 개최했다고 22일 밝혔다.

STEPI는 과학기술특별주간으로 지정된 지난 15일부터 18일까지 튀니지의 수도 튀니스에서 튀니지 고등교육과학연구부(MHESR)와 국가과학연구진흥청(ANPR)과 함께 ▲과학기술정책 고위급 STI 정책대화 ▲기술 가치평가 워크숍 ▲기술이전 성과확산 워크숍 등을 진행했다.

조황희 과학기술정책연구원장은 "튀니지는 지정학적으로 아프리카 진출을 위한 거점 국가이자

-Press Release in Korea (Science Economy News, date released: July 22, 2019)

한국과학경제
Science Economy

HOME > 정책 > 과학

# 한국-튀니지 수교 50주년...STEPI, 과학기술혁신 협력 강화

한국과학경제 | 승인 2019.07.22 22:10

한-튀니지 고위급 STI 정책대화에 참석한 양국 관계자들이 단체 사진을 촬영하고 있다. 앞줄 왼쪽 세번째가 조황희 과학기술정책연구원 원장.

【한국과학경제=최경제 기자】 과학기술정책연구원(STEPI)은 주튀니지 한국대사관(대사 조구래)과 한국과 튀니지 수교 50주년을 기념하고 과학기술혁신 분야 협력을 확대하기 위한 다양한 행사를 현지에서 개최했다.

과학기술특별주간으로 지정된 7월 15일부터 18일까지 튀니지의 수도 튀니스에서 STEPI는 튀니지 고등교육과학연구부(MHESR)와 국가과학연구진흥

- STEPI Press Release in Korean (http://www.stepi.re.kr/)

| STEPI 과학기술정책연구원 | 보 도 자 료 | | 국민에 쓰임새 있는 과학기술정책, STEPI가 만들어갑니다. |
|---|---|---|---|
| | 배포 일시 | 2019. 7. 19.(금) 총 4매(본문 2, 붙임 2 ) | |
| 담당 부서: 대외전략·홍보팀 | 담 당 자 | · 팀장 김영환, 연구원 천항기<br>· ☎ (044)287-2182, 2214 | |
| 담당 부서: ODA사업단 | 담 당 자 | · 팀장 김은주, 연구원 안지용<br>· ☎ (044)287-2075, 2278 | |
| 보 도 일 시 | 배포 즉시 게재 바랍니다. | | |

## STEPI, 한국-튀니지 수교 50주년… 과학기술혁신 협력 강화

**- 15일(월)~18일(목), 과학기술특별주간 지정으로 다양한 행사 개최 -**

□ **과학기술정책연구원(이하 STEPI, 원장 조황희)**은 주튀니지한국대사관(대사 조구래)과 한국과 튀니지 수교 50주년을 기념하고 **과학기술혁신 분야 협력**을 확대하기 위한 다양한 행사를 현지에서 개최했다.

○ **과학기술특별주간**으로 지정된 7월 15일(월)부터 18일(목)까지 튀니지의 수도 튀니스에서 STEPI는 튀니지 고등교육과학연구부(MHESR)와 국가과학연구진흥청(ANPR)과 함께 ▲과학기술정책 고위급 STI 정책대화 ▲기술 가치평가 워크숍 ▲기술이전 성과확산 워크숍 등을 진행했다.

□ 현지 시간으로 15일(월), 튀니지 고등교육과학연구부에서 진행된 **'한국-튀니지 과학기술정책 고위급 STI 정책대화'**에는 **카릴 아미리(Khalil Amiri) 과학연구 특임장관**을 비롯하여 20여명의 튀니지 주요 부처 고위급 공무원들이 참석하여 한국의 **과학기술혁신 경험과 노하우를 공유하고 향후 양국의 과학기술혁신 협력 과제에** 대한 의견을 나눴다.

○ 이와 함께 튀니지 주요 기술이전관계자들을 대상으로 **한국과학기술정보연구원(KISTI)이 개발한 기술 가치평가 시스템(STAR-Value System)을 소개하고**, 다양한 기술 가치평가 방법론을 전수하는 **'기술 가치평가 워크숍'**도 진행되었다.

□ 17일(수)에는 **'국가기술이전 시스템 개선을 위한 정책 자문 사업에 대한 성과확산 워크숍'**이 진행되었다.

○ 카릴 아미리(Khalil Amiri) 과학연구 특임장관의 개회사와 최영락 전 STEPI 원장의 환영사, 주튀니지한국대사관 조구래 대사의 축사로 시작된 성과확산 워크숍에는 튀니지 정부 대표단 및 기술이전 주요 관계자 70여명이 참석하여 튀니지 국가 기술이전시스템 구축에 대한 결과를 공유하는 시간을 가졌다.

○ 국가기술이전 시스템 개선 정책 자문의 성과로 튀니지 고등교육과학연구부는 튀니지 기술이전 및 사업화 법 제도 개선을 위한 실행계획을 수립하고, 25개 기술이전센터의 역량강화를 체계적으로 지원하기 위해 산학연정 기술이전사업화 네트워크를 발족하였다.

□ **조황희 과학기술정책연구원장**은 "튀니지는 지정학적으로 **아프리카 진출을 위한 거점 국가**이자, 과학기술 분야에서 **한국의 주요 파트너 국가**이다."라며 "앞으로도 양국 간 **과학기술 분야의 협력**을 강화하고 튀니지를 중심으로 **아프리카의 지속가능한 과학기술에 정책자문** 등 의미 있는 지원을 하는데 힘쓰겠다."라고 말했다.

□ 한편, STEPI는 2009년부터 한국국제협력단(KOICA)과 함께 튀니지 정부 정책결정자들을 비롯한 산학연 전문가 200여명을 대상으로 **과학기술정책 역량 강화**를 위한 초청연수와 현지 교육훈련을 제공하기도 했다.

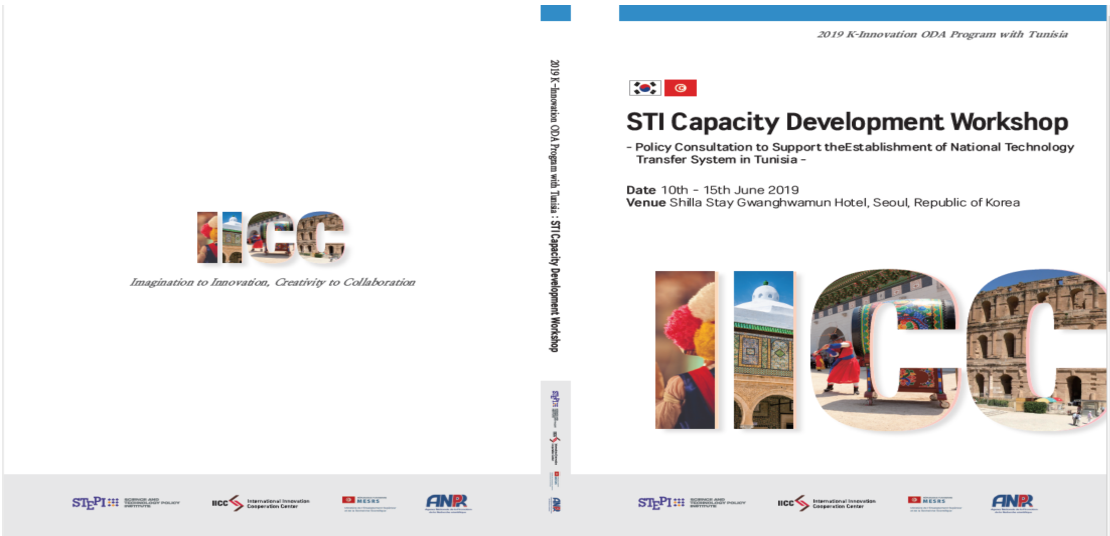
○ 이어 지난 2018년부터 **2년간 국제기술혁신협력사업(K-Innovation) 연구 과제**를 통해 **튀니지 국가 기술이전시스템 구축을 위한 정책 자문 사업**을 진행하고 있다.

-

■ **Workshop Materials**

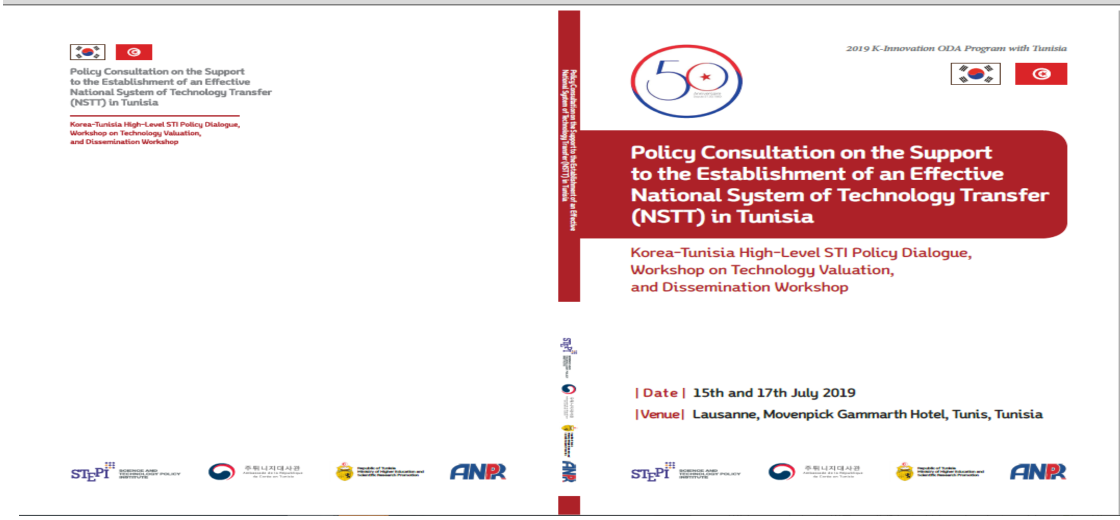
2019 STI Capacity Development Workshop (Policy Consultation to Support the Establishment of a National Technology Transfer System in Tunisia)

<워크숍 자료집>




2019 Korea-Tunisia High-Level STI Policy Dialogue, Workshop on Technology Valuation and Dissemination Workshop

<워크숍 자료집>



# 4. Future Collaborations

## ■ PCPs for follow-up projects

The K-Innovation Project with Tunisia is a two-year project, and there were a number collaboration demand and project ideas discovered during the last two years thanks to the vigorous support and keen interest of the counterparts of Tunisian partners. The team developed and submitted several project concept papers through KOICA.

The ideas and brief information on the PCPs are attached to this page.

1. Project Concept Paper for the Korea-Tunisia Africa Innovation Center

KOICA

RÉPUBLIQUE TUNISIENNE
MESRS
Ministère de l'Enseignement Supérieur et de la Recherche Scientifique

Korea Tunisia Africa Innovation Center
(4 years/ 8.5 Million USD)

The Republic of Tunisia

Project Concept Paper

Sep. 29. 2019

| Applicant Information | |
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| SECTION 1. BASIC PROJECT INFORMATION | | |
|---|---|---|
| **1.1** | **Country** | Tunisia |
| **1.2** | **Title** | Korea Tunisia Africa Innovation Center |
| **1.3** | **Location(s)** | Manouba region-Tunis Tunisia |
| **1.4** | **Duration** | 48 months (2021-2024) |
| **1.5** | **Budget (total)** | US$ 8.5 million |
| **1.6** | **Objectives** | Contribution to Science, Technology, and Innovation (STI)-based development in Tunisia and Africa through the capacity building of STI policymakers and practitioners |
| **1.7** | **Beneficiary** | University, Industry, Research Institute, and STI Ministry in Tunisia and other African countries |
| **1.8** | **Implementing organization** | Ministry of Higher Education and Scientific Research, Tunisia |

2. PCP for a Benchmark Training Program on Querying the Establishment of a Policy Think Tank for Science, Technology, and Innovation (STI) in Tunisia

KOICA

RÉPUBLIQUE TUNISIENNE
MESRS
Ministère de l'Enseignement Supérieur et de la Recherche Scientifique

A Benchmark Training Program on Inquiring the Establishment of a Policy Think Tank for Science, Technology and Innovation (STI) in Tunisia (2020/Budget)

Tunisia

**Program Concept Paper**

(* Capacity Improvement and Advancement for Tomorrow)

27, September, 2019

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Discussion contents regarding the establishment of Tunisia-STEPI

- Establishing the Tunisia-STEPI think tank: Strengthening policy capacity to develop science and technology in Tunisia is a matter of urgency, but it is difficult because there is no policy research organization.

If an organization such as Korea's STEPI is established within the University of Tunisia and support is provided for five years, it will lay the foundation for the development of science and technology in Tunisia.

- Major details of the Tunisia STEPI project
  - Establishment of science and technology policy organization and support for fostering researchers
  - Assistance in establishing national science and technology development in Tunisia
  - Tunisia's national science and technology information system, national R & D business system, national science and technology roadmap
- Tunisia's STEPI business budget: 3 billion for 3 years (1 billion annually)

# CHAPTER 5

## Conclusion

Policy Consultation on the Support for the Establishment of an Effective National System of Technology Transfer (NSTT) in Tunisia

# Chapter 5. Conclusion

## 1. Policy Implications from the Korean Experience

### 1.1 Challenges in the Development of of University TLO in Korea

#### 1.1.1 Inducement of Co-growth of Leading TLOs and Late TLOs

The university's TLO support program, which began in 2006, improved overall TLO capacity. There is a clear difference between the performance of commercialization and the competency of the technology between the university that has not received business support and the university that has been supported. Based on the Korean experience, 18 leading university TLOs account for 70% of all university transfer revenues, and most of the leading universities utilize professional staff such as patent attorneys.

The TLO, which is in the development stage above the stabilizer period, establishes the conditions for planning and promoting the business independently. The TLO in the late stage of the start-up phase can be assisted by patent lawyers or technology trade institutions. For the TLOs in the early stages of the startup, which are difficult to operate independently, it is necessary to construct an E-TLO (External TLO) and to promote systematic technology transfer and commercialization by utilizing the experts of technology transfer and commercialization experience and experts of external support organizations. Although these universities demand technology transfer due to the increase of intellectual property rights, it is difficult to carry out the systematic work such as valuation of the assets owned and the future technology transfer.

[Table 5-1] Support type with business range of University TLO

| Support Type | Supported | Business Range |
|---|---|---|
| Standalone | TLOs of universities with excellent technology transfer and commercialization capability, large R & D capacity and high technology, TLO operation by itself | Centered on patent packaging, standard patent creation, following R&D (gap funding program) and prototype production, and overseas marketing |
| Consortium I (professional institution-centered) | Universities with somewhat lower TLO capabilities. After selecting a specialized agency for the commercialization of technology trading, the participating universities are invited. | The technical trading commercialization organization supports the commercialization and marketing of TLO. |
| Consortium II | It is composed of universities with relatively low technology transfer and commercialization capability and low R & D scale and technology. A consortium of universities with similar domains or similar characteristics. | Sharing know-how and professional manpower among participating universities.<br>(IP management, technology discovery, technology transfer, and marketing) |
| External-TLO | Establishment in management institutions for small-scale universities where it is difficult to organize their own TLO organization and secure professional manpower. | IP and technology transfer specialists to carry out a similar role as the existing technology transfer organization. Part of the labor cost of the staff dedicated to participating universities. |

### 1.1.2 Fostering technology transfer and commercialization experts

In order to maximize the university's technology transfer and commercialization performance, it is important to expand and cultivate TLO personnel. That is because "people" do everything to manage the college's intellectual assets and transfer profits by technology transfer and commercialization. In 2006, the average number of people in university TLOs was 4.6, and the percentage of full-time TLOs reached 67%. Nonetheless, the average number of people in charge of technology privatization and technology transfer is only 2.2, or 48% of the total. The Korea Research Foundation analyzed the 2011 business plan of 30 universities participating in the 2nd "TLO Support Program for Universities," showing that the ratio of non-regular workers to regular workers is 47:53.

Since technology transfer and commercialization work requires many years of experience and expertise in related fields, systematic specialist training system is needed. A "professional contract worker" system is needed for this purpose. Some leading universities are operating contracts for technology transfer and commercialization that pay salaries and renew contracts according to the technology transfer and commercialization results. Because it takes a lot of time and money to train specialists, it is more economical to scout and utilize experts who have been trained at leading universities.

From a professional point of view, there is surely a job at a late college where one can feel the rewards of TLO growth and earn more money. Through the transfer of personnel with such expertise, the university's technology transfer and commercialization capacity will be improved as a whole. Excellent TLO staff are also expected to be favored as experts.

In order to enhance the competence of university TLOs, practical training of preliminary professional license holders such as lawyers can be utilized. According to the law amended in May 2011, law school graduates (law school) graduates must pass the lawyer's examination but must have completed six months of practical training at law enforcement agencies before they can work as lawyers. Nonetheless, institutions that accept these practical exercises are reportedly lacking. On the other hand, university TLOs have legal requirements such as signing of contracts, possession of intellectual property rights, and infringement problems, but they are not able to secure lawyers due to lack of financial resources. Therefore, if a university TLO is included in a practical training institute such as an attorney, this problem will be solved somewhat. In addition, through their practical training, they will be able to improve understanding of the university's intellectual property management, technology transfer, and commercialization activities, and the pool of specialists that can be utilized by universities will also increase.

### 1.1.3 Improving the profitability of university TLOs

In order for a university TLO to be financially independent without relying on government support, a review of the royalty allocation system is required. The royalty income of the total budget of university TLO is more than 55%, but only 9.3% of university TLOs can use the royalties as operating expenses.

In accordance with the "Regulations on the Management of National Research and Development Projects" and the "Law on the Transfer of Technology and Promotion of Commercialization" and its Enforcement Decree, more than 50% of the royalty income of universities should be used as compensation for researchers. As a result, universities use 52.5% of technology transfer revenues as researcher compensation.

Most universities in Korea stipulate that 60~80% of royalty income should be used as compensation for the inventor in order to induce university professors' job inventions (Song, Wanchul, 2010). Stanford University in the USA deducts 15% of royalty income and distributes it to the college, department, and inventor. The deducted amount is fully used for the labor and patent costs of the Stanford University TLO. Considering the fact that the inventor compensation standard of major universities in the United States is 30% of the royalty income, the ratio of inventor compensation in Korea is relatively high. Improving the profitability of the university TLO requires lowering the level of inventor compensation.

### 1.1.4 Activating the role of technology market participants for commercialization

It is generally said that there are three technology market participants for commercialization:

- Technology suppliers (professors, researchers),
- Intermediaries (including Korea Technology Exchange, trading institutions, trading RTTCs)
- Technology users (individuals and companies).

Even though Korea reached the high level of technology commercialization compared to the DCs, most of the technology deals in the country still have a lot of direct transactions, and the utilization of intermediary agencies is very insufficient (Technology Transactions: Direct Transactions 78.3%, Technology Intermediators 6%, Relevant Transit Transactions 8.5%, Others).

The specialized agency is expected to play an increasing role of acting on the patent transfer and commercialization through integrated management by specialized agencies to search for demand-provider relationships through reduction in transaction costs and enhancement of negotiating power based on wide network and improve the efficiency of technology transfer. The role of professional technical intermediaries in technology transactions must grow to expand technology commercialization from the public to the private in a competitive manner and based on the market principle.

# 2. Suggestions for TTO Activation and Development in Tunisia

Based on Korea's experience and our judgments, the suggestions for enhancing the feasibility of technology diffusion are presented as applicable implementations to supplement Tunisia's technology policy. They are expected to contribute to the acceleration and facilitation of technology diffusion in Tunisia.

## 2.1 Situation of TTOs in Tunisia

In 2012, ANPR of Tunisia has established the network of TTOs in Universities and Research centers through specific conventions. A total of 25 TTOs are available in this network as Centers of Research /Universities and Technopoles /Technical Centers.

This initiative was a bold step as it stated that the TTO has technical duties in the transfer of technology in Tunisia. One of the advantages of this initiative is that it showed the willingness of universities and Public Research Organizations (PROs) to pay attention to technology transfer. Obviously, TTOs in Tunisia played a role in raising awareness of the importance of technology transfer at universities.

Based on the preliminary survey and investigation, they recognized that the TTOs in Tunisia are in a situation of weakness and difficulties. Most of the TTOs are identified to be lacking resources, budget, procedures, guides, contracts, etc.;

- For University Sfax and Sousse, they are active with their establishment but have no visibility with ANPR despite the contract between them.
- Lack of resources for organization
- Difficulty in the identification of laboratories results
- Lack of infrastructure and clusters bringing together different stakeholders
- Budget limitation

- Lack of information about financing mechanisms

As weaknesses for this implementation, not all TTOs have worked with the same motivation. TTO missions require multiple experts to provide the proper services for the supplier and demander. One person in the Agency cannot handle all the tasks of TTO. The agency itself does not have sufficient resources to ensure assistance to all partners. Moreover, the collaboration between the research sectors and industry sectors requires a national and regional coordination structure facilitating systematic exchange between supply and demand.

## 2.2 Challenges for the Development of University TTOs in Tunisia

A developing country should compare the best experiences of an advanced country for TTO development and extract the unique model for implementation considering the Tunisian situation. In that sense, the Korean experience clearly demonstrates a good example of technology transfer structures of TTO, which were established by the initial investment and strong support in the beginning stage of implementation for the TLO policy.

### 2.2.1 Initiate the enactment to promote technology transfer and TTO

In the early stages of development of the TTO, as in Korea, it is necessary to establish a legal infrastructure to promote technology transfer on a legal basis in the country. A national system for the construction and support of the TTO should be established in such legal infra.

### 2.2.2 Set up the basic fund and minimum full-time manpower

It is most urgent for universities and research centers to focus policies and incentives on TTO revitalization, and the country should select the excellent TTOs through appropriate evaluation procedures to support the budgets that allow the TTOs to have minimum operations and full-time workforce.

### 2.2.3 Application of Work Manual and Process

A significant number of TLOs have difficulty in carrying out technology transfer business due to insufficient basic operating system such as work manual and standard contract form. Complementing and improving the work manuals should be produced and distributed. Enhancement of user convenience should also be implemented by providing services based on the Internet.

### 2.2.4 Fair Evaluation and Strengthening of Performance Management of TTO

After a country selects and supports excellent TTOs, appropriate assessment and evaluation should be made through a reasonable, fair evaluation system. TLOs could be evaluated as performance-focused, such as the number of technology transfers and royalty income and support for differences in project expenses. After the appropriate evaluation, the carrot and stick principle should be followed to ensure fair follow-up and accountability that reflect the results of the evaluation. To this end, the online community should be activated, and real-time management of technology transfer and related information by TTO should be carried out.

Just because a country makes a law for a specific purpose does not mean that it is the rule of law. The development of a country depends on the level of compliance of the people who follow and enforce the law and the level of implementation of the national system that operates the law.

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